

Shatter Clustering: Trajectory Step Threshold (t_observables) Analysis

Sweep over Default vs $t_{\text{observables}} \in \{5, 50, 1000 \text{ (Max)}\}$ in `grow\_ensemble`

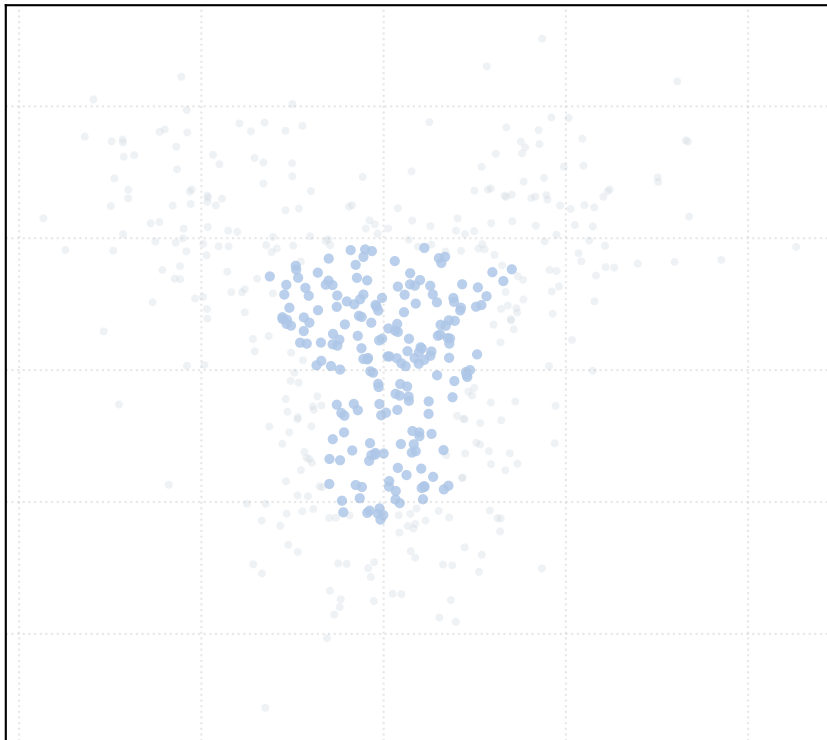
Experimental Configuration

- Total Evaluated Datasets: 30
- Tested $t_{\text{observables}}$ Values: Default (-1), 5, 50, 1000 (Max)
- Total Executions: 120 (4 runs per dataset)
- Fixed Default Parameters: $\alpha_{\text{MAD}} = 0.5$, $\alpha_{\text{accept}} = 0.5$, $t_{\text{percentile}} = 0.15$, $r = \text{adaptive}$
- Target Output PDF: `benchmark_report t_observables.pdf`

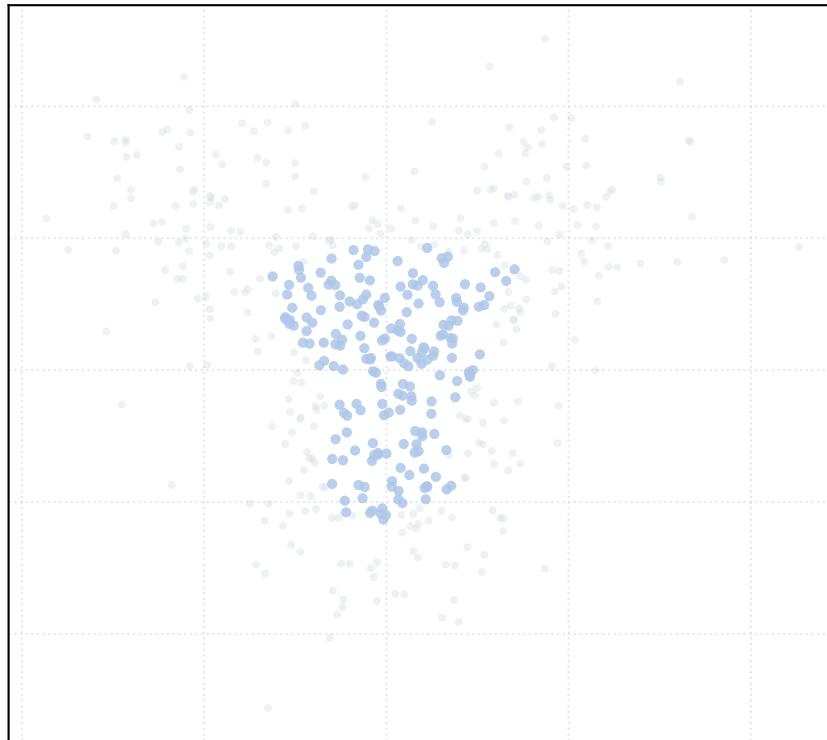
Parameter	Value / Range	Algorithmic Role
t_observables	{Default, 5, 50, 1000 (Max)}	Trajectory step count threshold in grow_ensemble
alpha_MAD	0.50 (Default)	MAD multiplier for surface growth compatibility
alpha_accept	0.50 (Default)	Log2 fold-change acceptance cutoff
t_percentile	0.15 (Default)	Top 15% densest seeds
mean_dist_quant	0.15 (Default)	Quantile for adaptive search radius r
min_intersection	1 (Default)	Overlapping vectors to trigger merge

Dataset: bifurcation_2way_noise_high | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

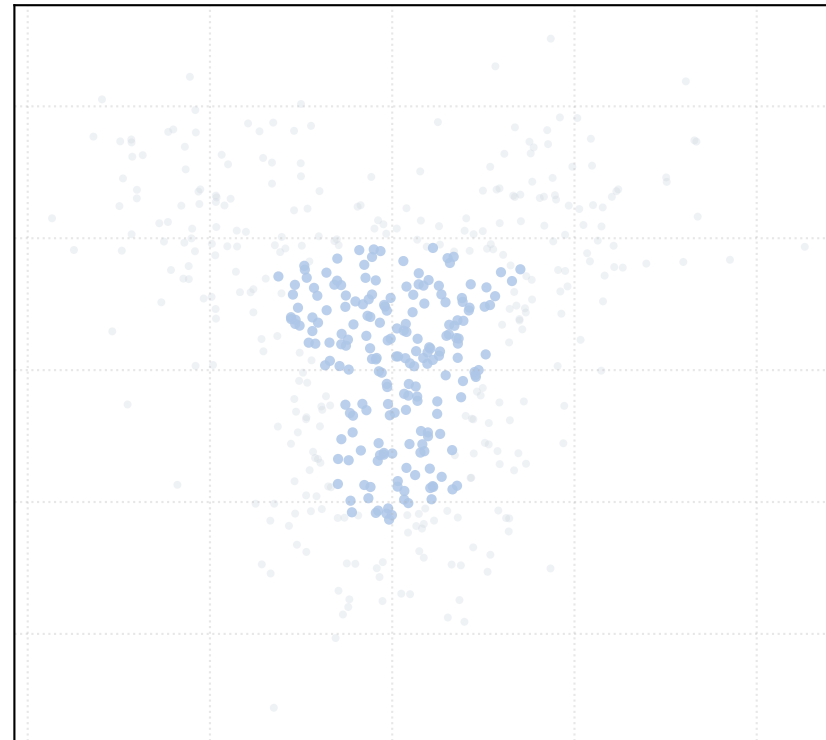
t_obs = Default [1 Clusters]



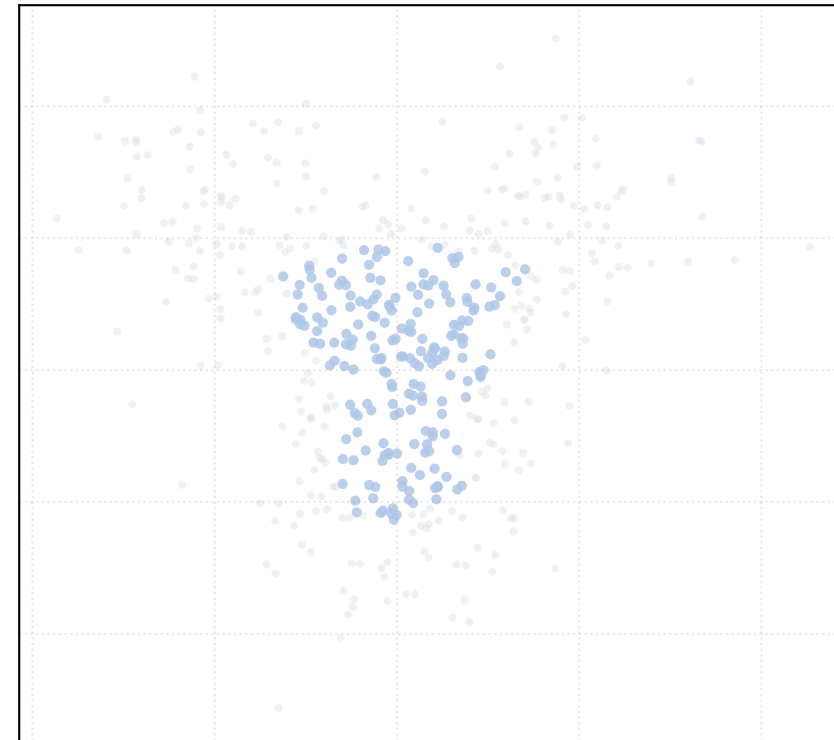
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

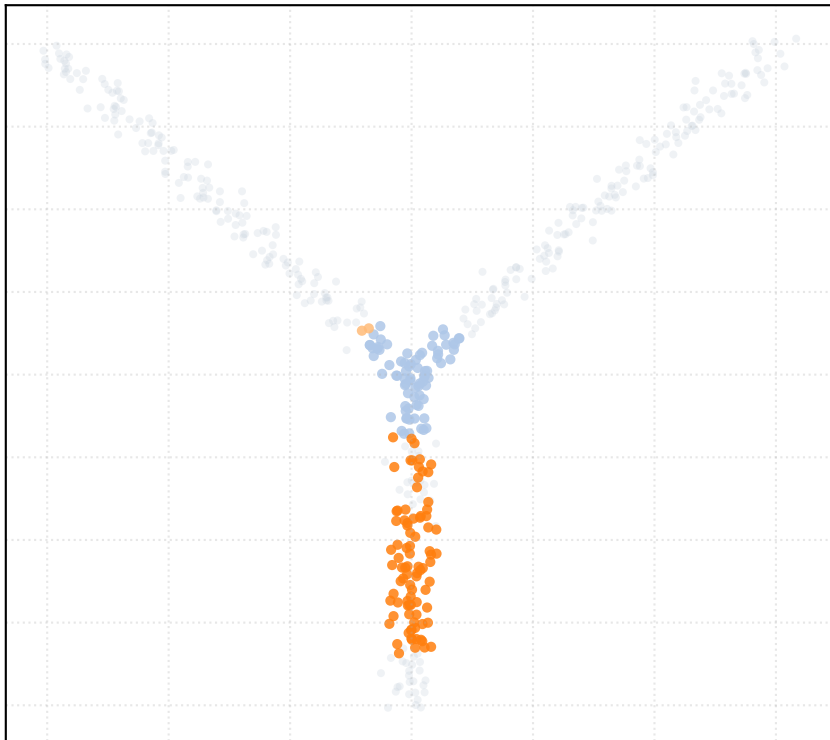


t_obs = 1000 (Max) [1 Clusters]

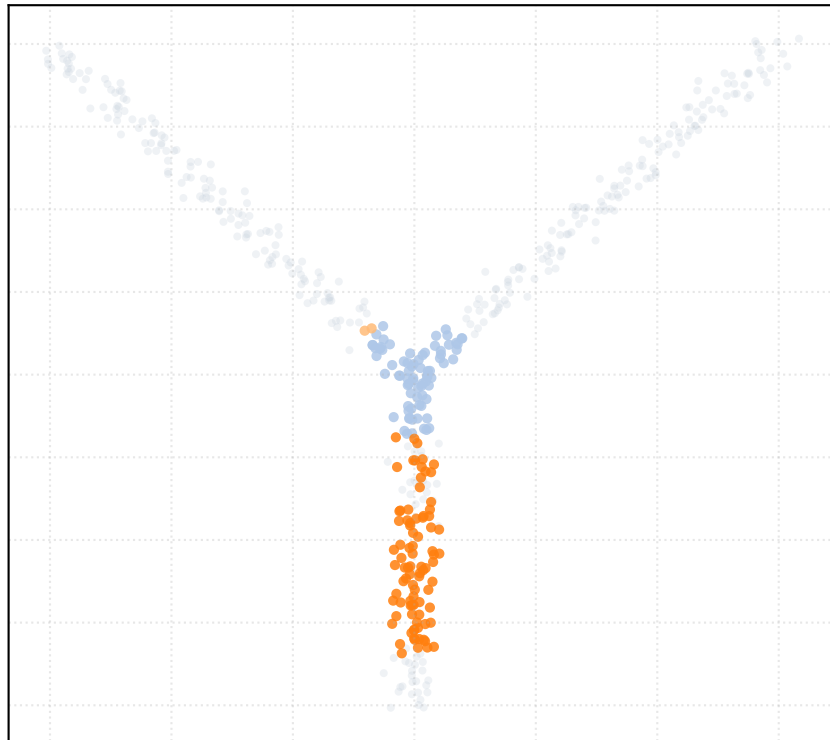


Dataset: bifurcation_2way_noise_low | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

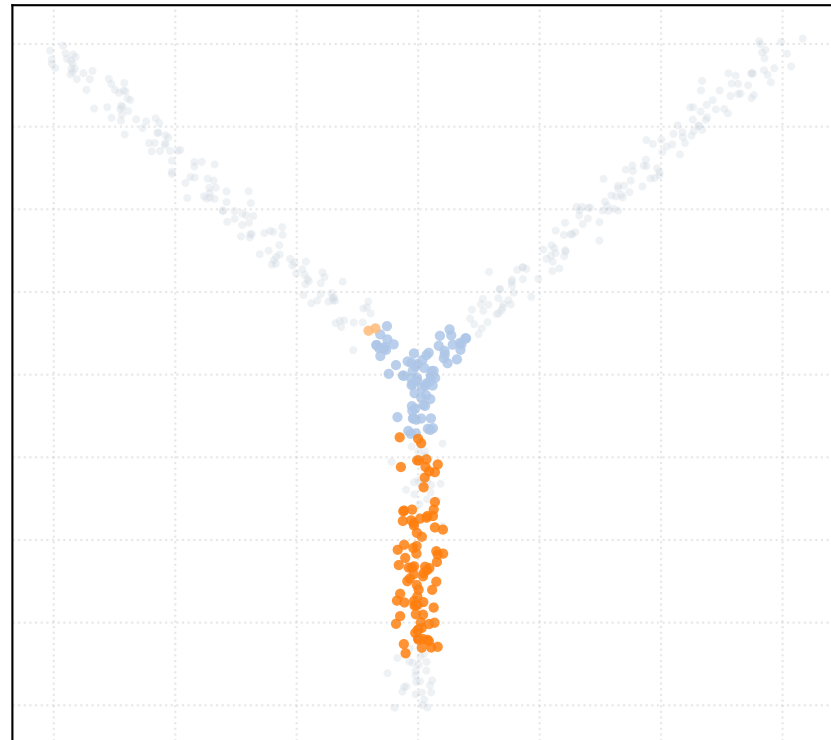
t_obs = Default [3 Clusters]



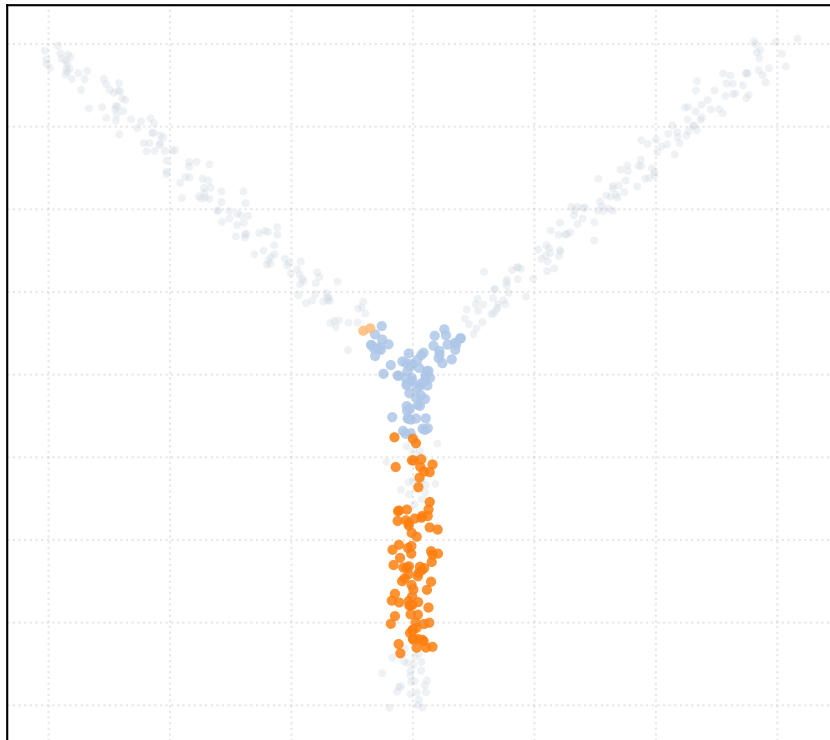
t_obs = 5 [3 Clusters]



t_obs = 50 [3 Clusters]

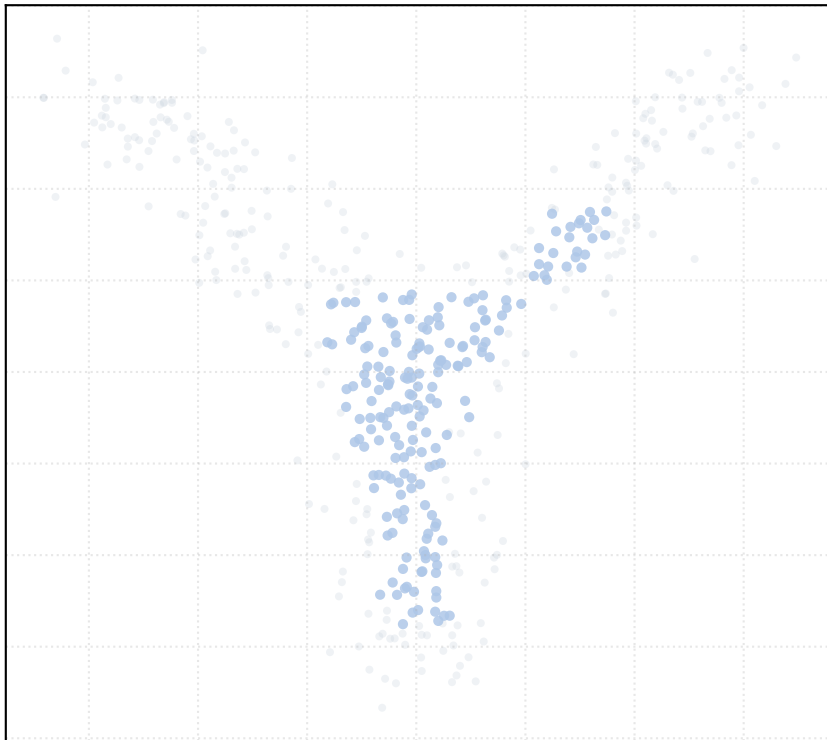


t_obs = 1000 (Max) [3 Clusters]

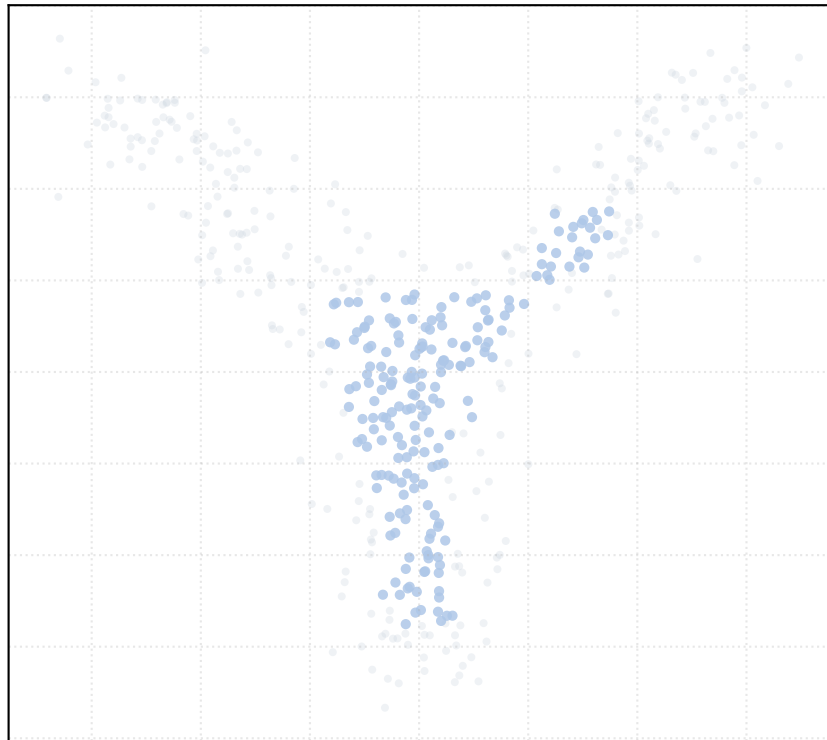


Dataset: bifurcation_2way_noise_medium | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

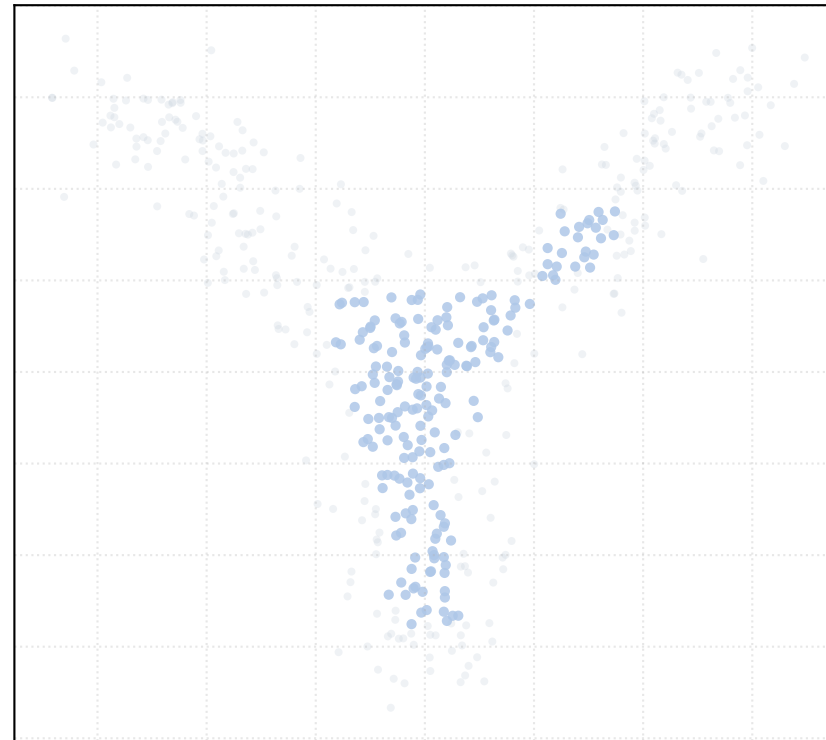
t_obs = Default [1 Clusters]



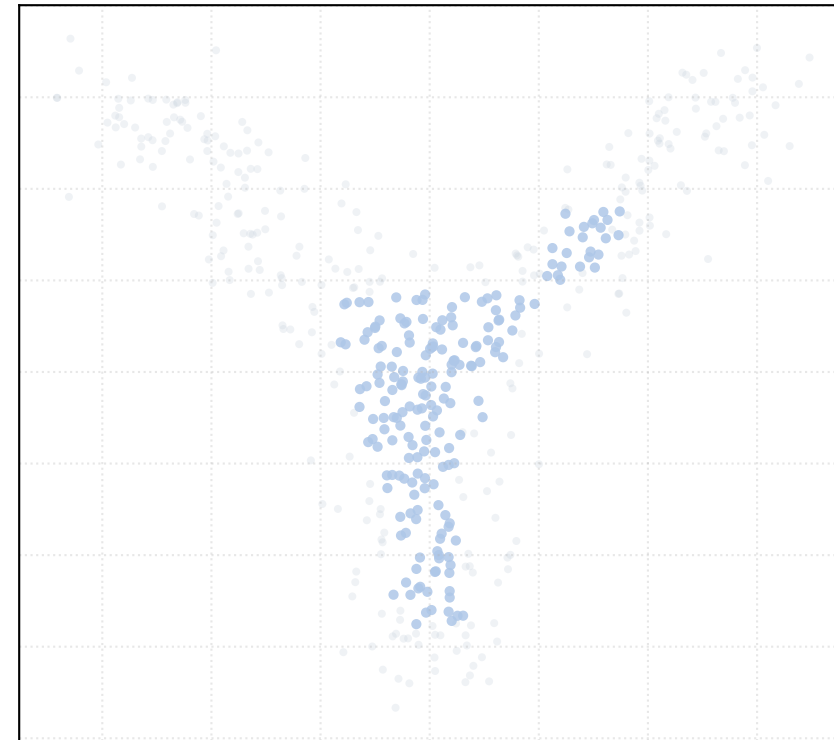
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

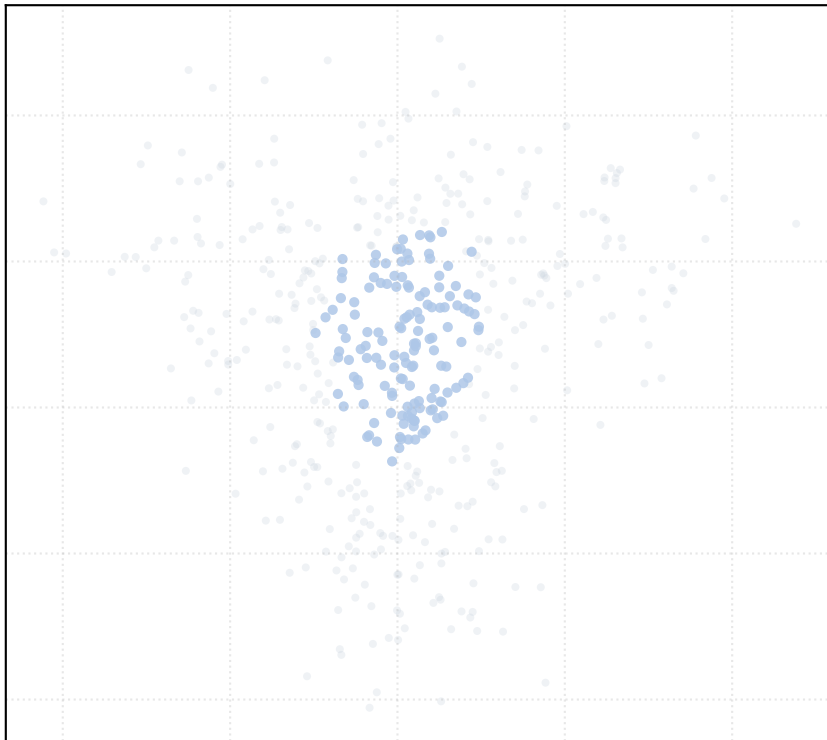


t_obs = 1000 (Max) [1 Clusters]

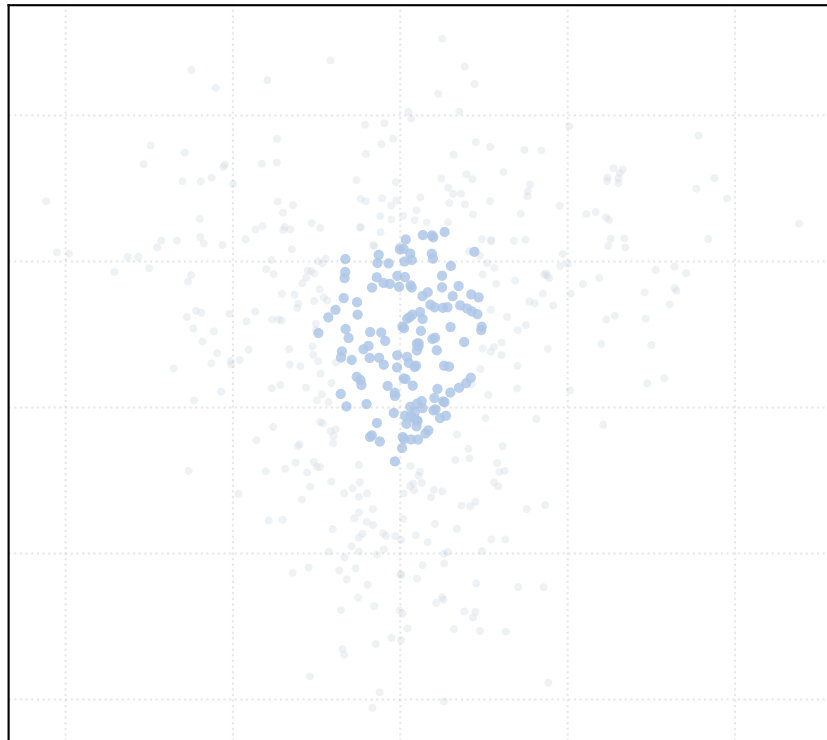


Dataset: bifurcation_3way_noise_high | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

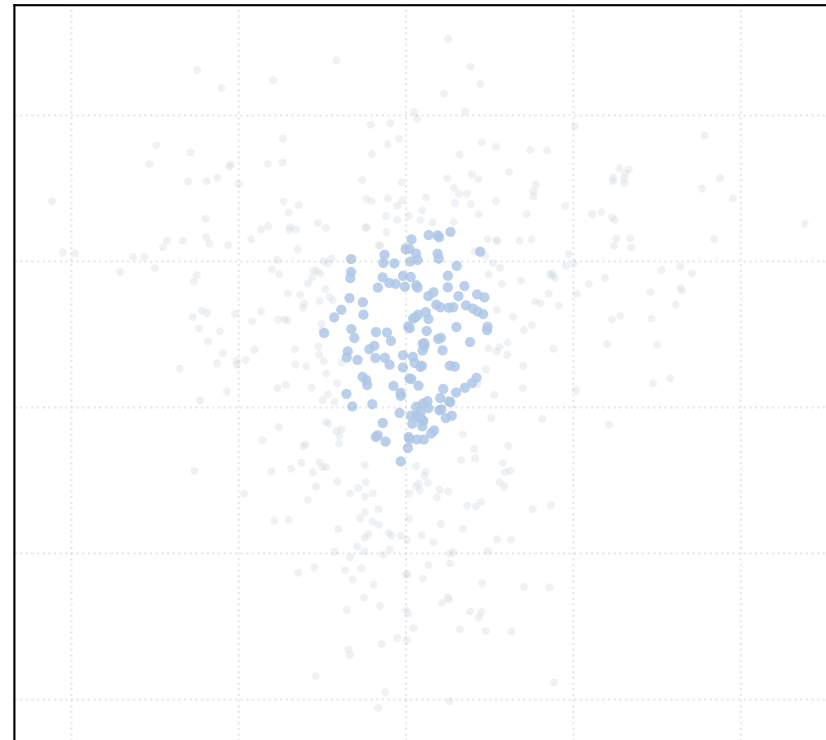
t_obs = Default [1 Clusters]



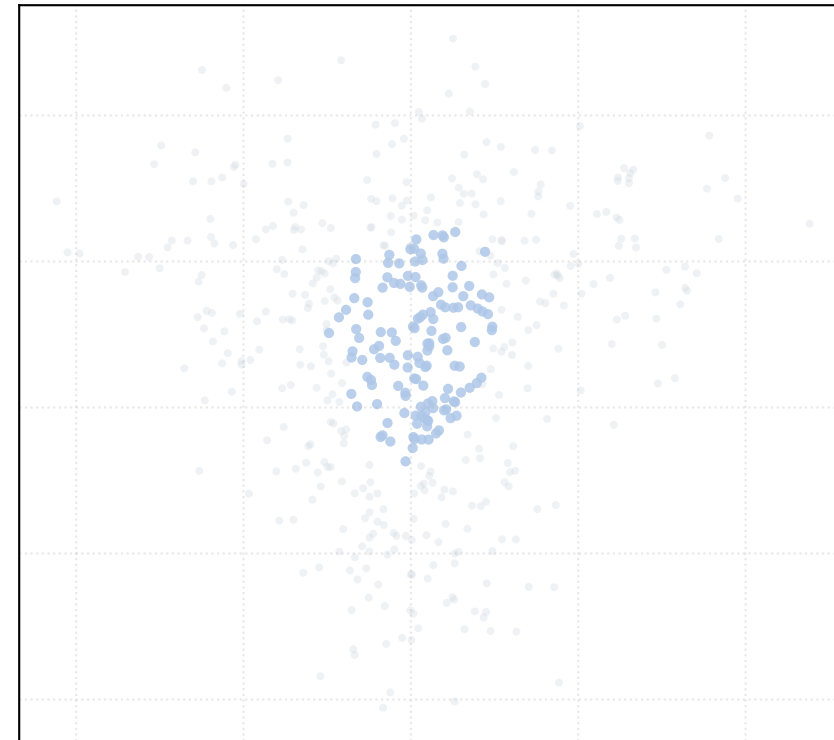
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

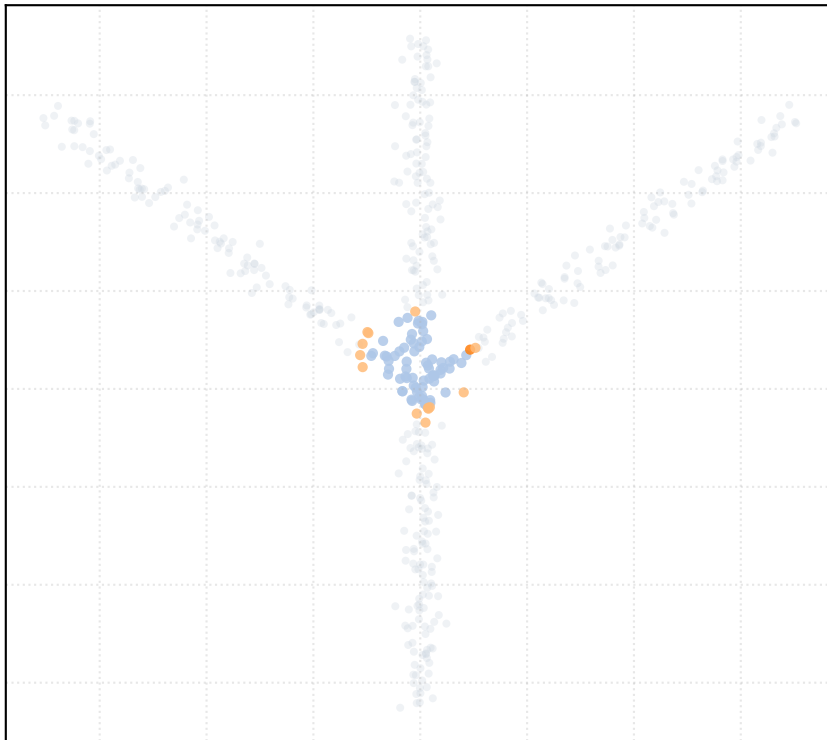


t_obs = 1000 (Max) [1 Clusters]

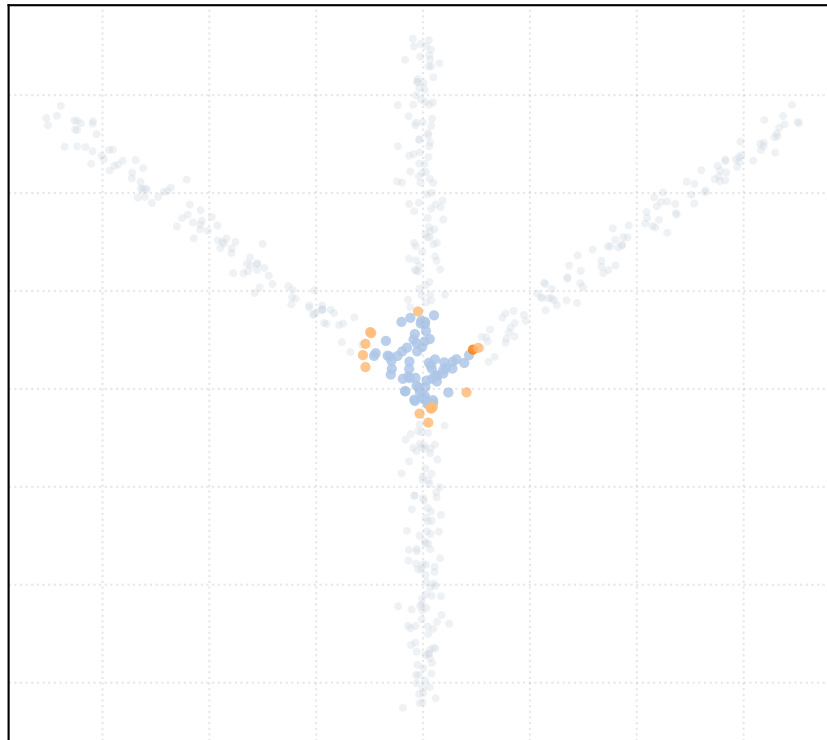


Dataset: bifurcation_3way_noise_low | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

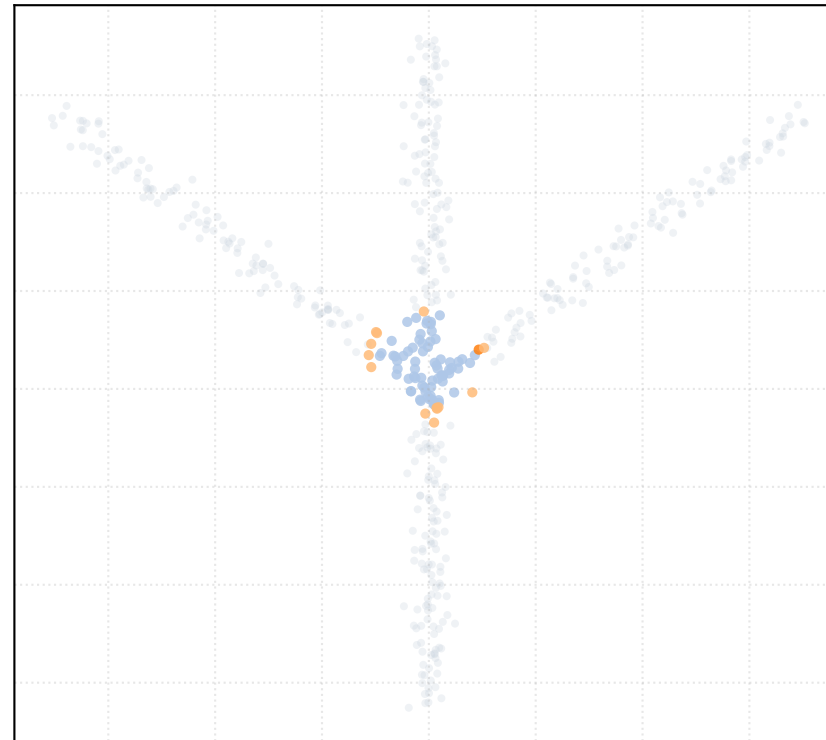
t_obs = Default [3 Clusters]



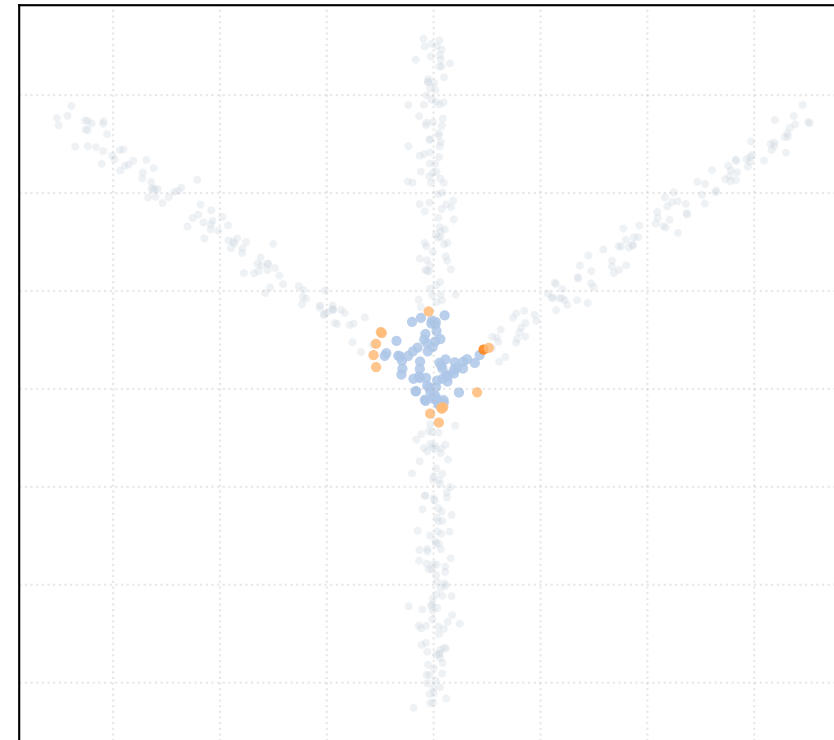
t_obs = 5 [3 Clusters]



t_obs = 50 [3 Clusters]

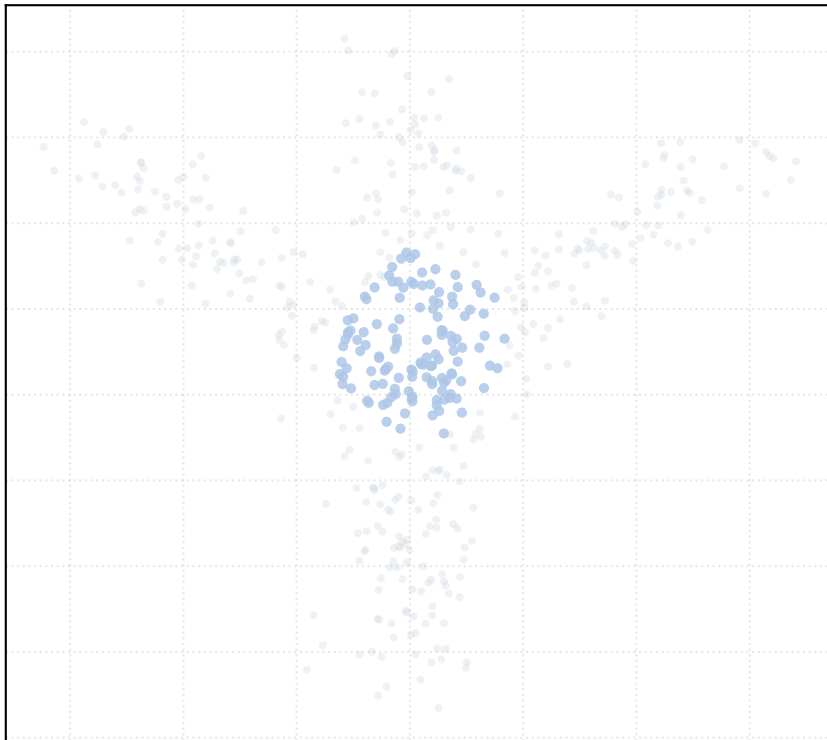


t_obs = 1000 (Max) [3 Clusters]

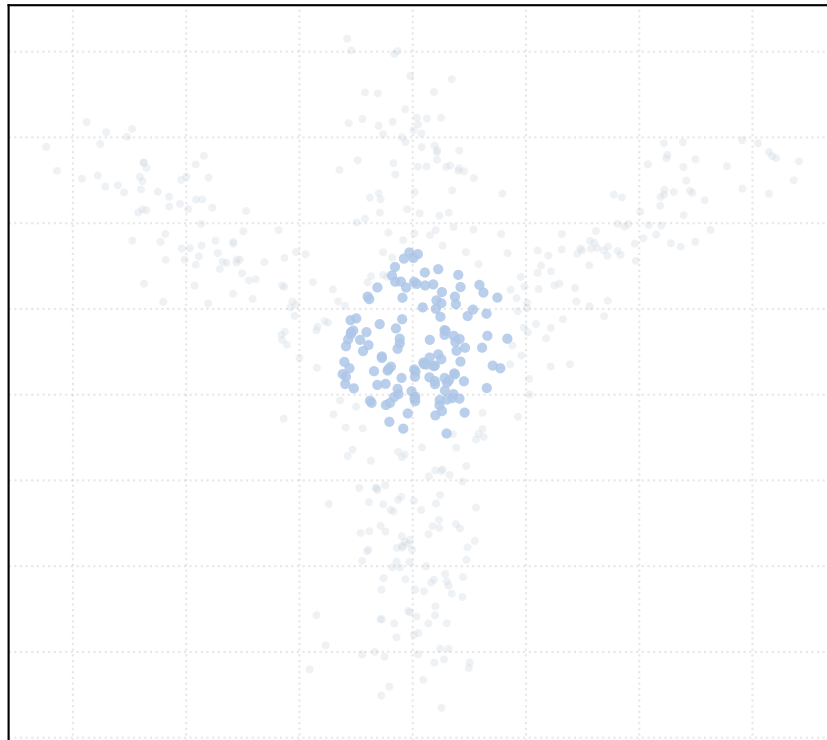


Dataset: bifurcation_3way_noise_medium | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

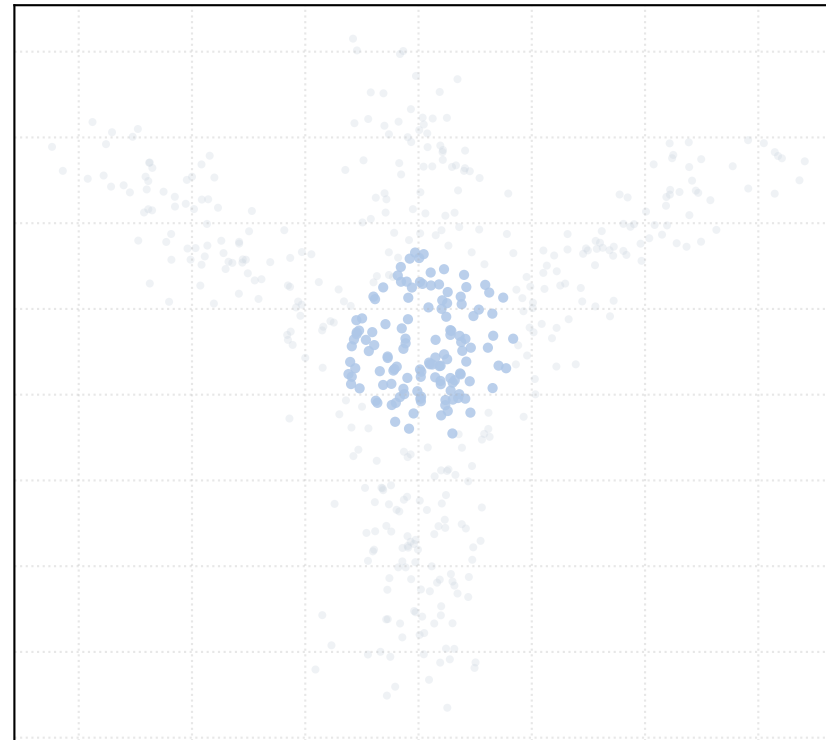
t_obs = Default [1 Clusters]



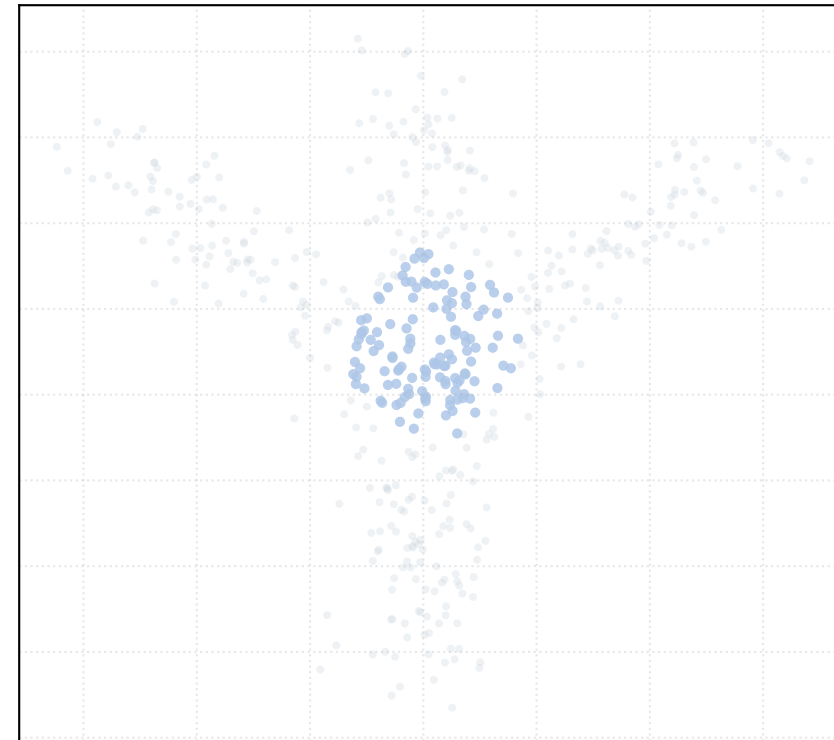
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

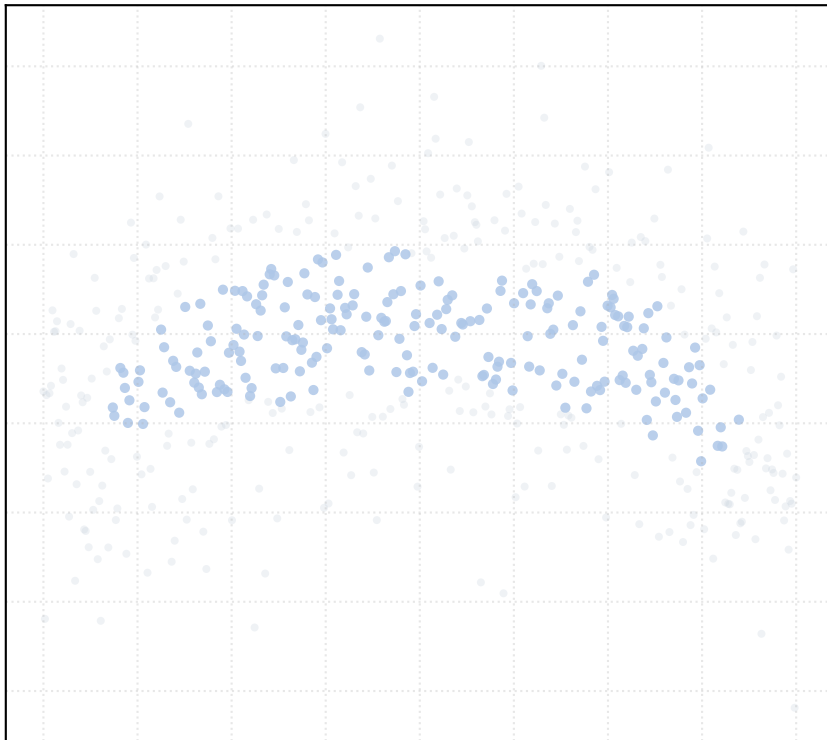


t_obs = 1000 (Max) [1 Clusters]

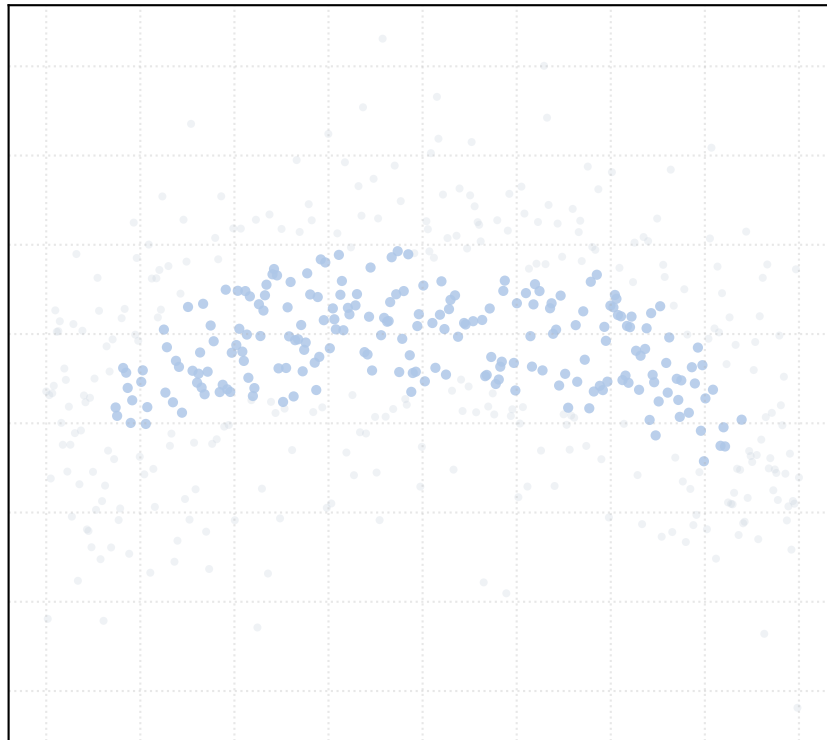


Dataset: circular_arc_noise_high | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

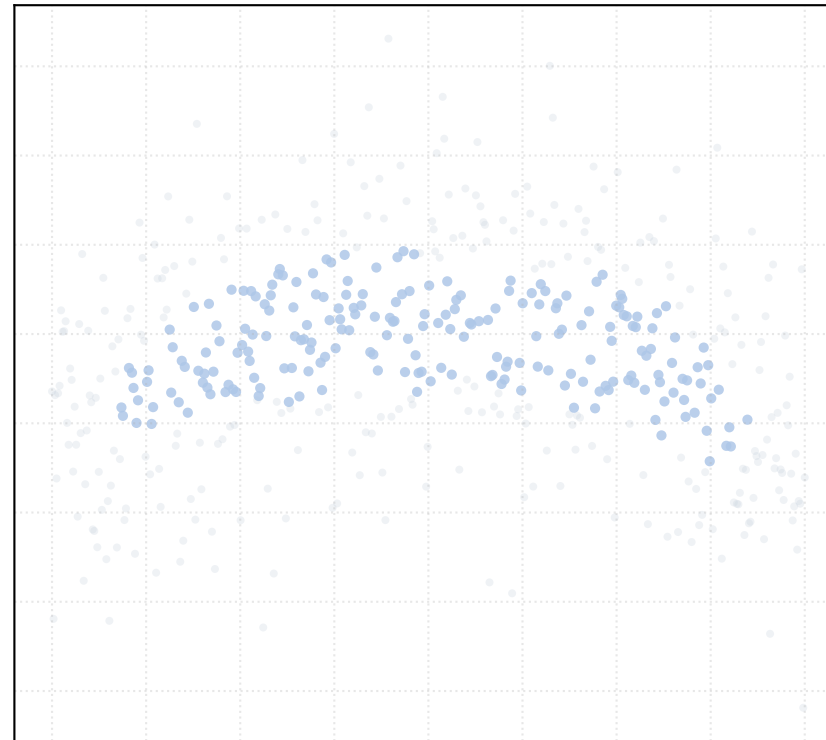
t_obs = Default [1 Clusters]



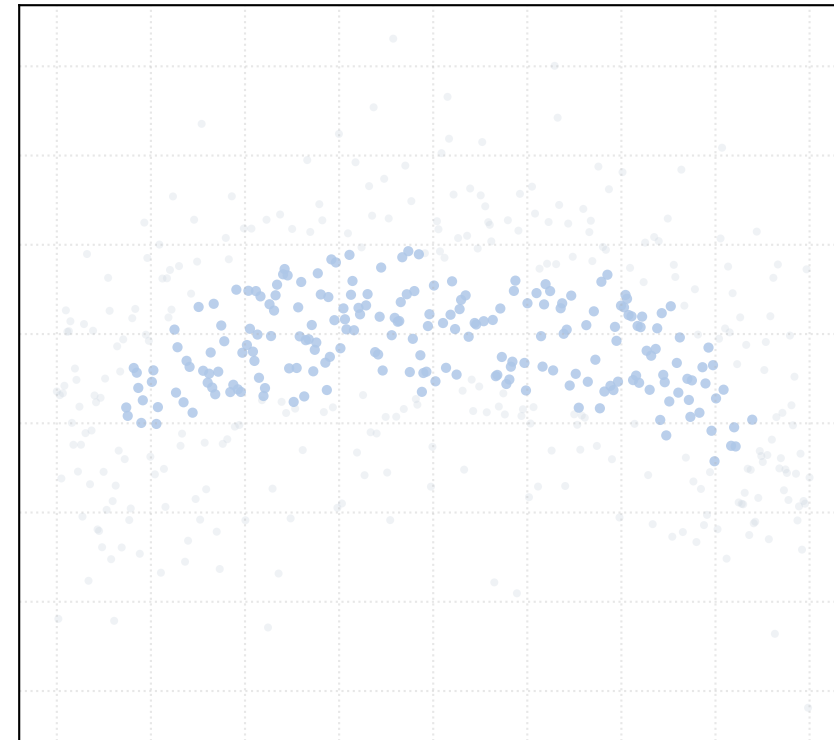
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

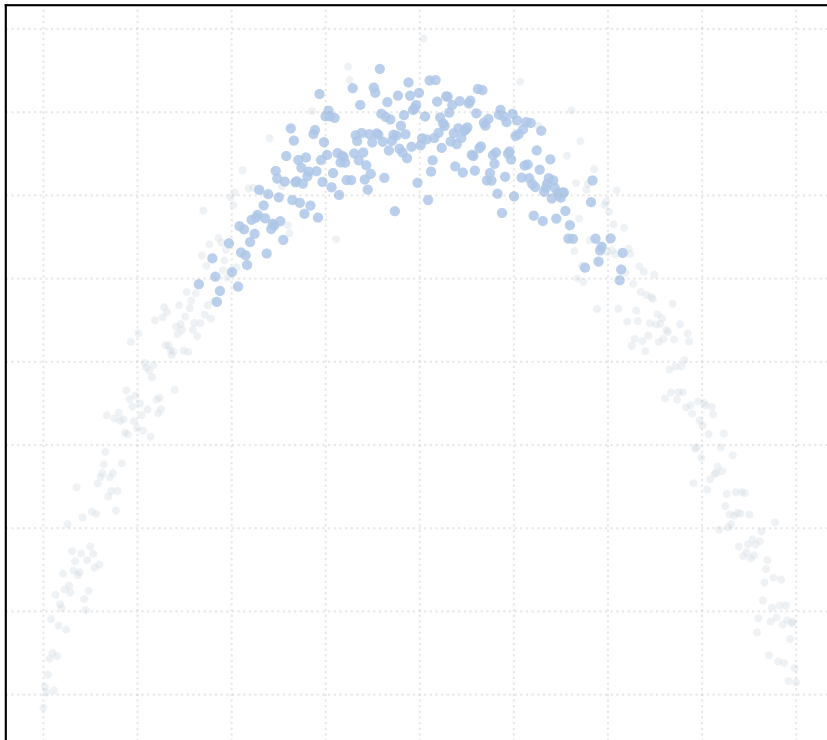


t_obs = 1000 (Max) [1 Clusters]

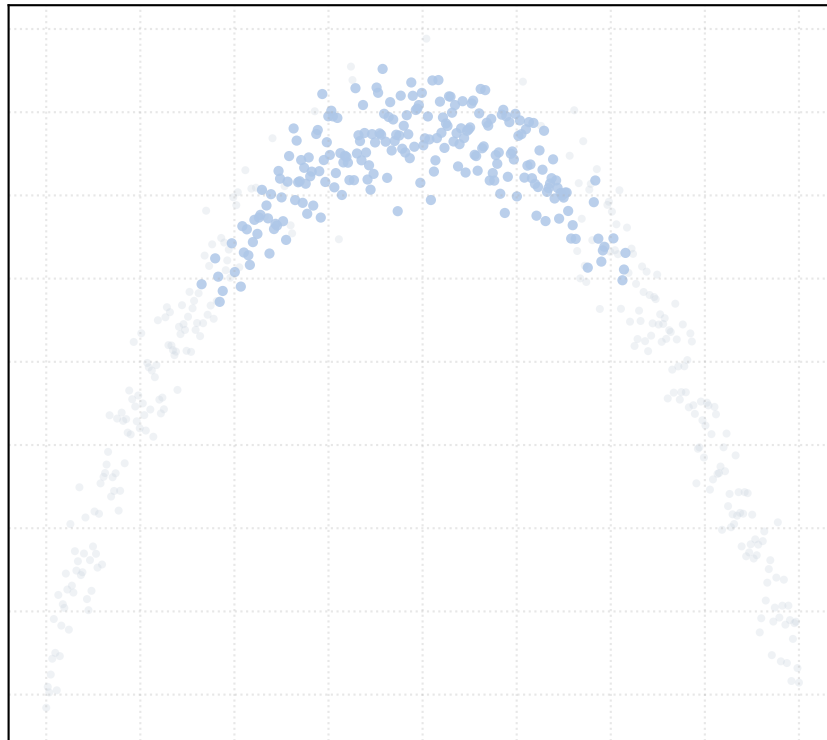


Dataset: circular_arc_noise_low | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

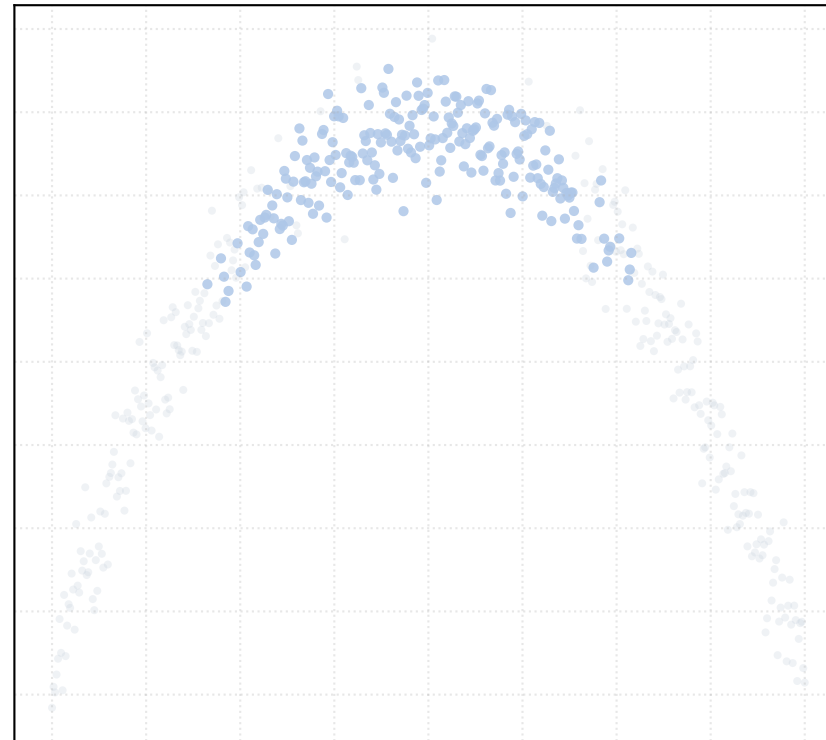
t_obs = Default [1 Clusters]



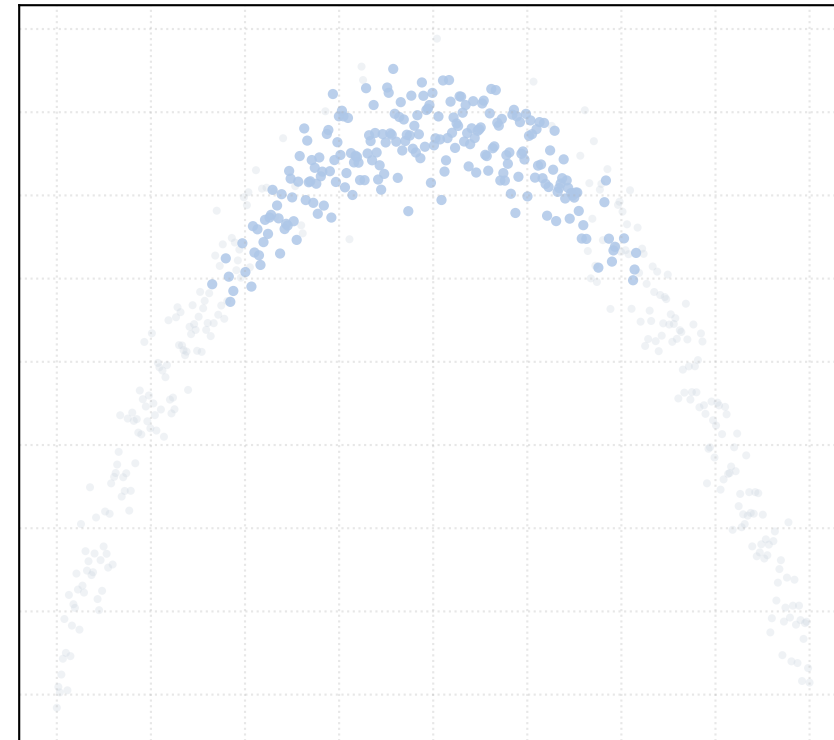
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

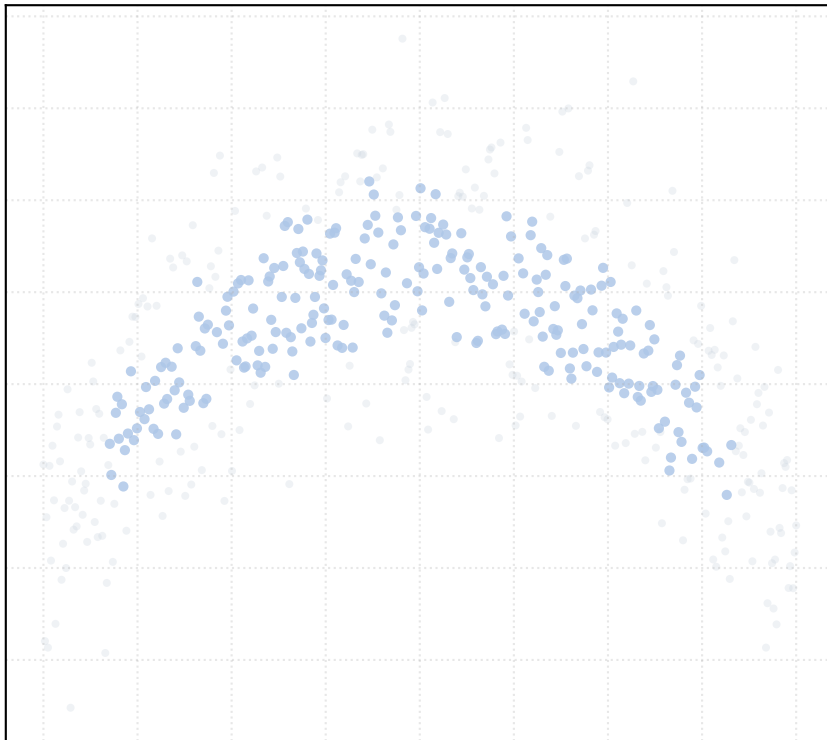


t_obs = 1000 (Max) [1 Clusters]

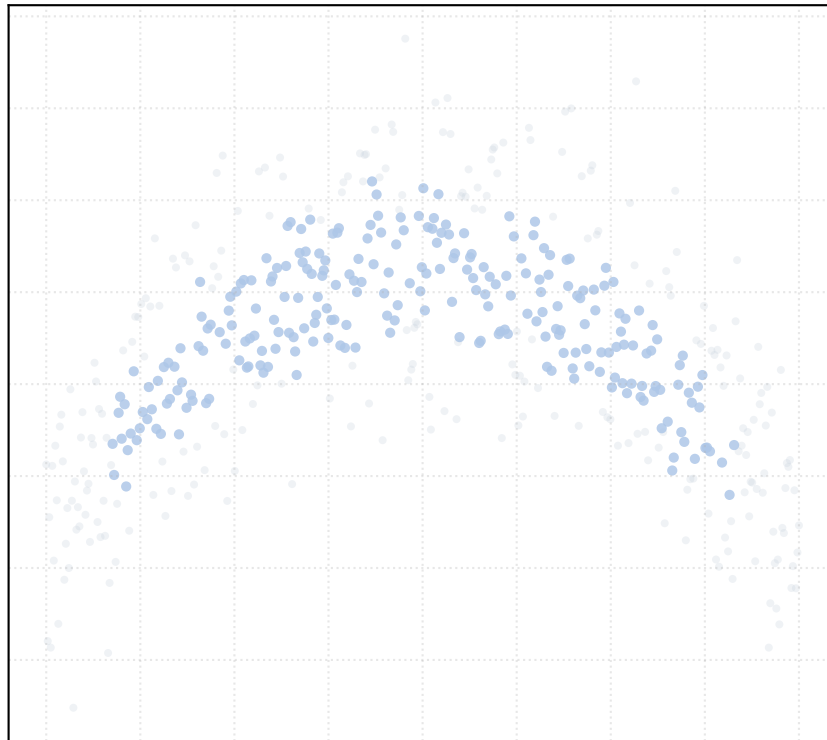


Dataset: circular_arc_noise_medium | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

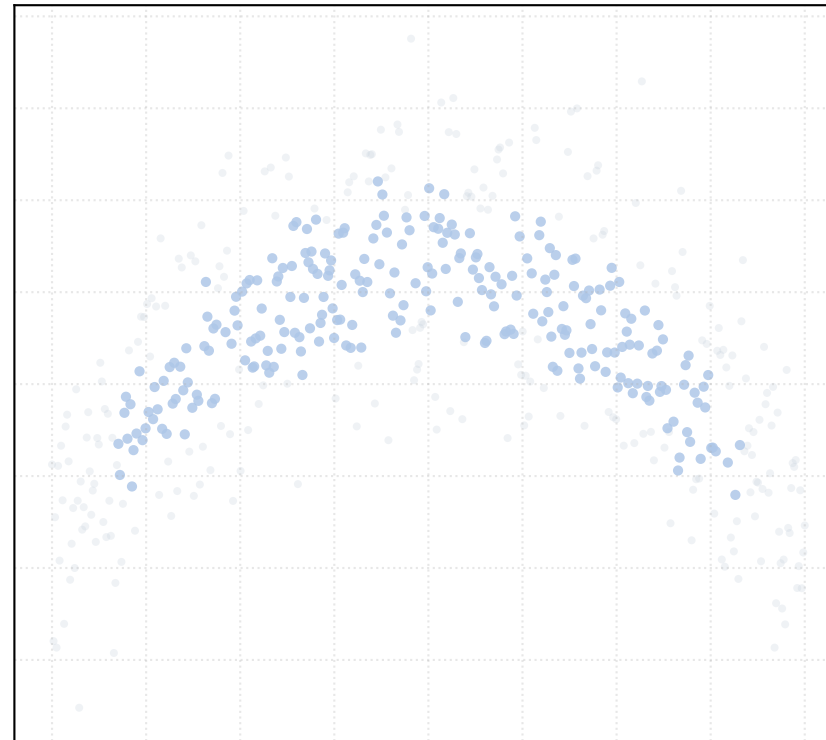
t_obs = Default [1 Clusters]



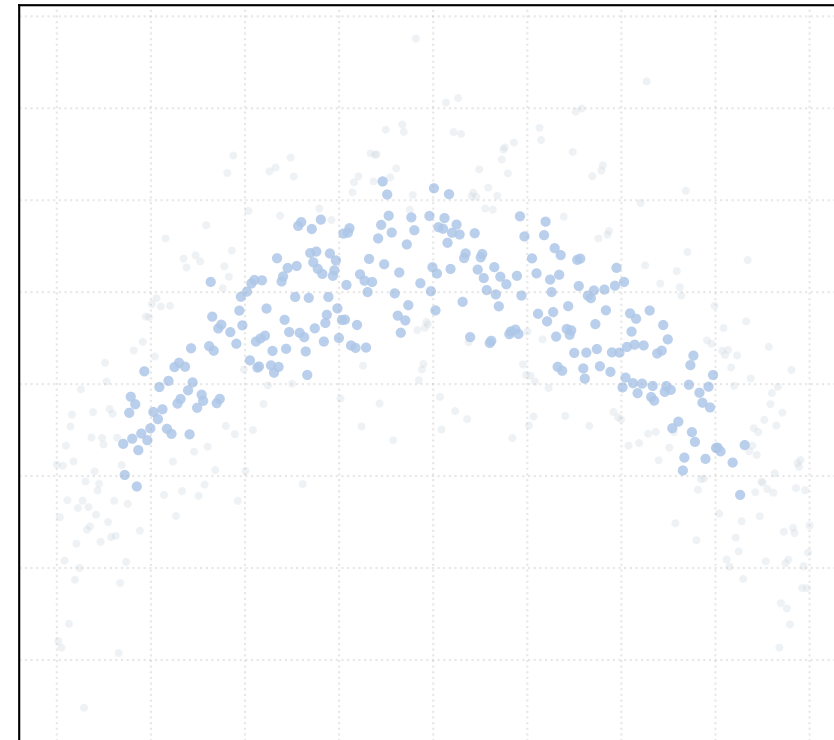
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

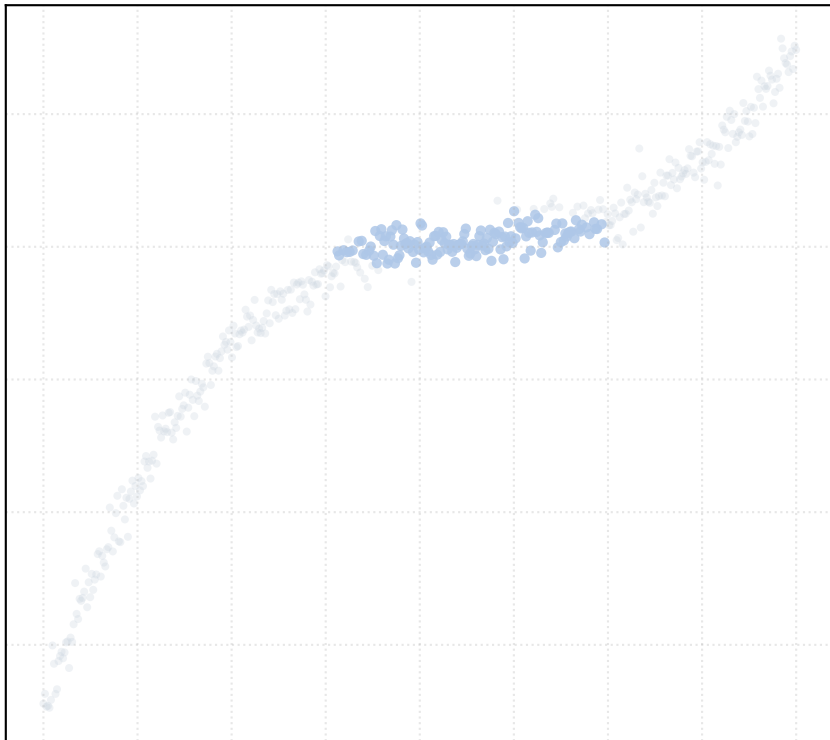


t_obs = 1000 (Max) [1 Clusters]

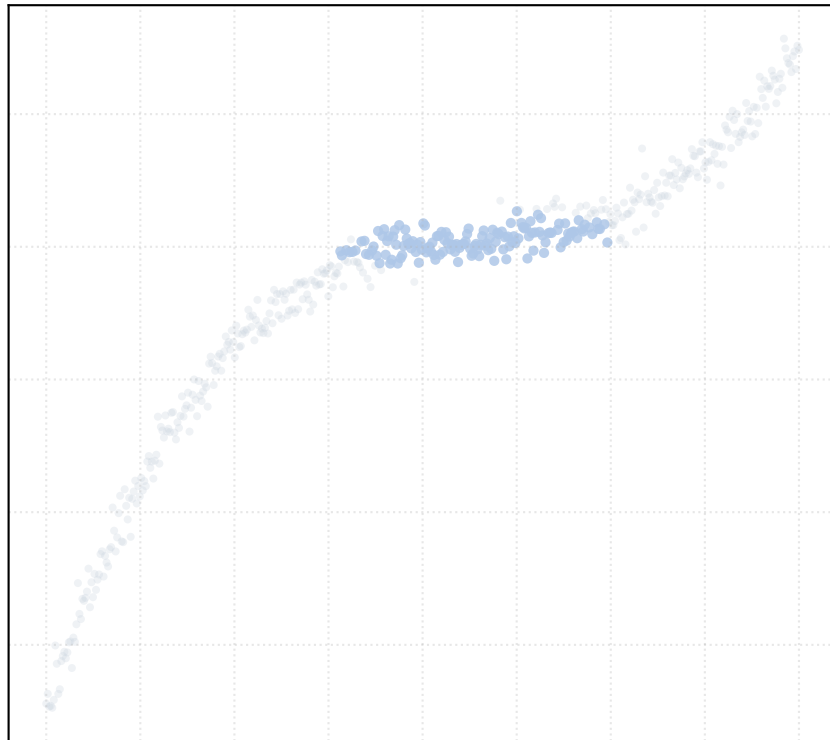


Dataset: cubic_noise_high | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

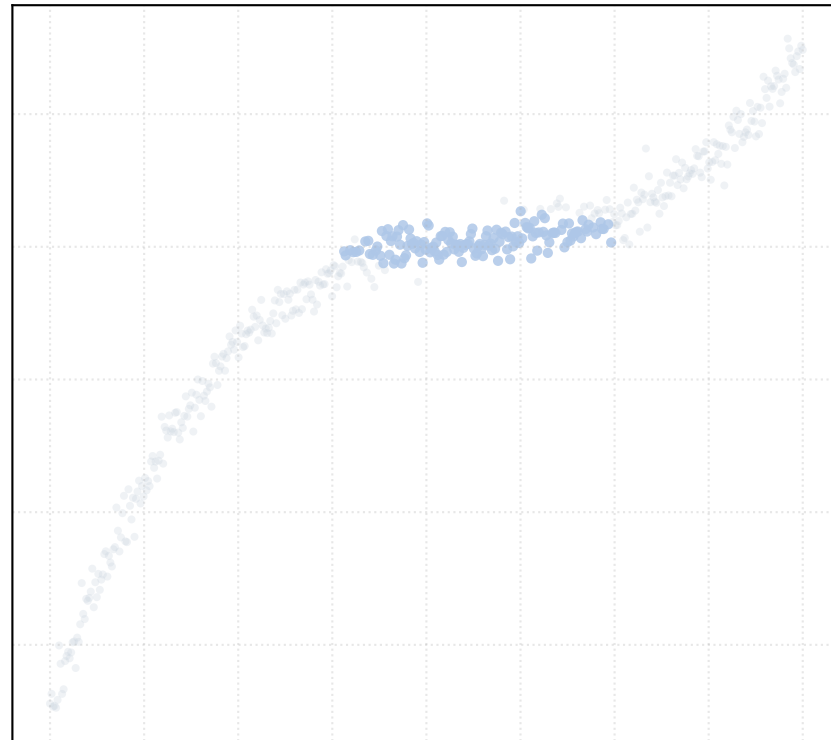
t_obs = Default [1 Clusters]



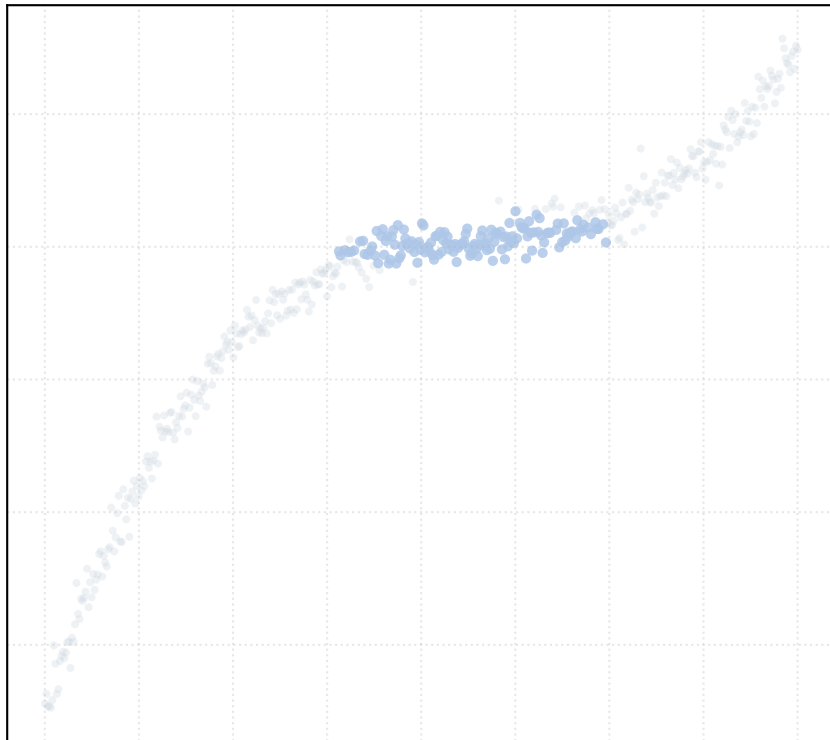
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

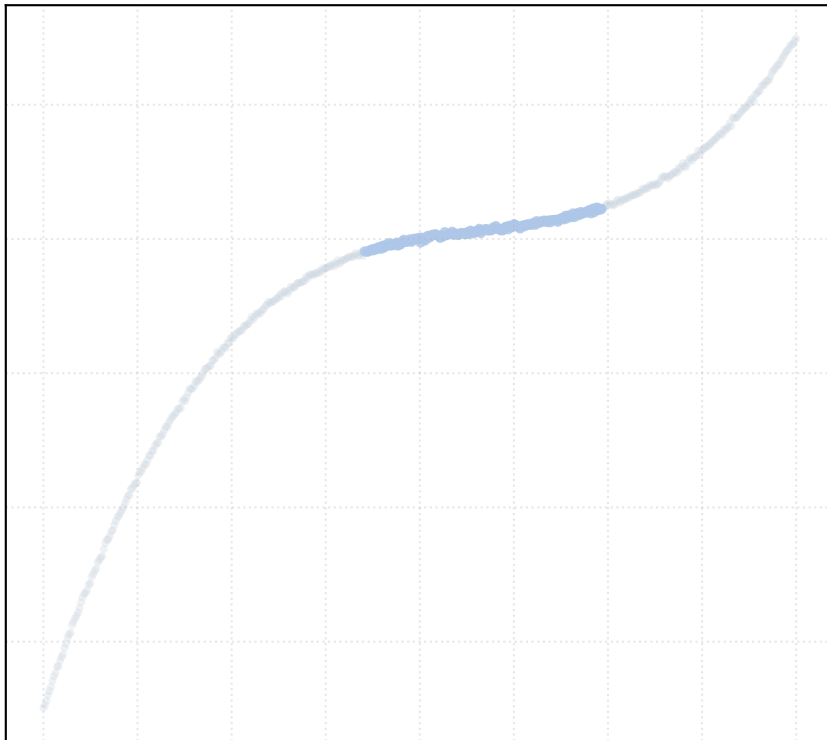


t_obs = 1000 (Max) [1 Clusters]

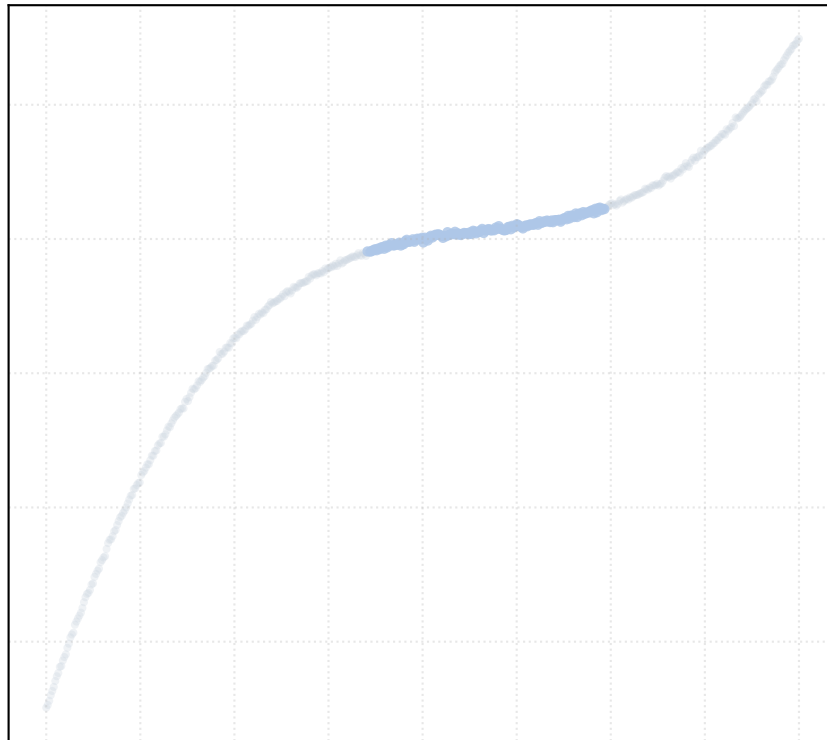


Dataset: cubic_noise_low | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

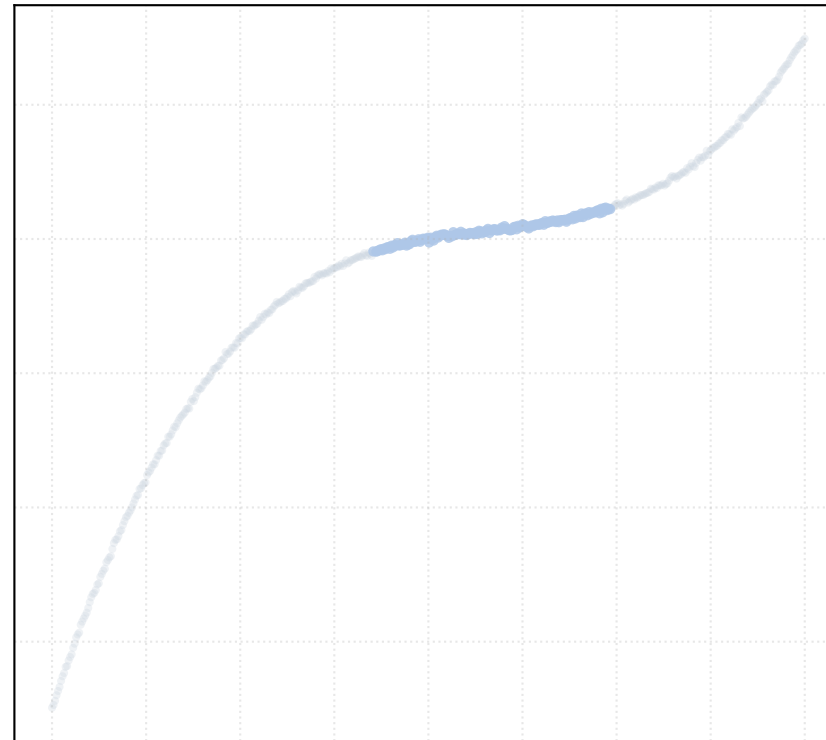
t_obs = Default [1 Clusters]



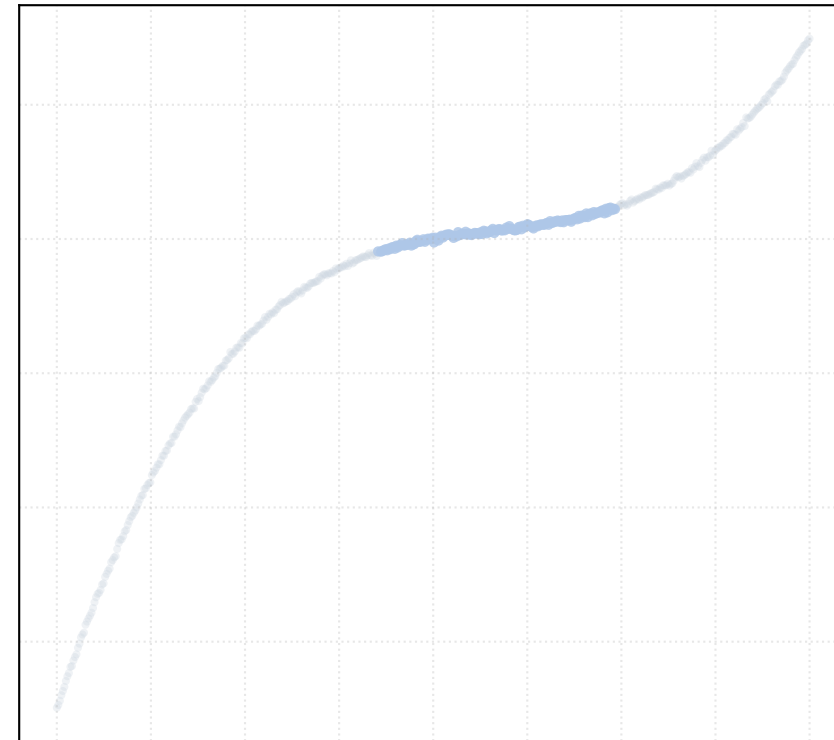
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

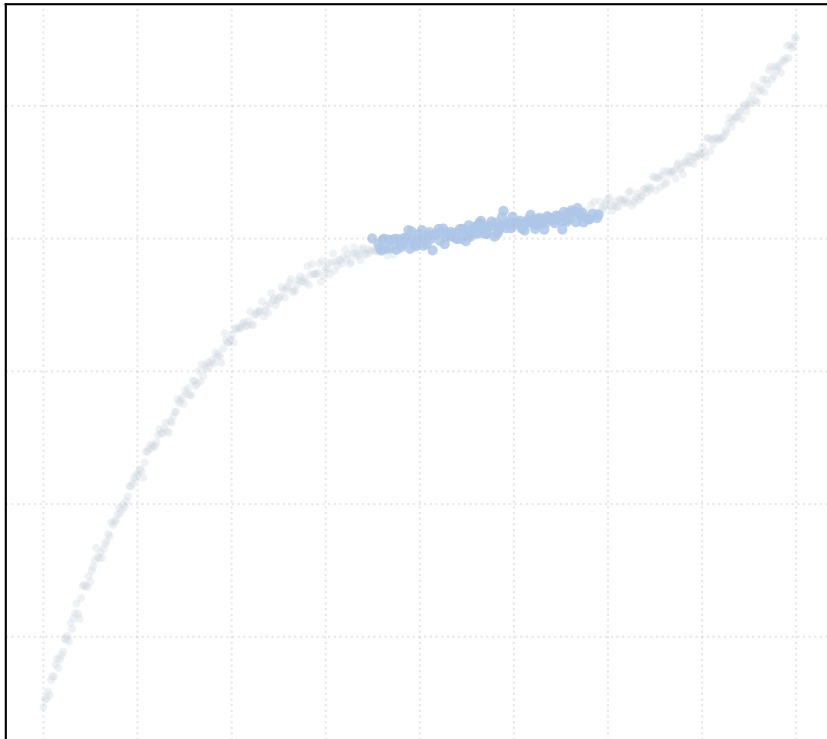


t_obs = 1000 (Max) [1 Clusters]

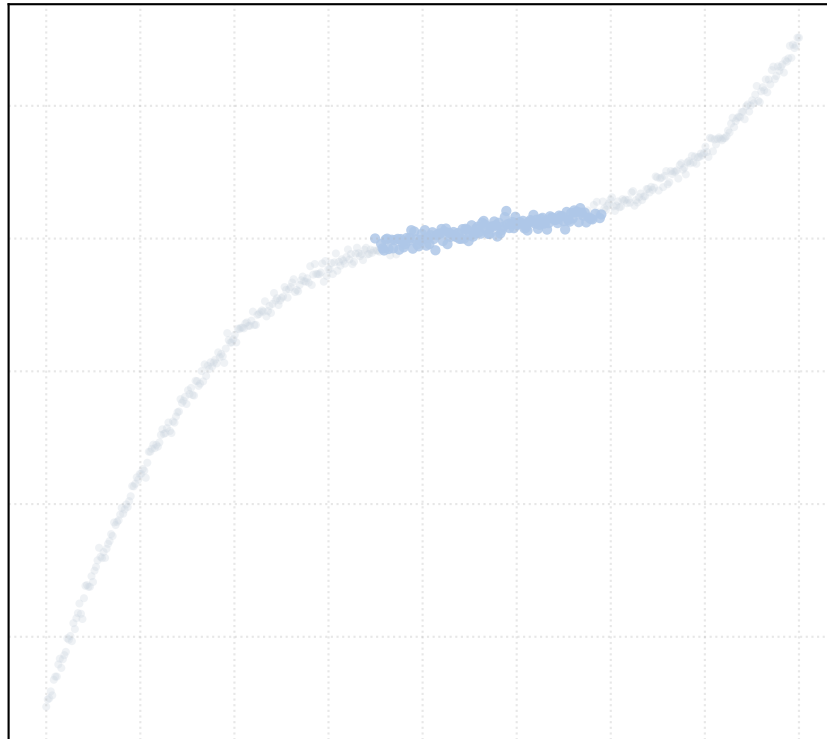


Dataset: cubic_noise_medium | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

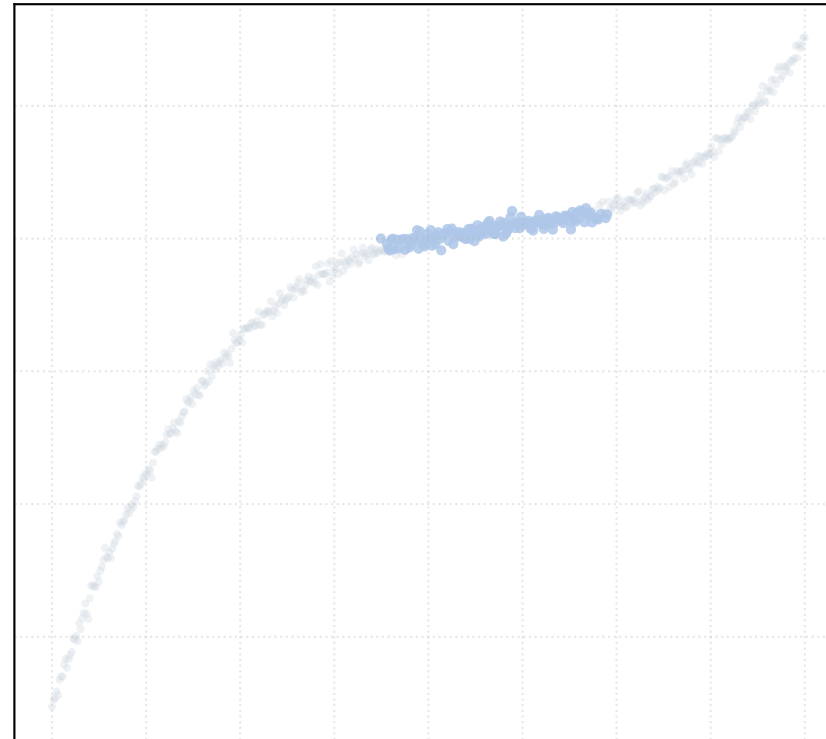
t_obs = Default [1 Clusters]



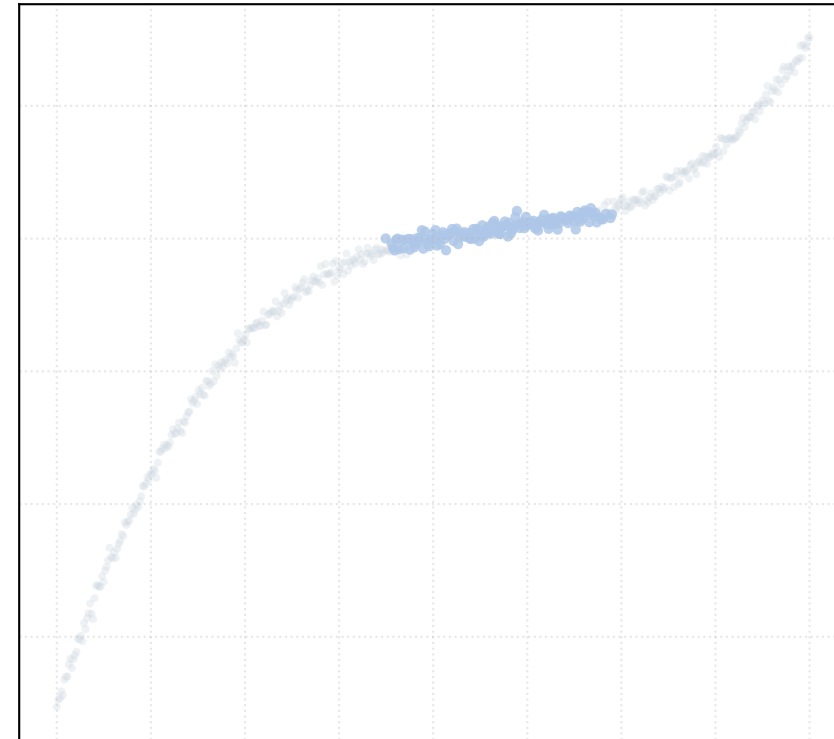
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

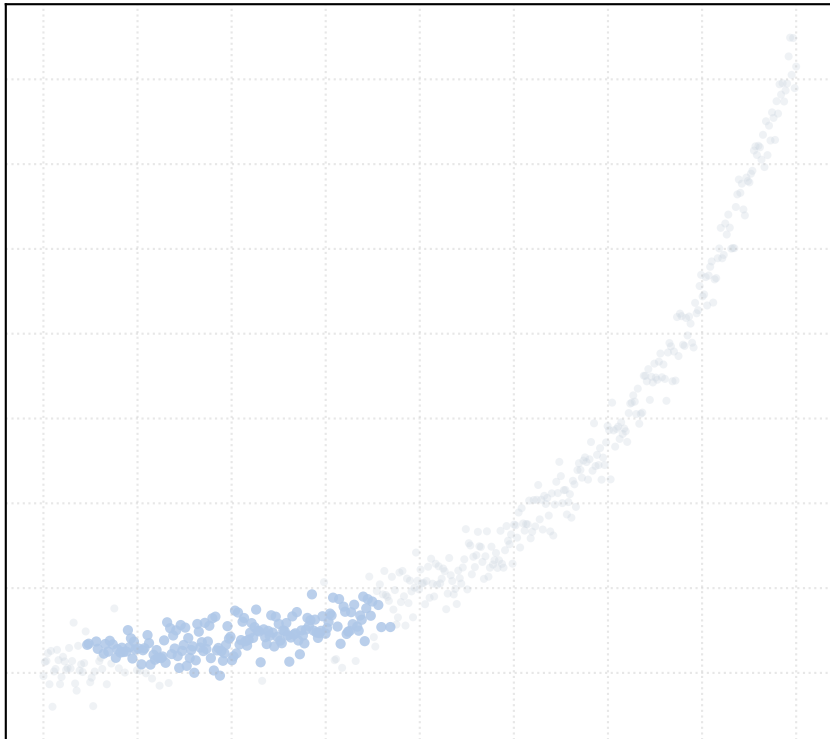


t_obs = 1000 (Max) [1 Clusters]

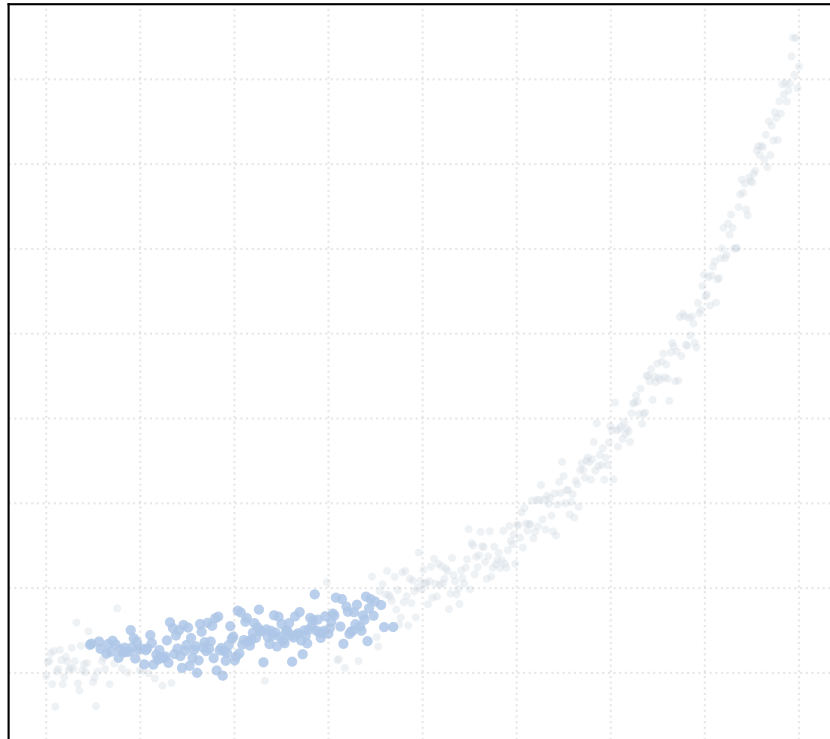


Dataset: exponential_noise_high | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

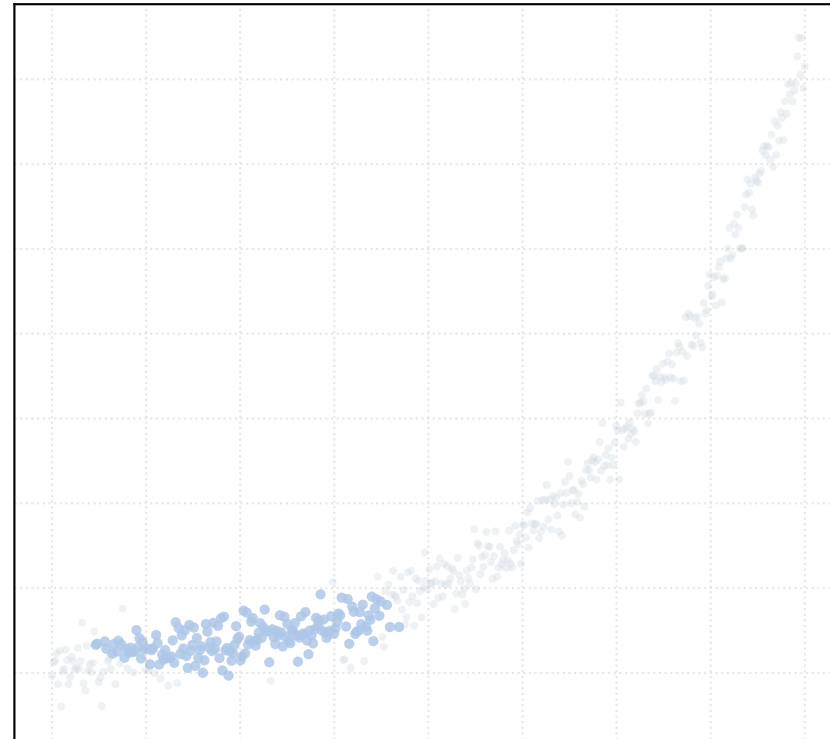
t_obs = Default [1 Clusters]



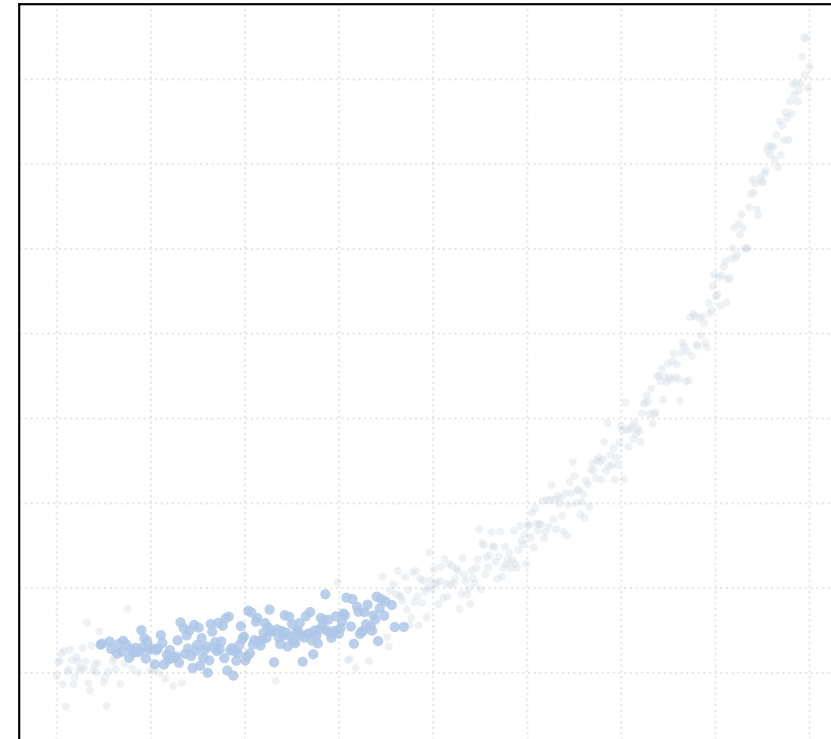
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

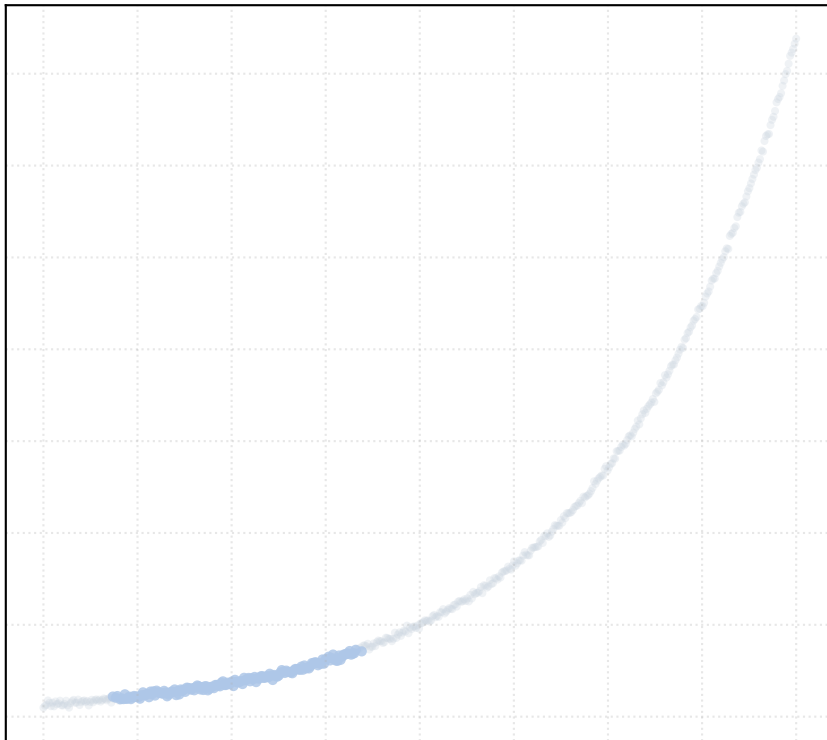


t_obs = 1000 (Max) [1 Clusters]

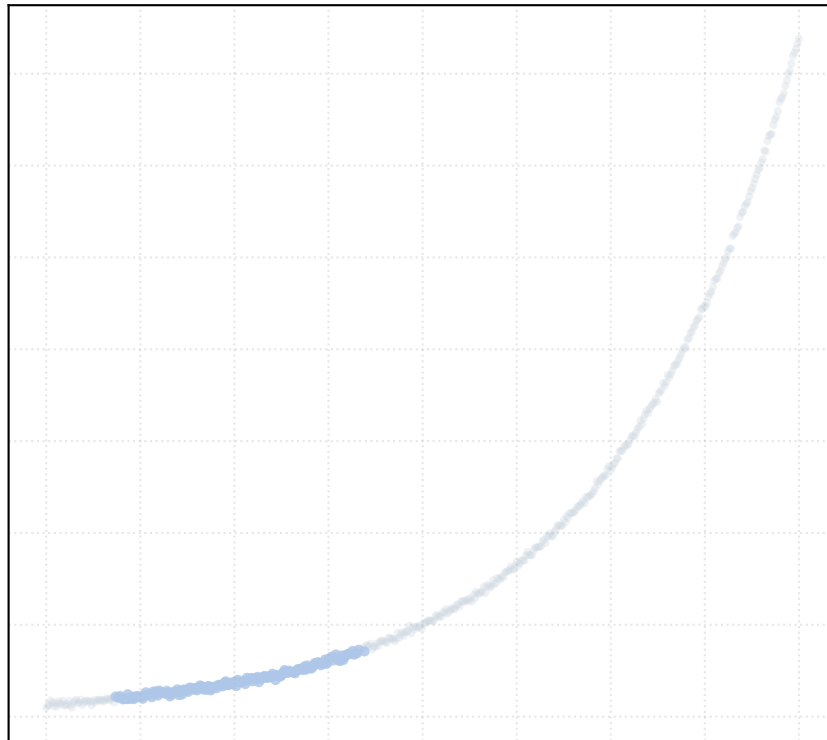


Dataset: exponential_noise_low | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

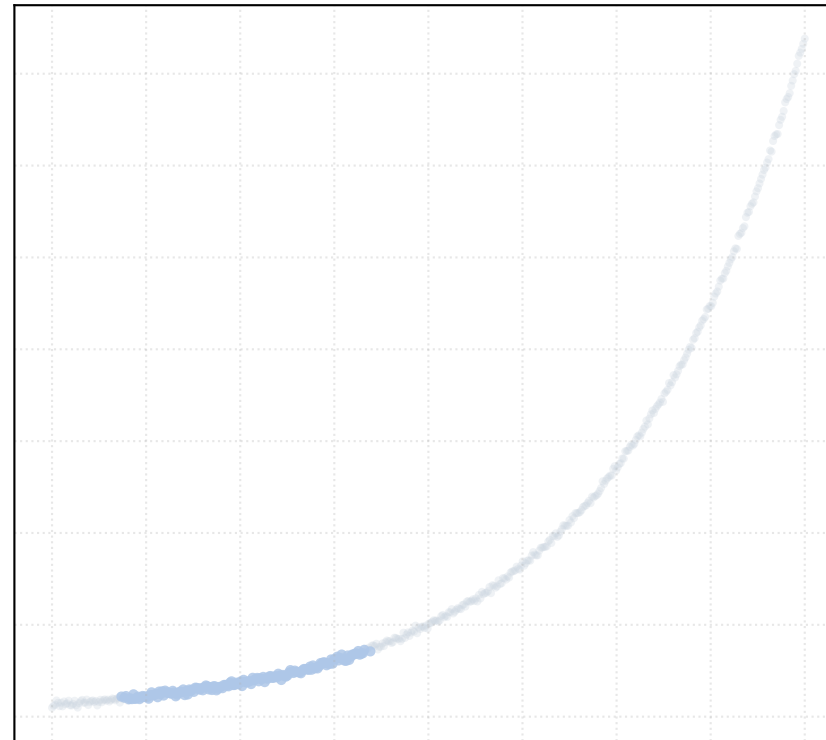
t_obs = Default [1 Clusters]



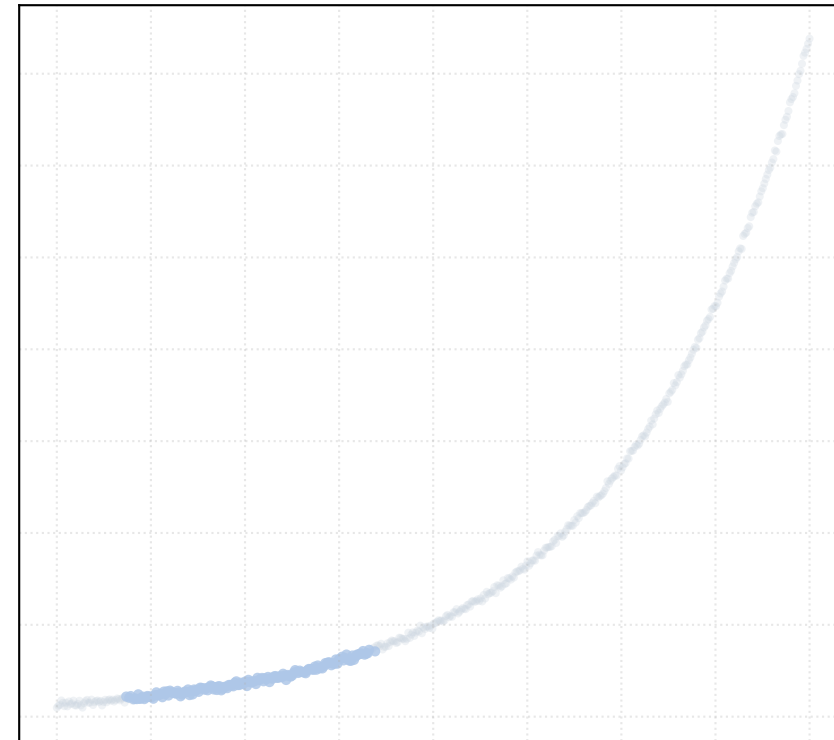
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

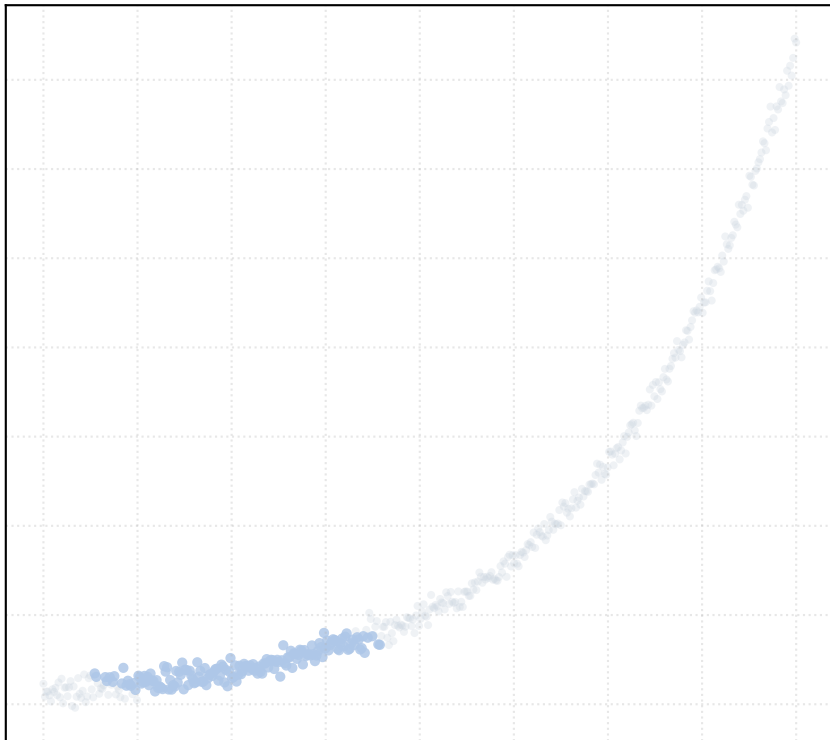


t_obs = 1000 (Max) [1 Clusters]

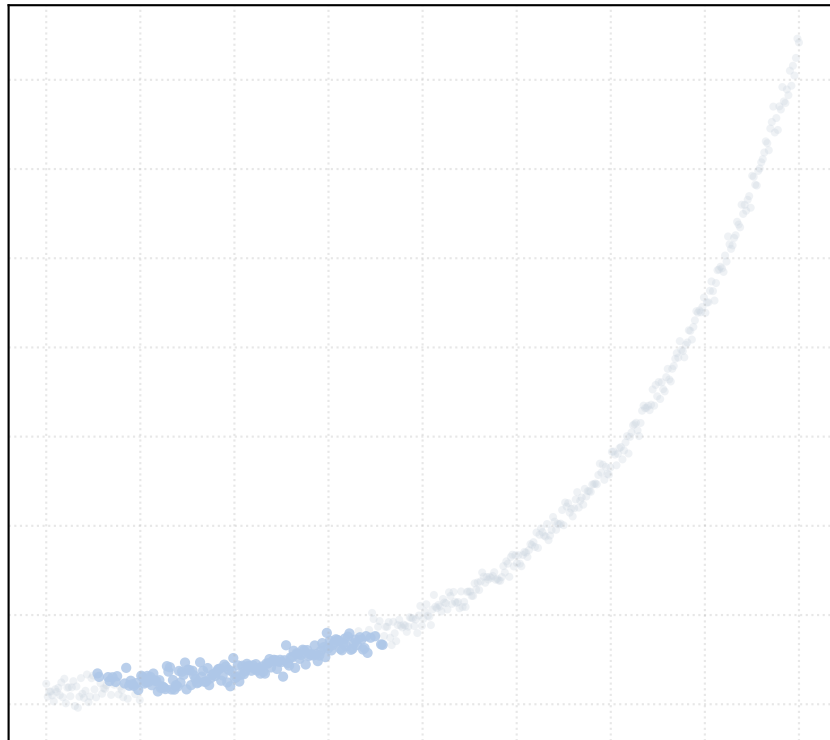


Dataset: exponential_noise_medium | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

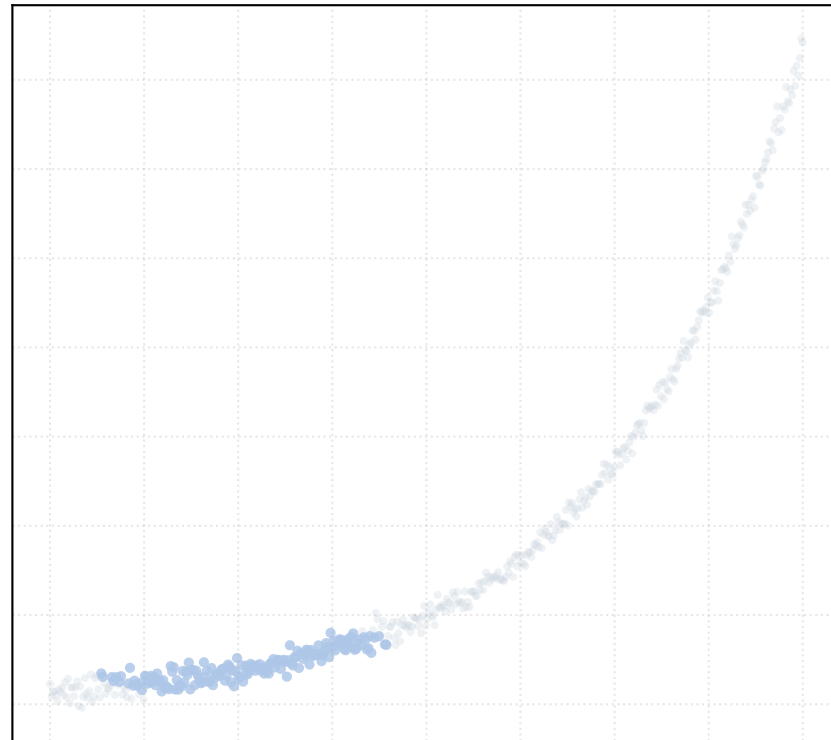
t_obs = Default [1 Clusters]



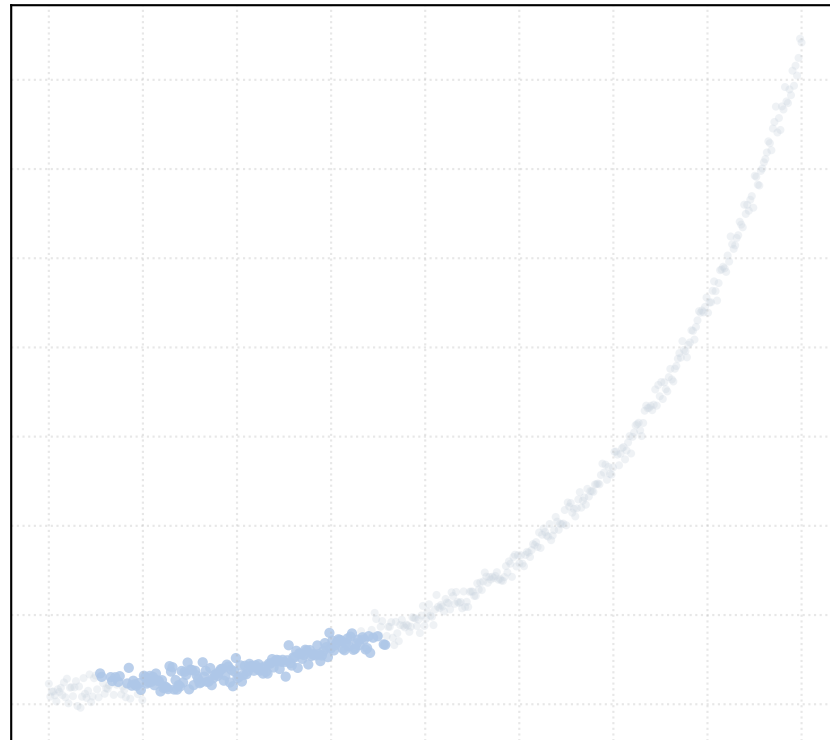
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

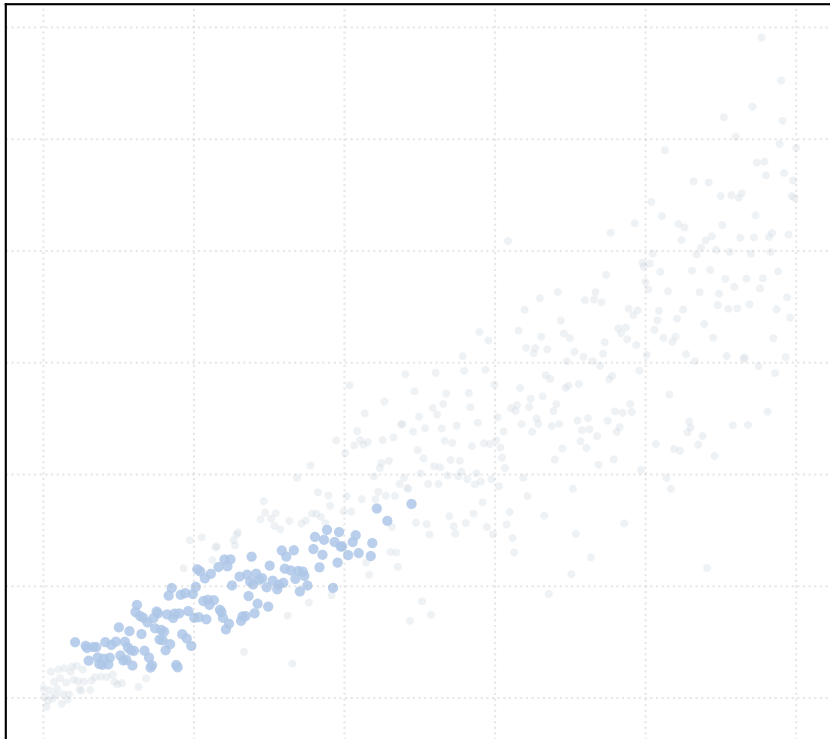


t_obs = 1000 (Max) [1 Clusters]

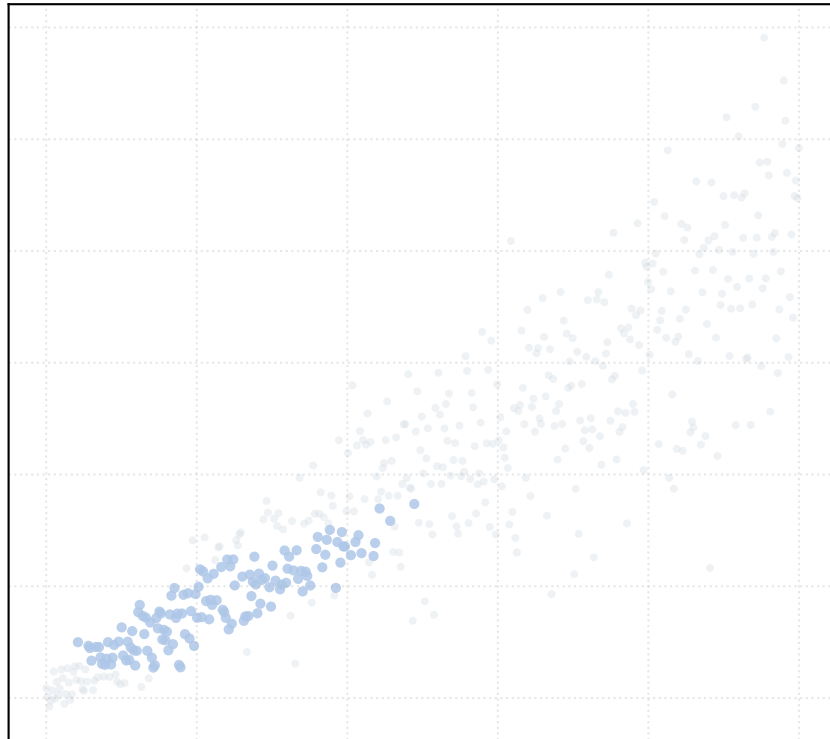


Dataset: heteroscedastic_noise_high | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

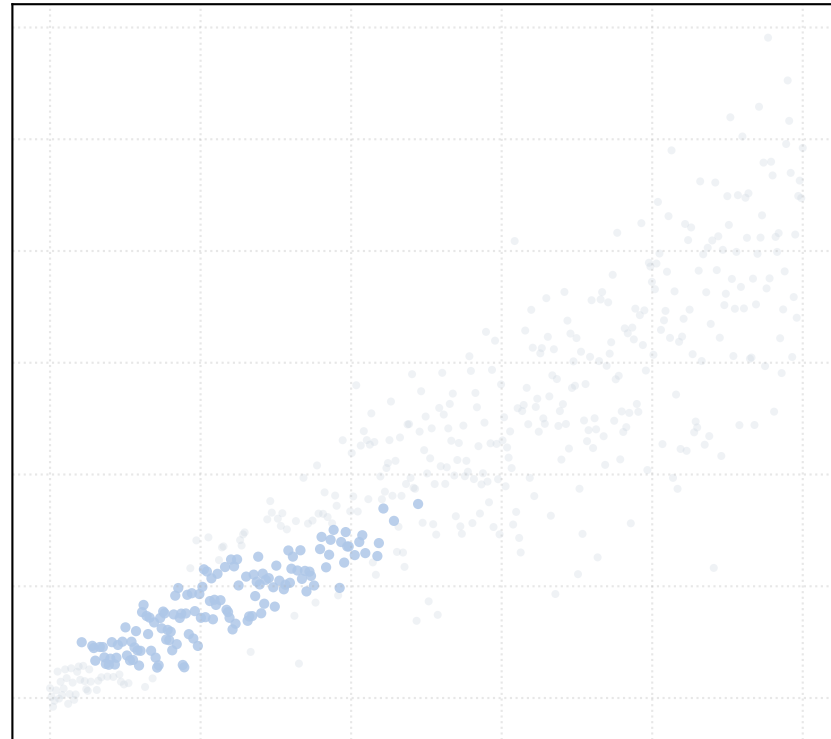
t_obs = Default [1 Clusters]



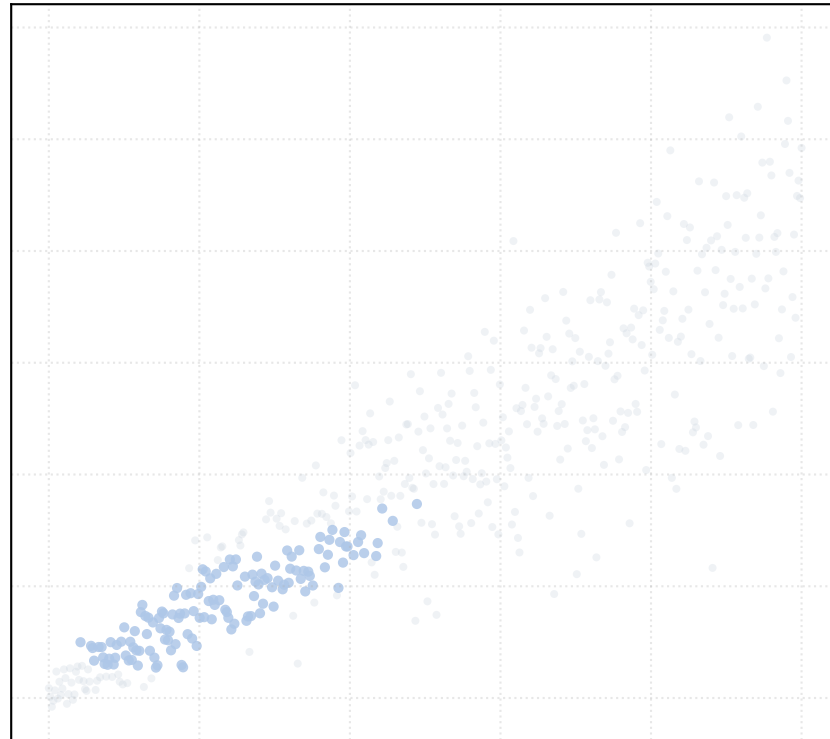
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

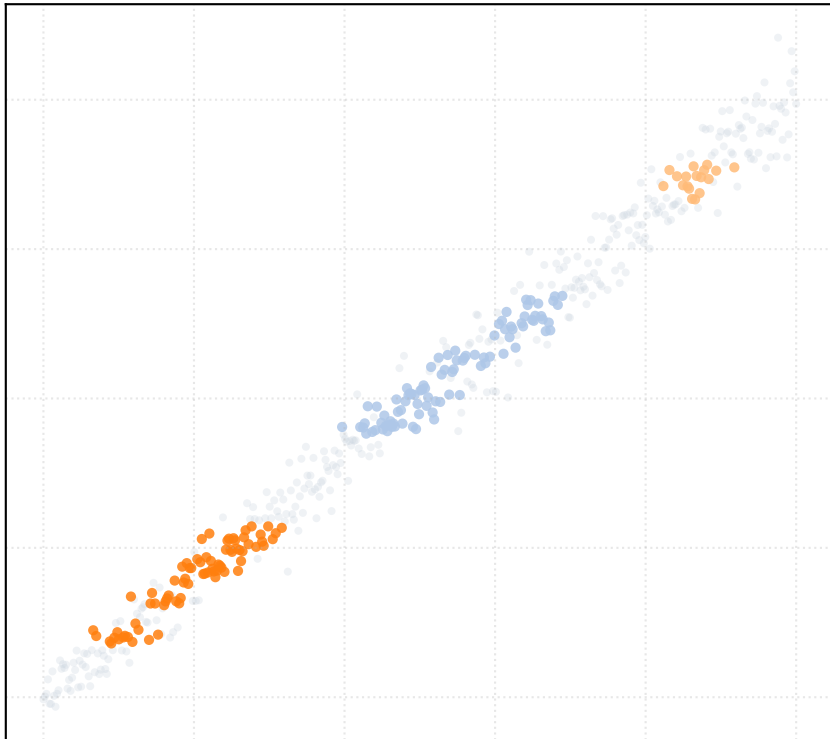


t_obs = 1000 (Max) [1 Clusters]

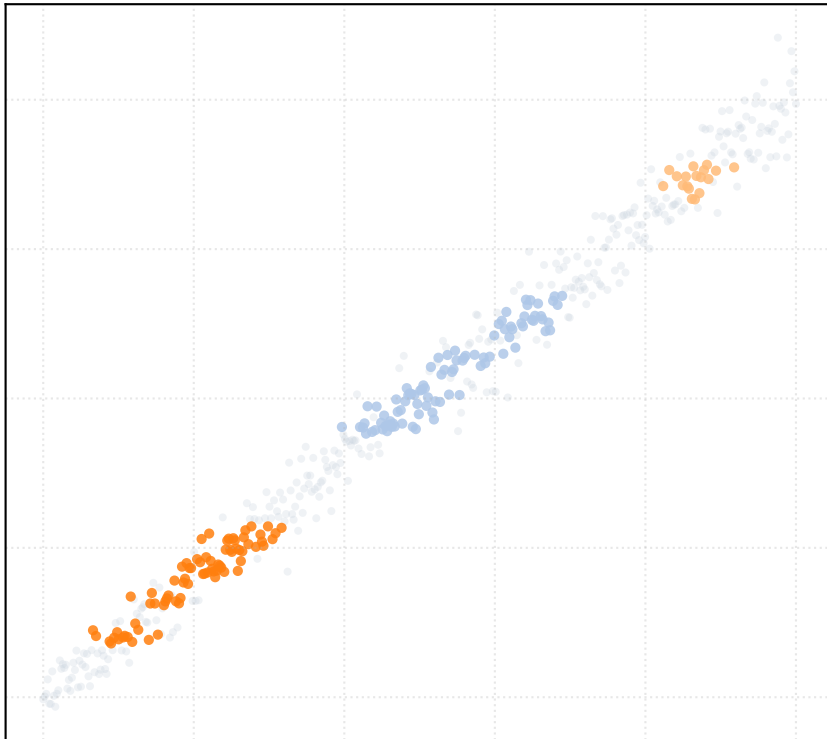


Dataset: heteroscedastic_noise_low | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

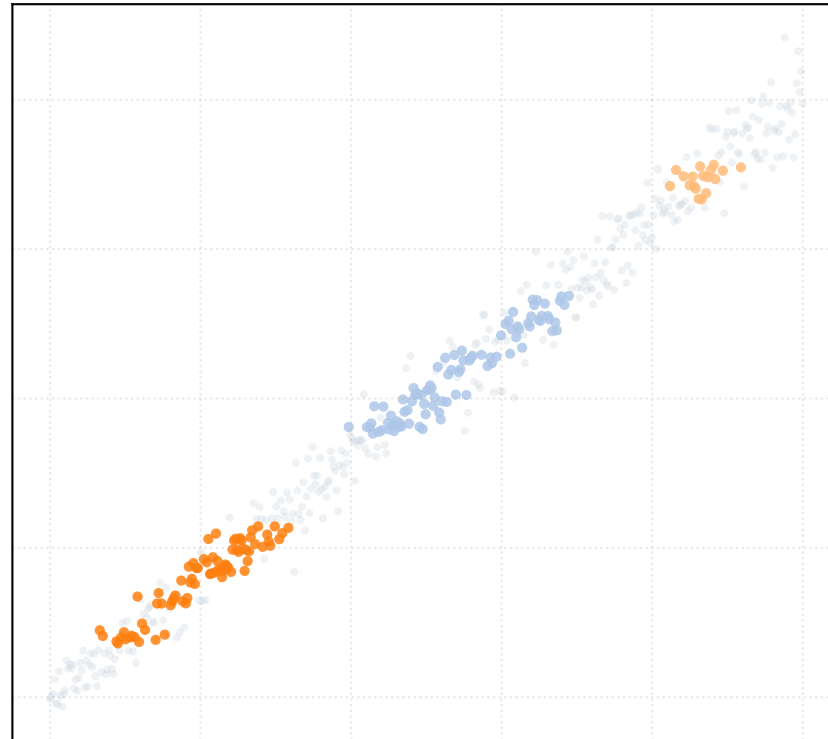
t_obs = Default [3 Clusters]



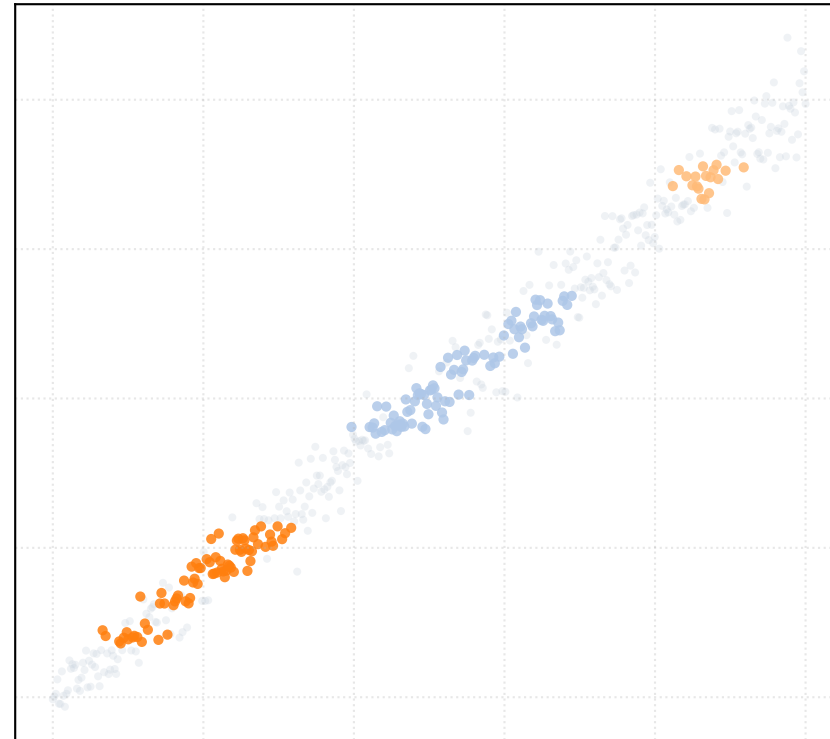
t_obs = 5 [3 Clusters]



t_obs = 50 [3 Clusters]

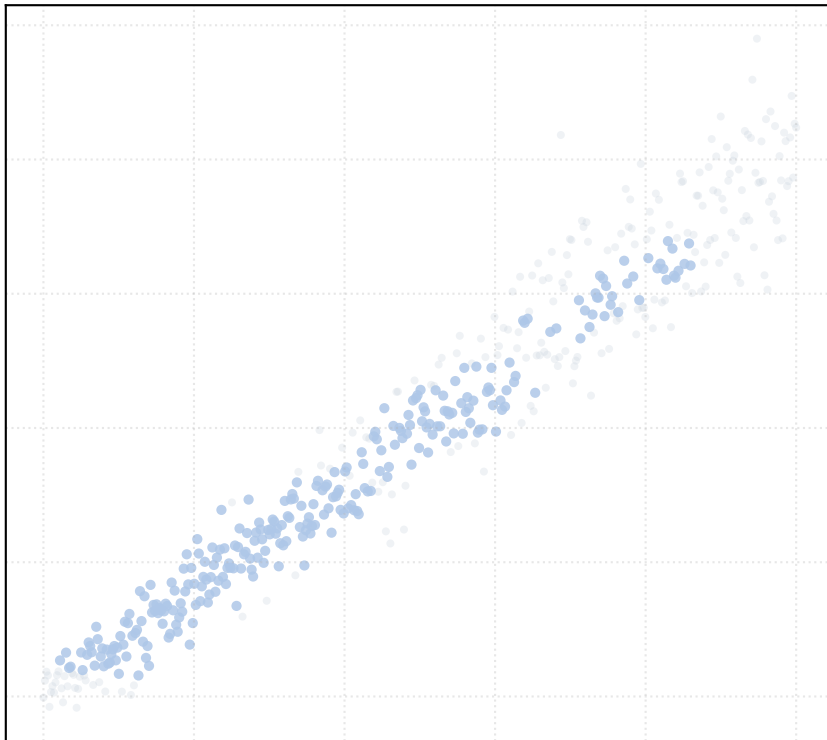


t_obs = 1000 (Max) [3 Clusters]

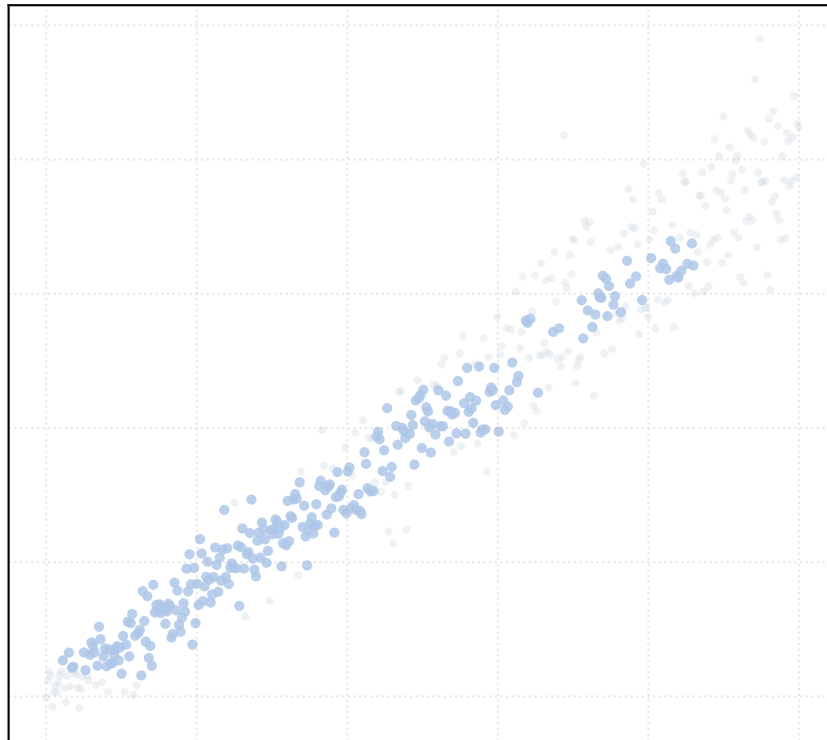


Dataset: heteroscedastic_noise_medium | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

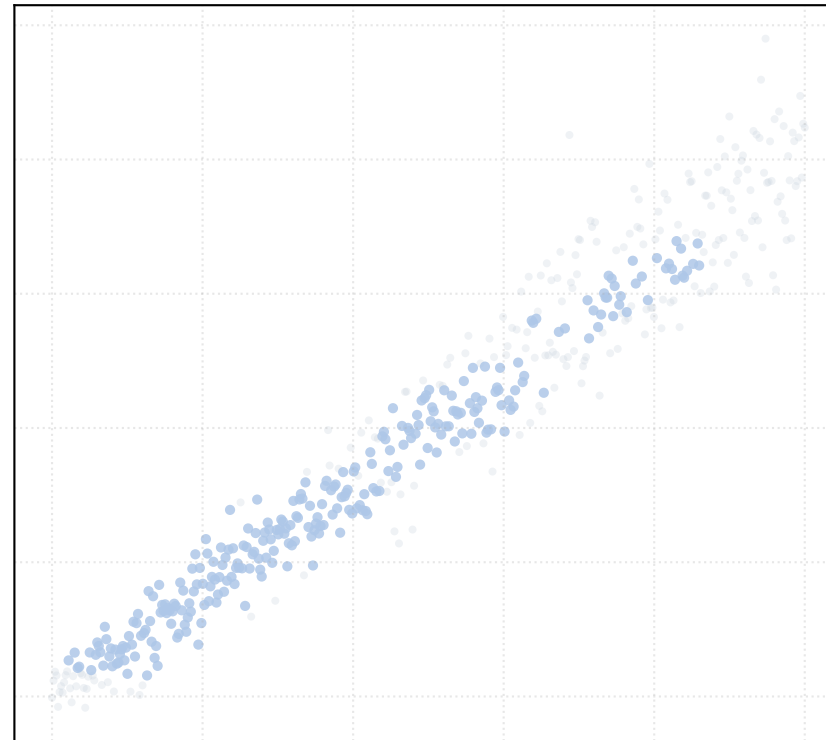
t_obs = Default [1 Clusters]



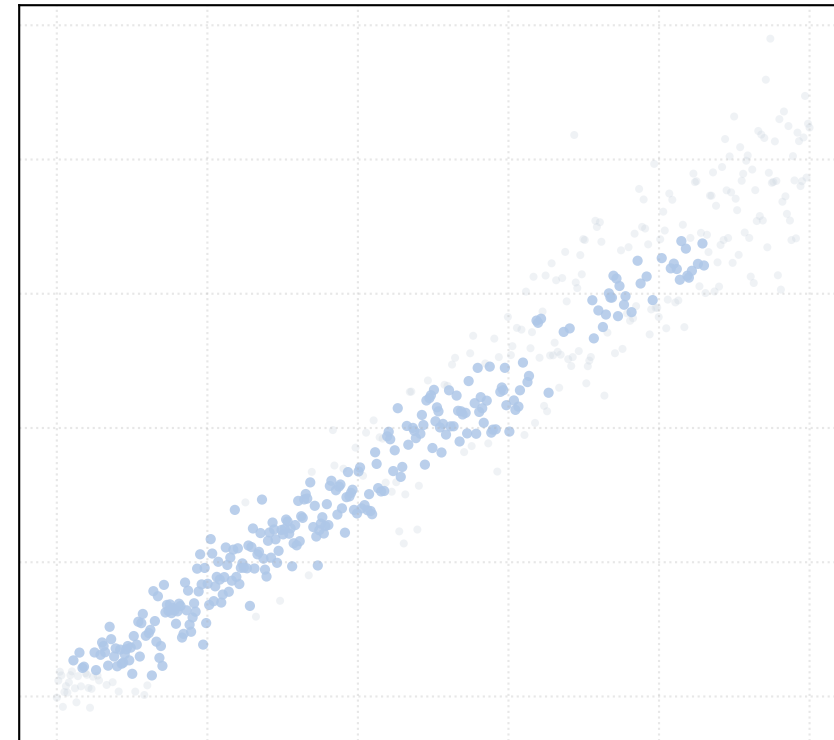
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

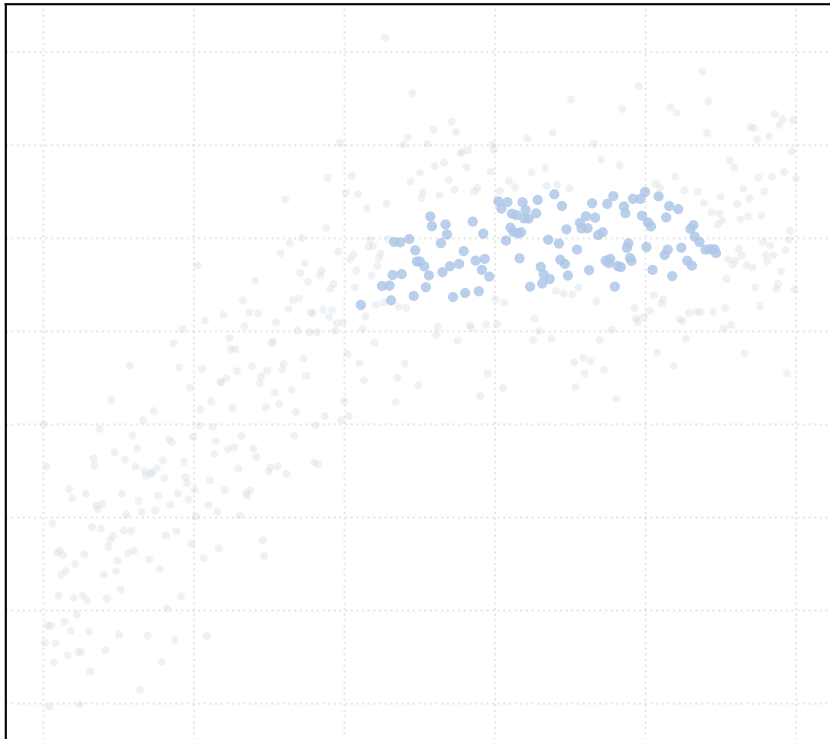


t_obs = 1000 (Max) [1 Clusters]

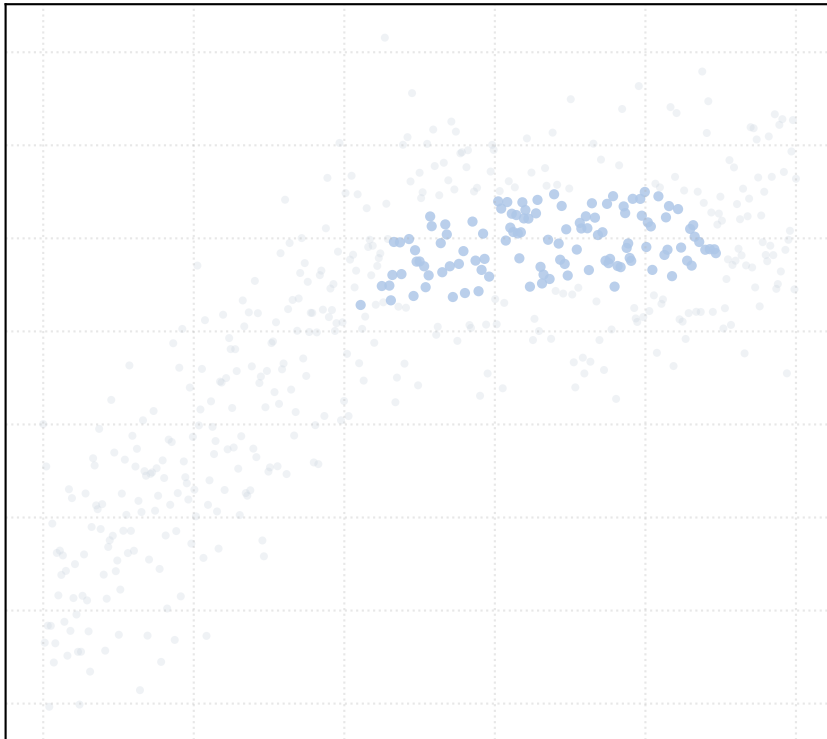


Dataset: kinked_curve_noise_high | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

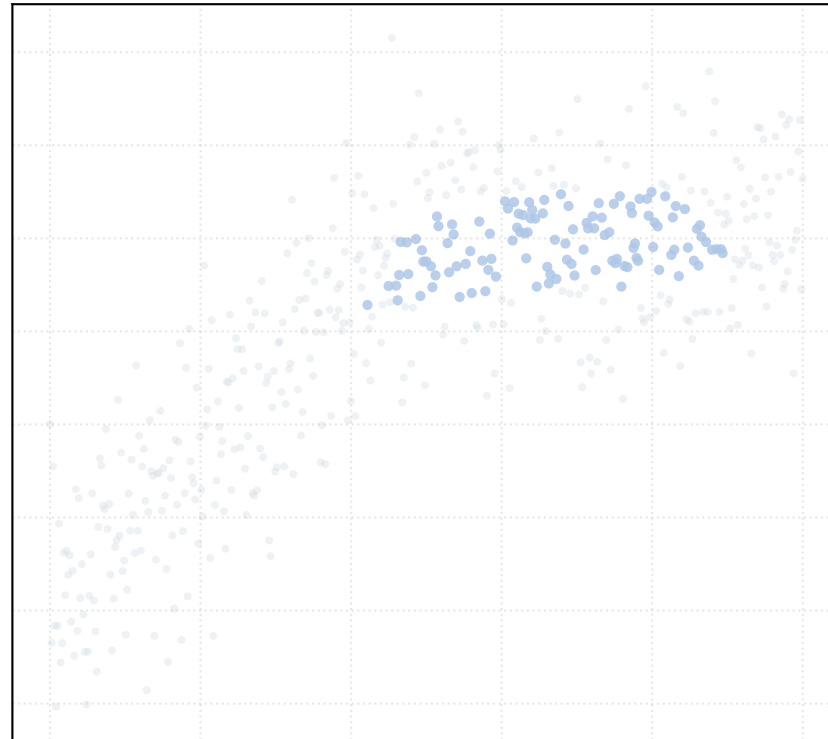
t_obs = Default [1 Clusters]



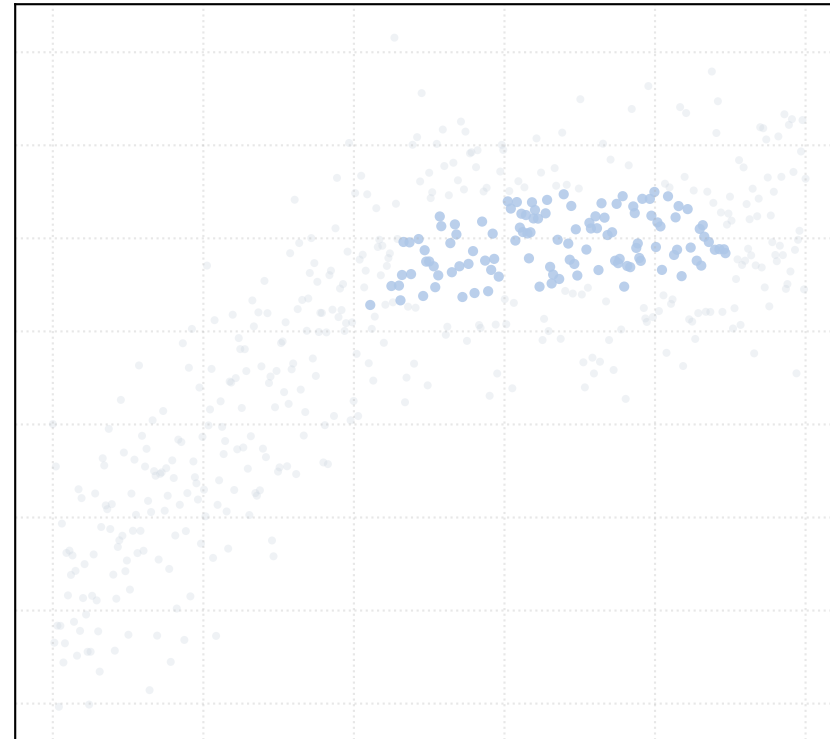
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

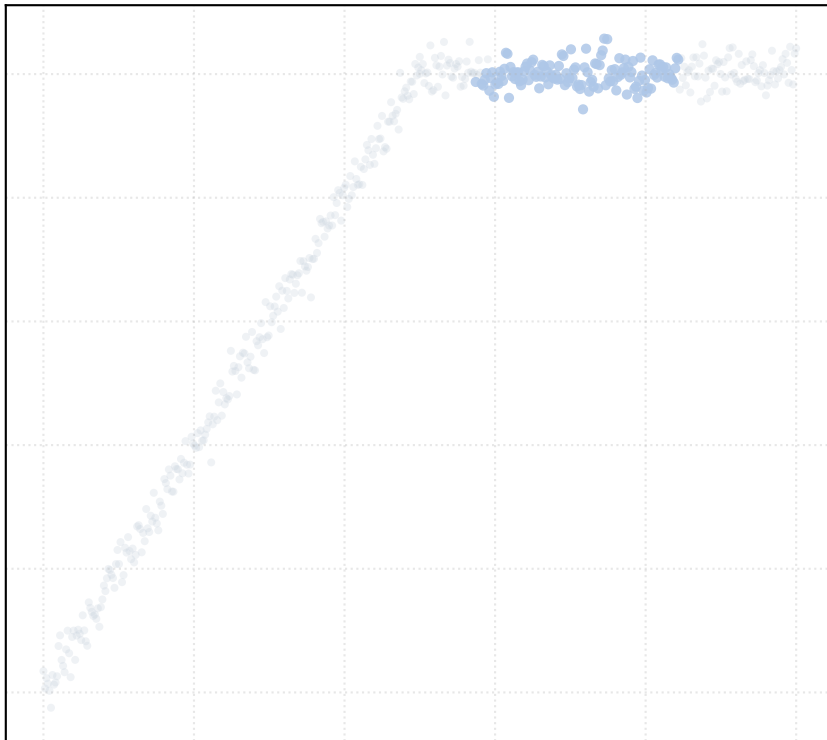


t_obs = 1000 (Max) [1 Clusters]

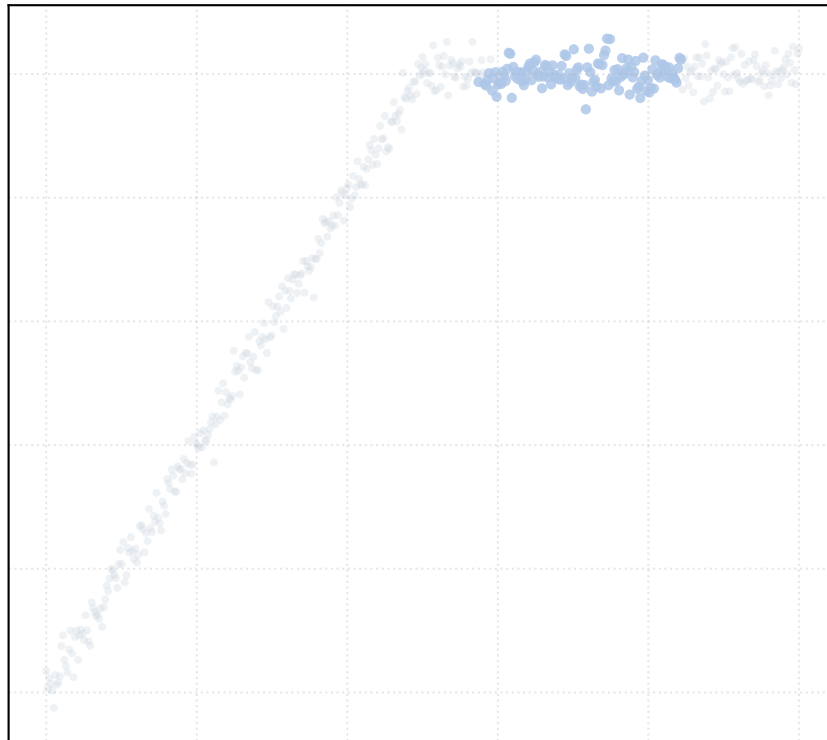


Dataset: kinked_curve_noise_low | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

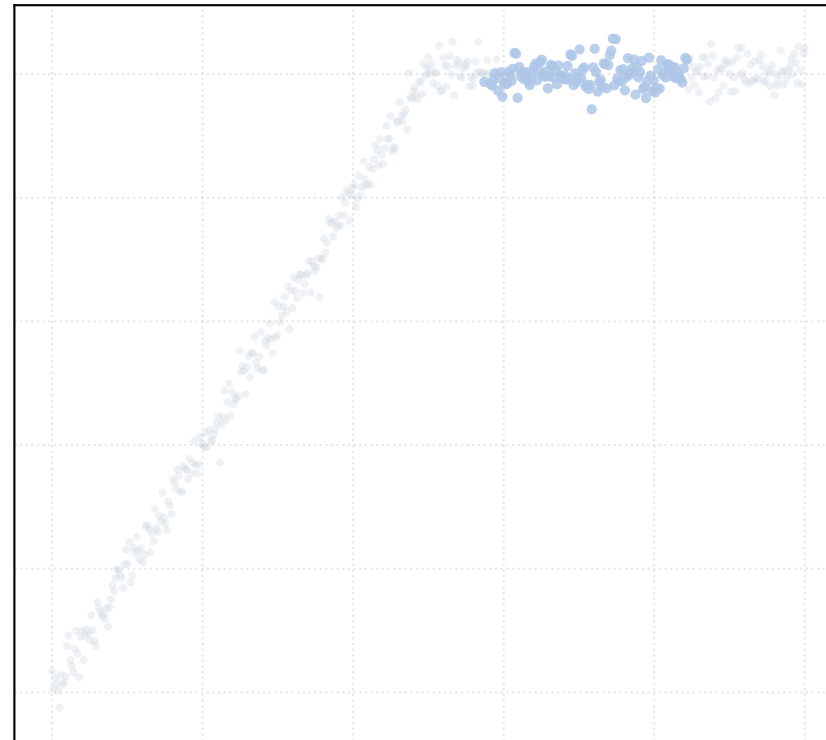
t_obs = Default [1 Clusters]



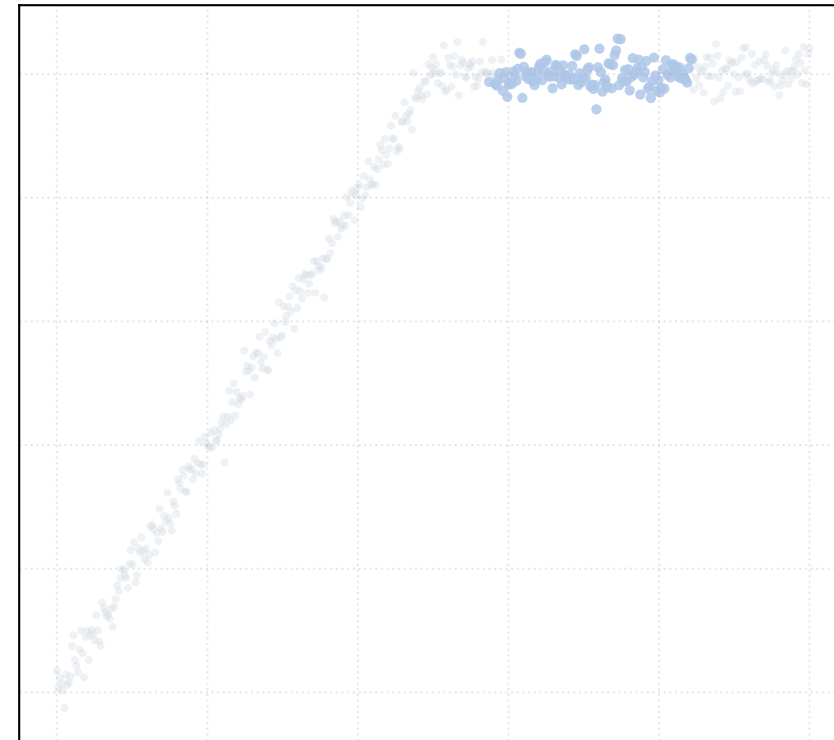
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

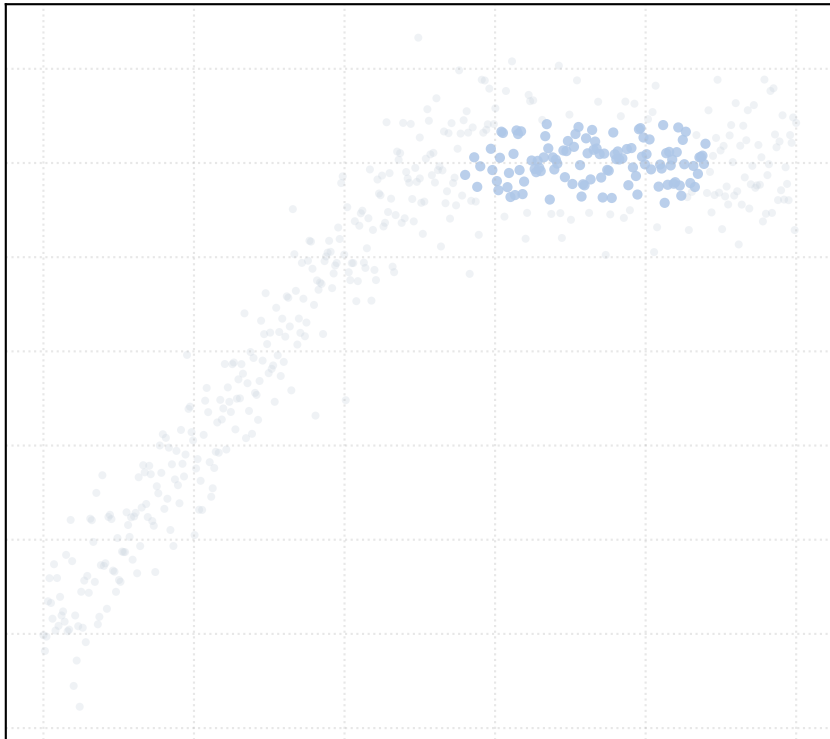


t_obs = 1000 (Max) [1 Clusters]

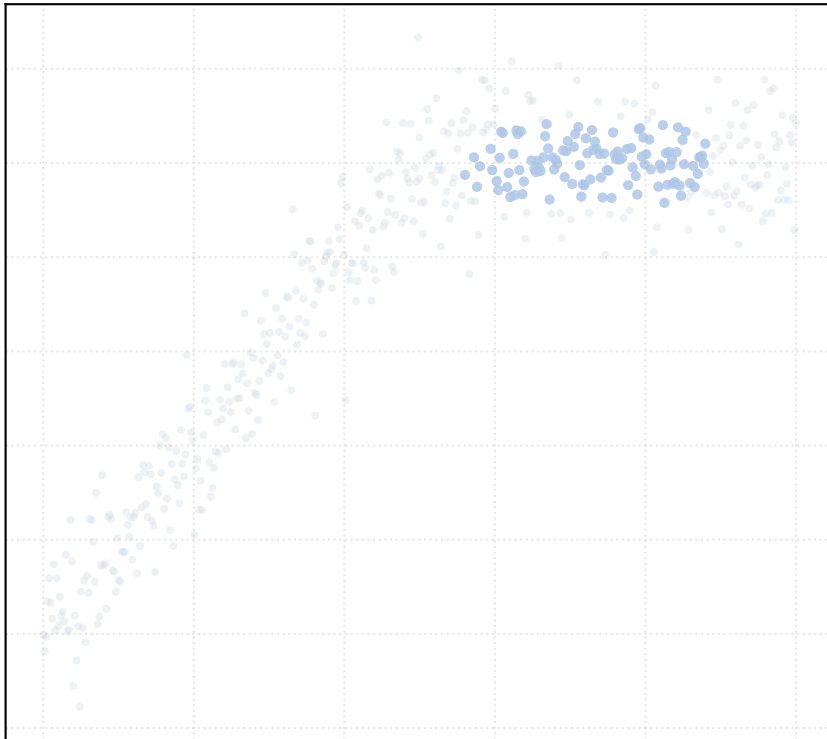


Dataset: kinked_curve_noise_medium | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

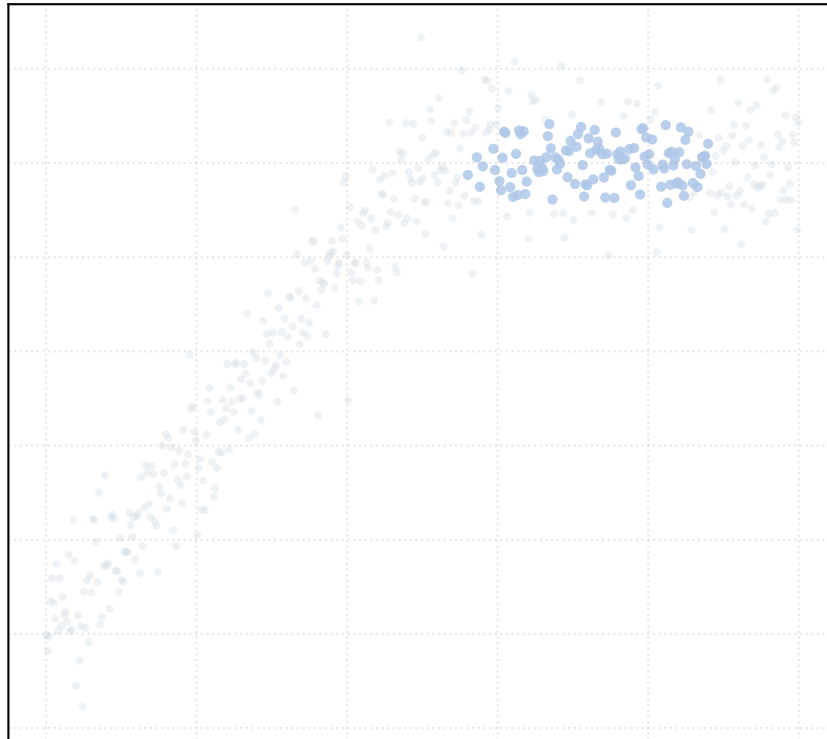
t_obs = Default [1 Clusters]



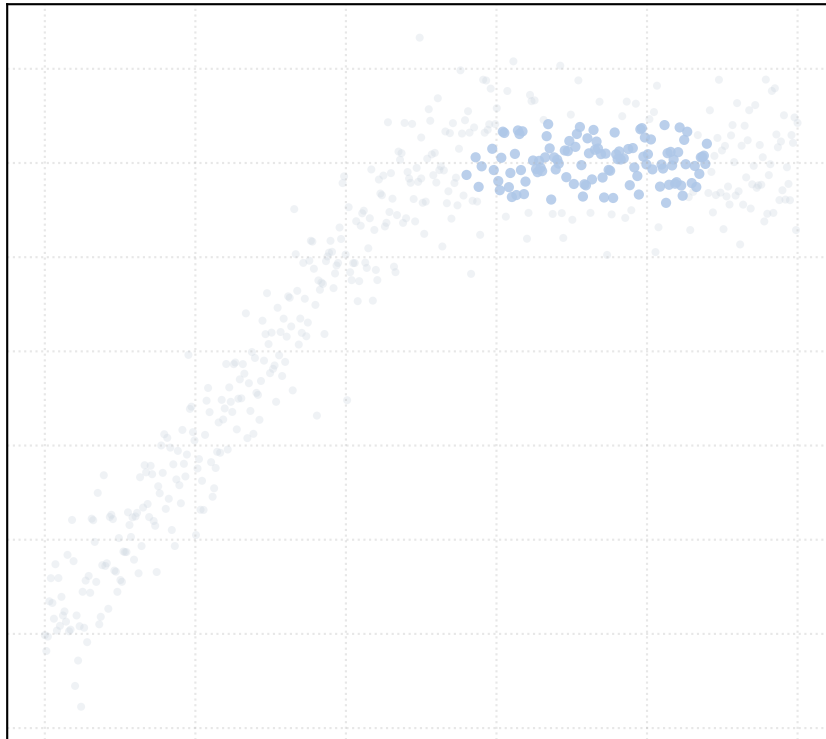
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

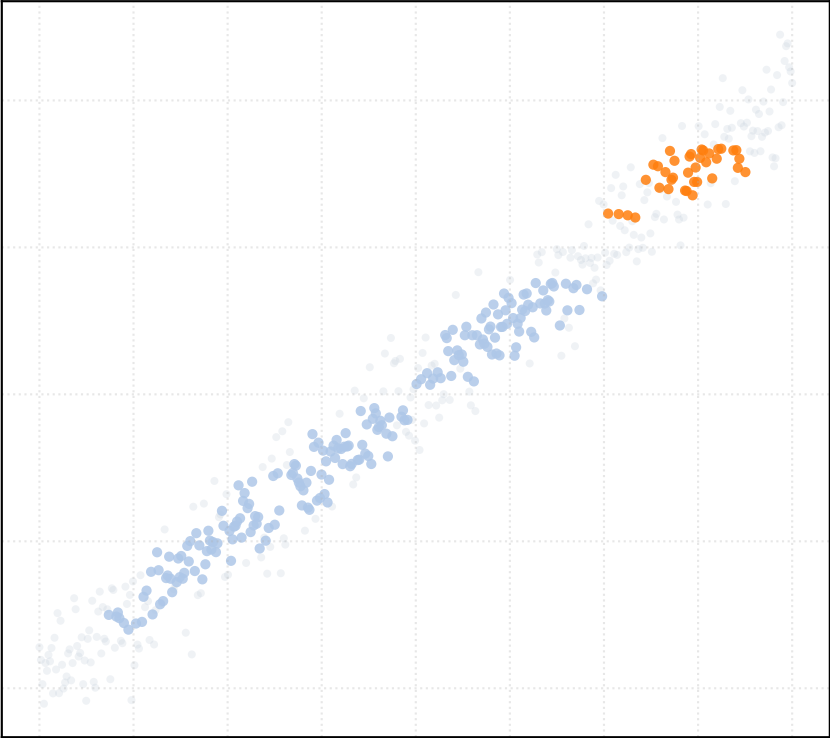


t_obs = 1000 (Max) [1 Clusters]

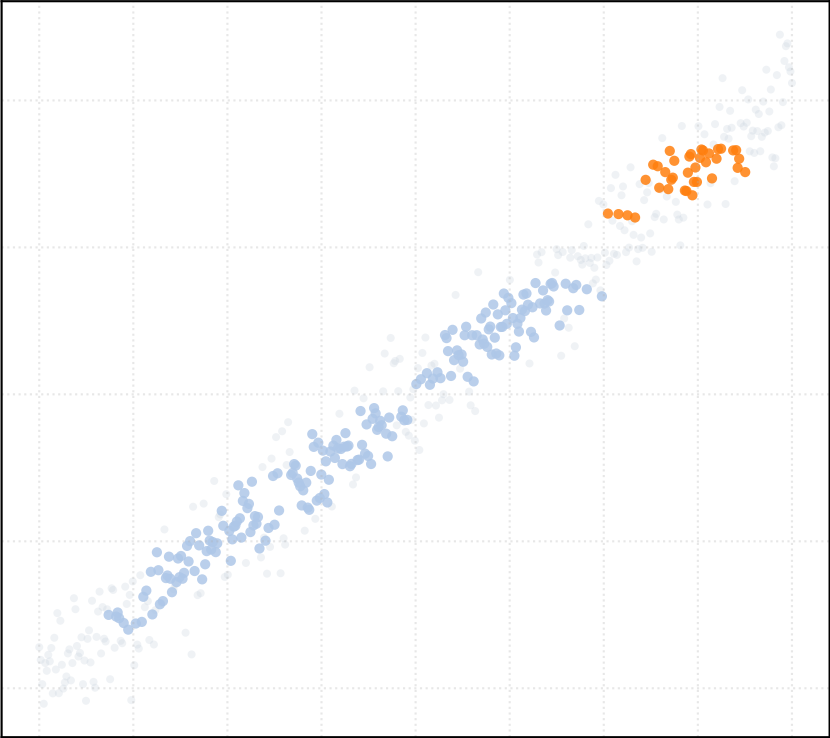


Dataset: linear_noise_high | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

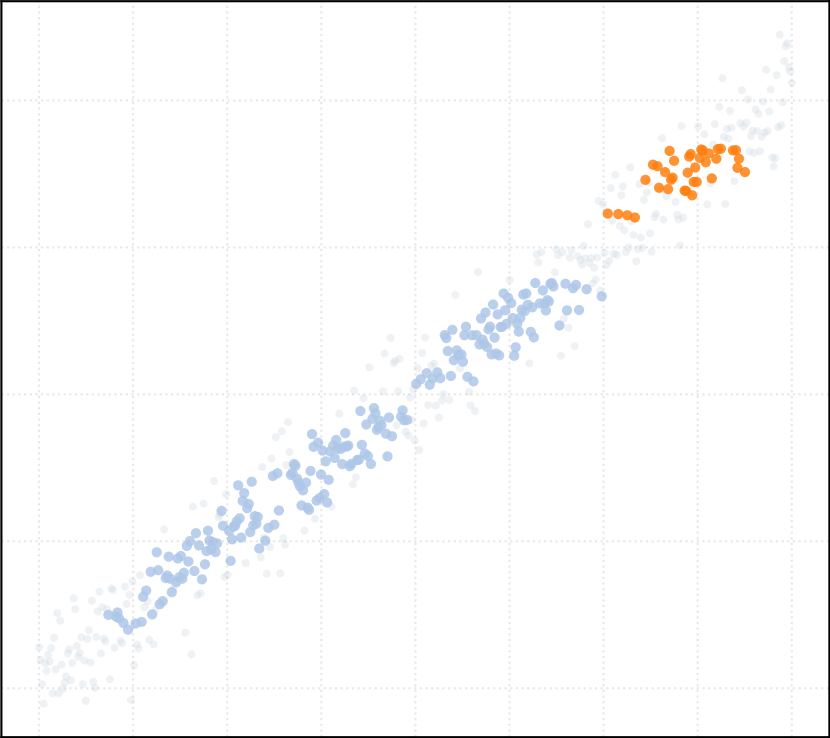
t_obs = Default [2 Clusters]



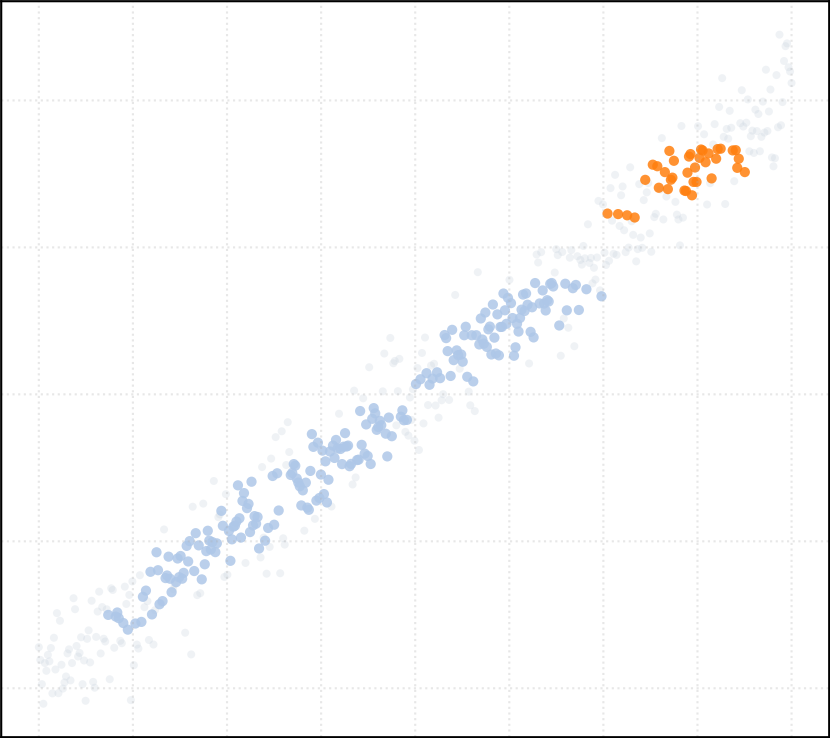
t_obs = 5 [2 Clusters]



t_obs = 50 [2 Clusters]

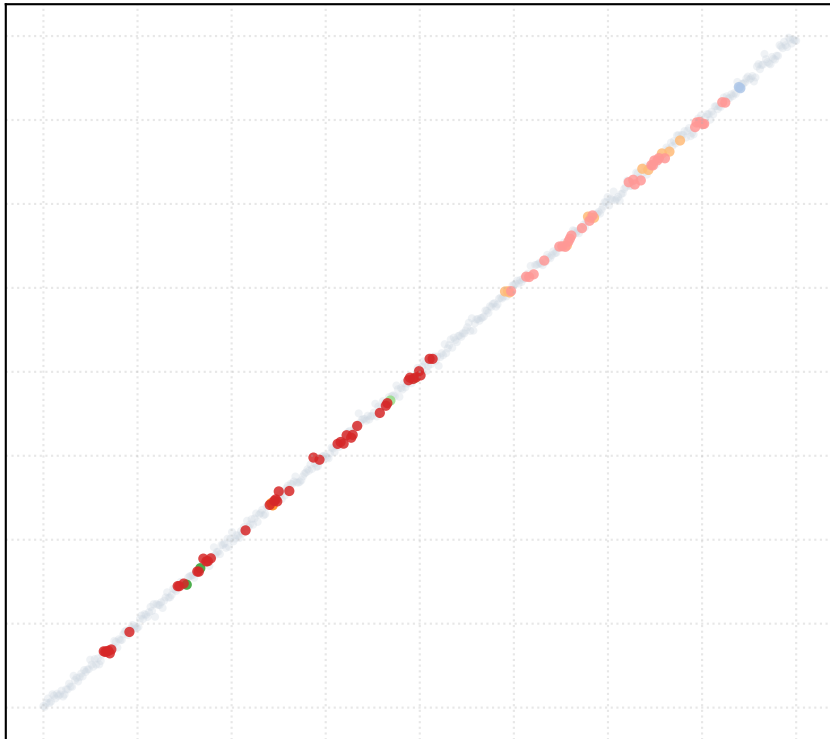


t_obs = 1000 (Max) [2 Clusters]

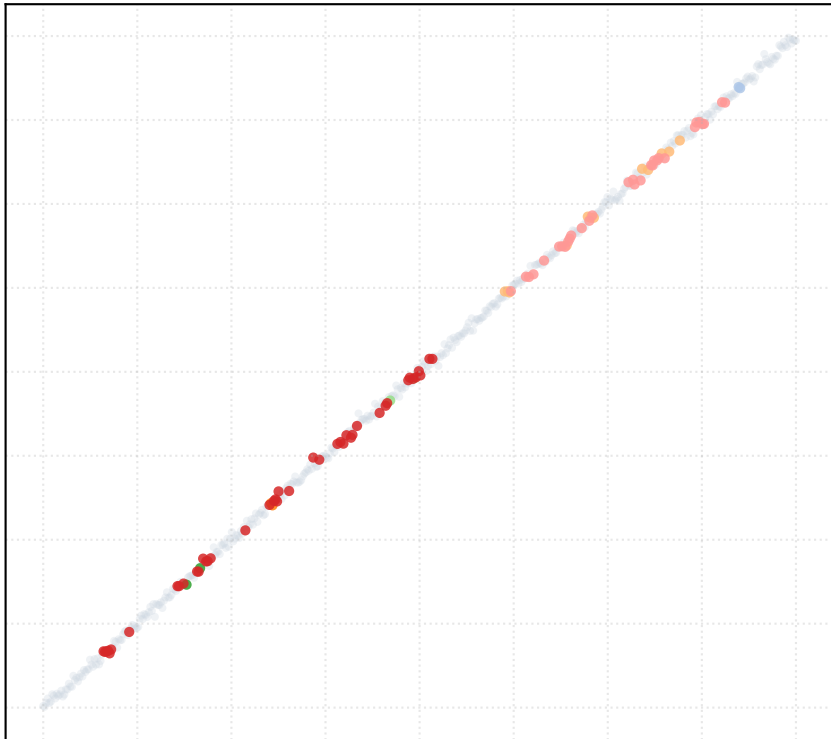


Dataset: linear_noise_low | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

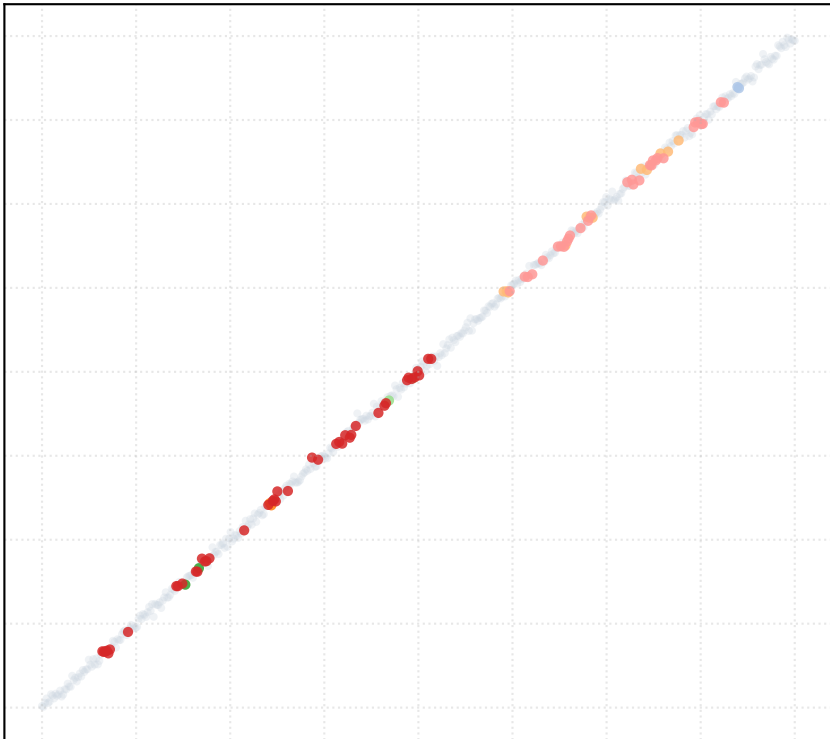
t_obs = Default [7 Clusters]



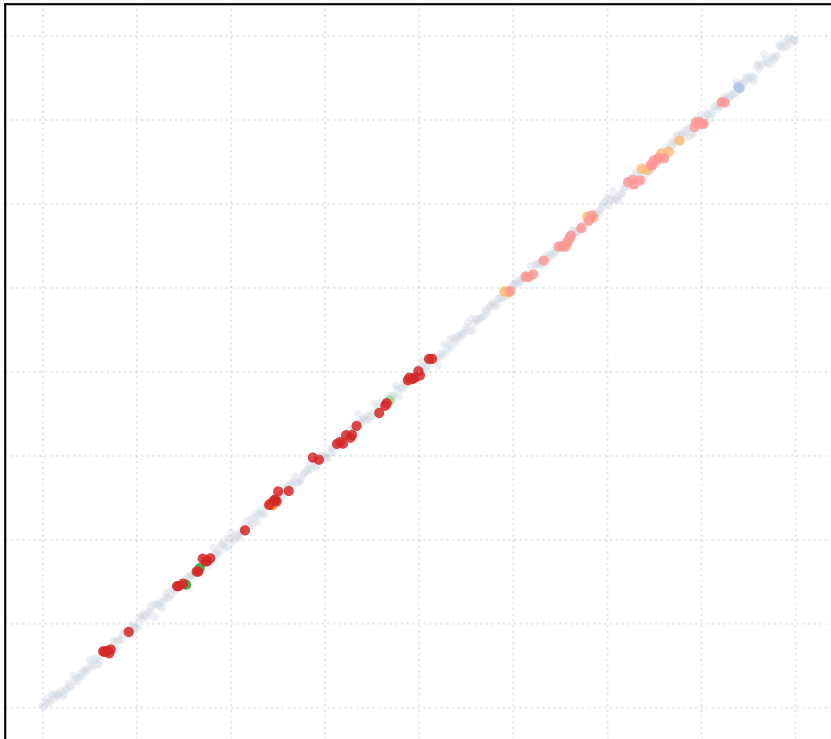
t_obs = 5 [7 Clusters]



t_obs = 50 [7 Clusters]

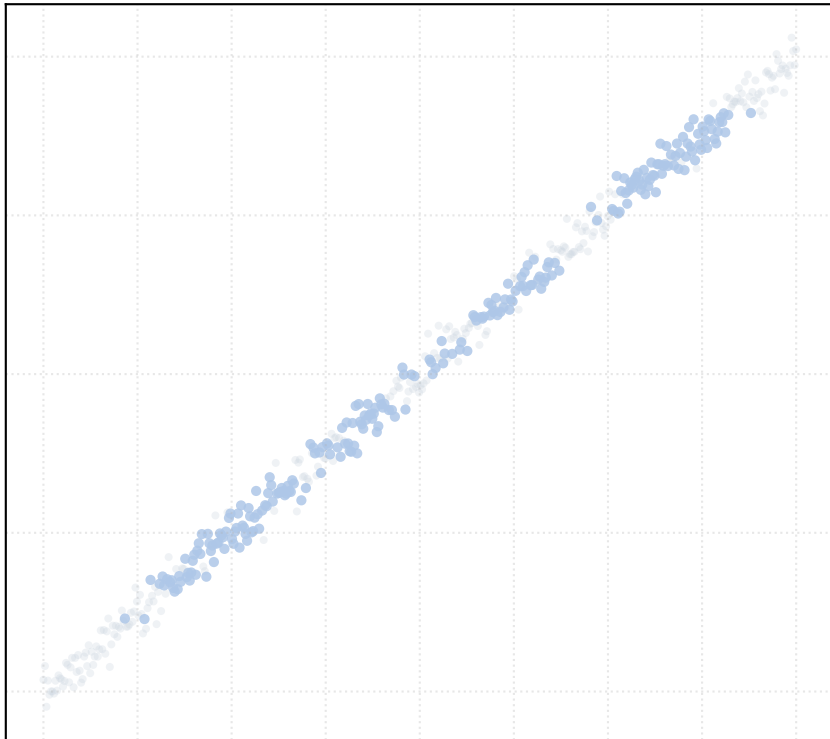


t_obs = 1000 (Max) [7 Clusters]

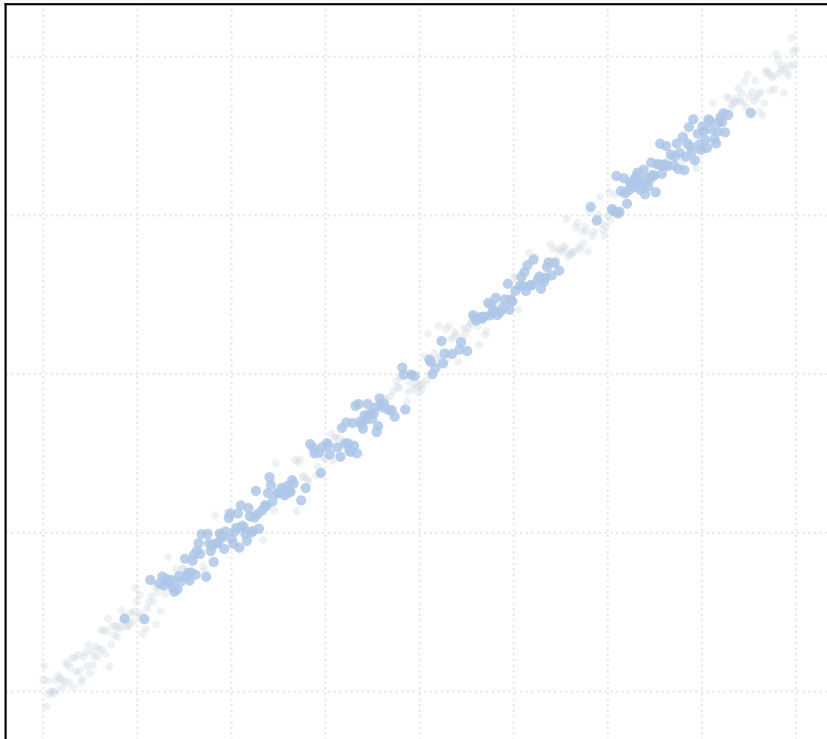


Dataset: linear_noise_medium | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

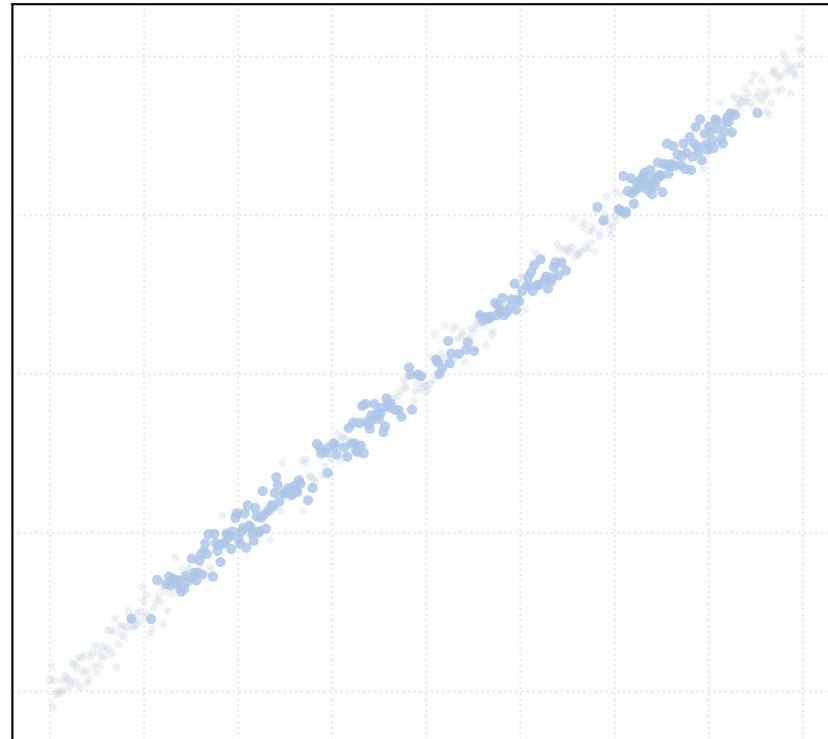
t_obs = Default [1 Clusters]



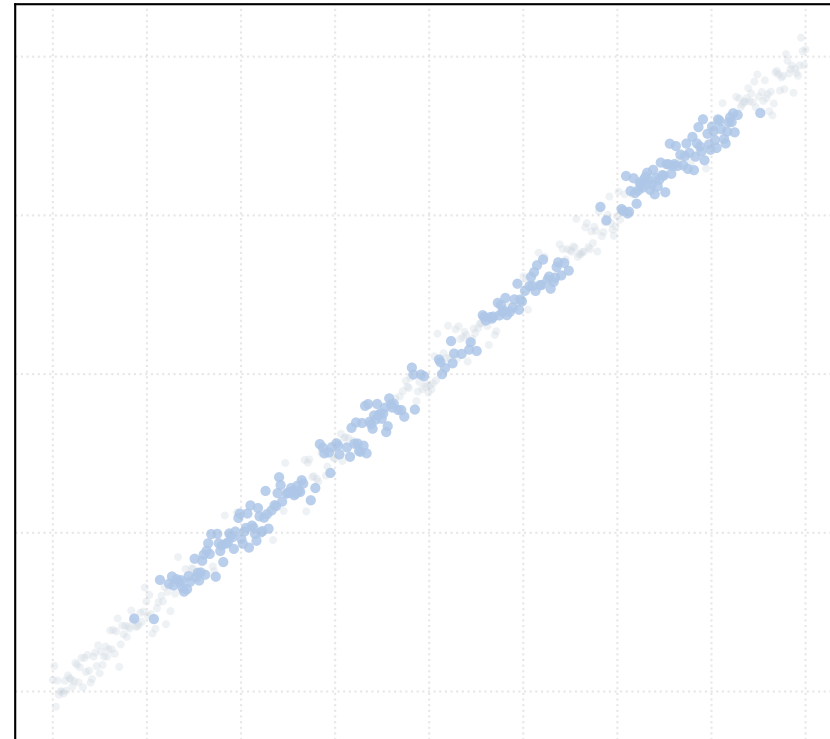
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

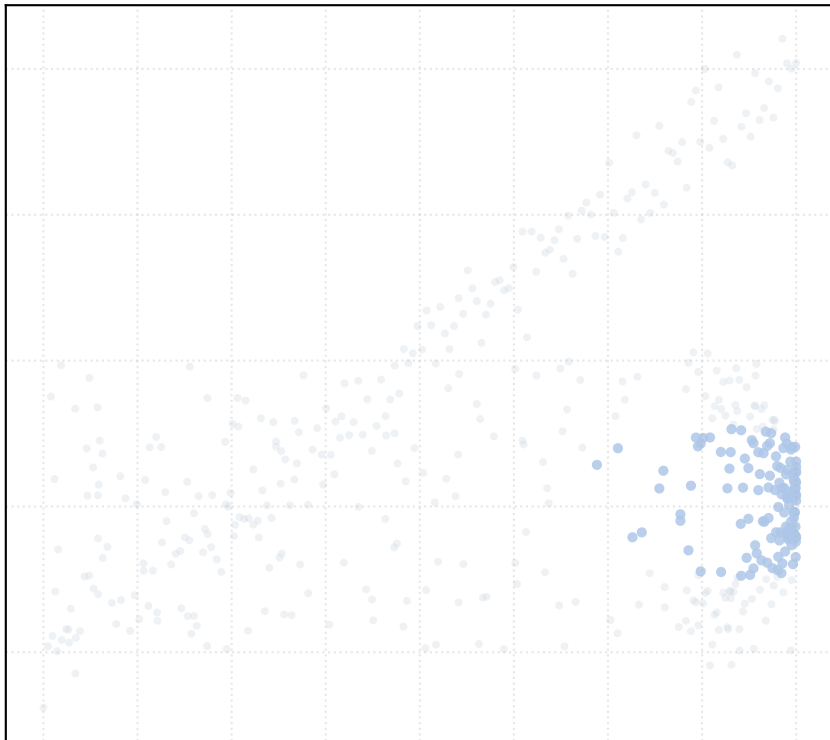


t_obs = 1000 (Max) [1 Clusters]

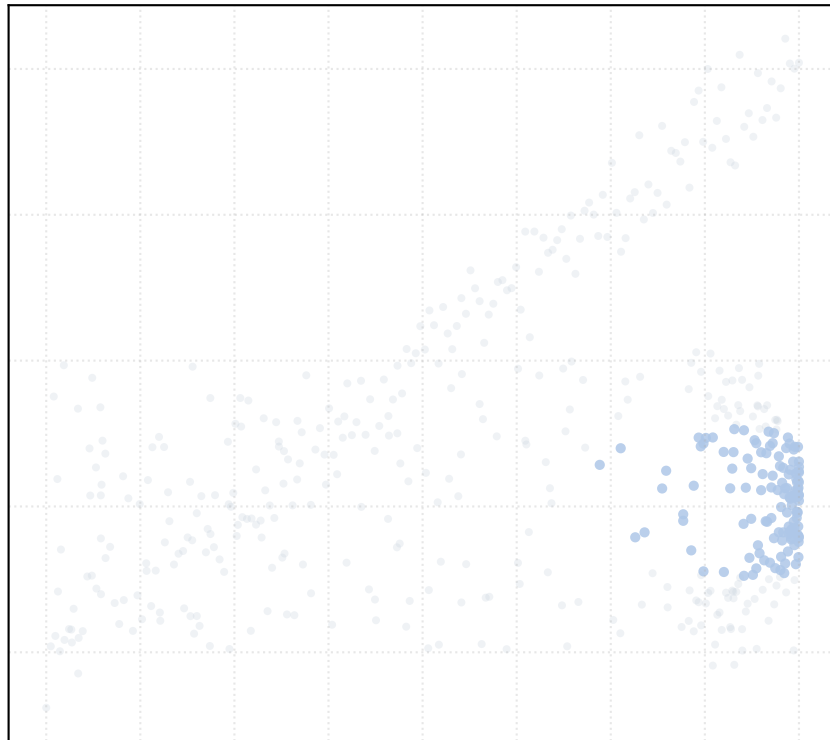


Dataset: mixed_generators_noise_high | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

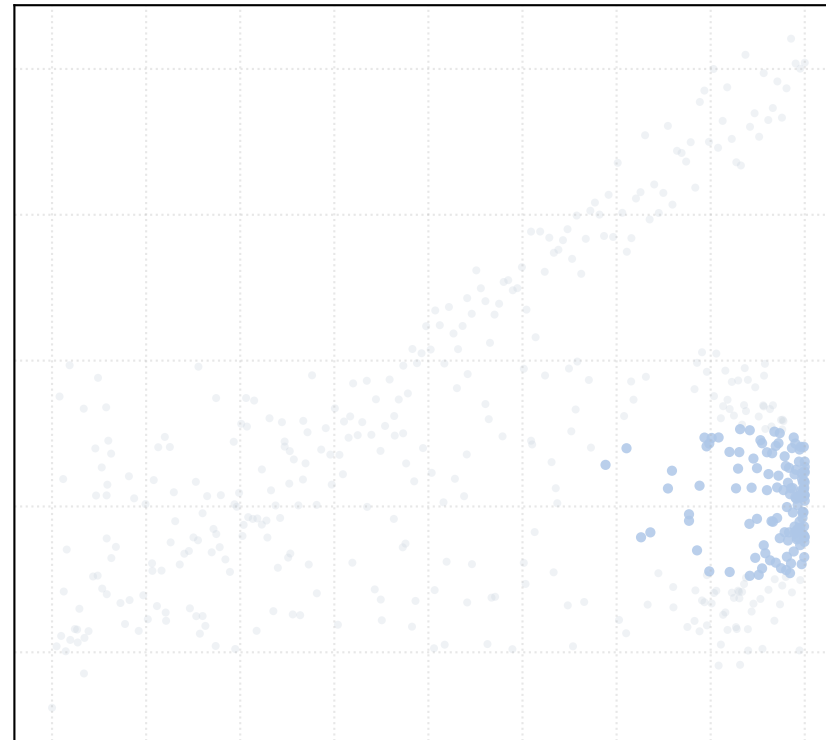
t_obs = Default [1 Clusters]



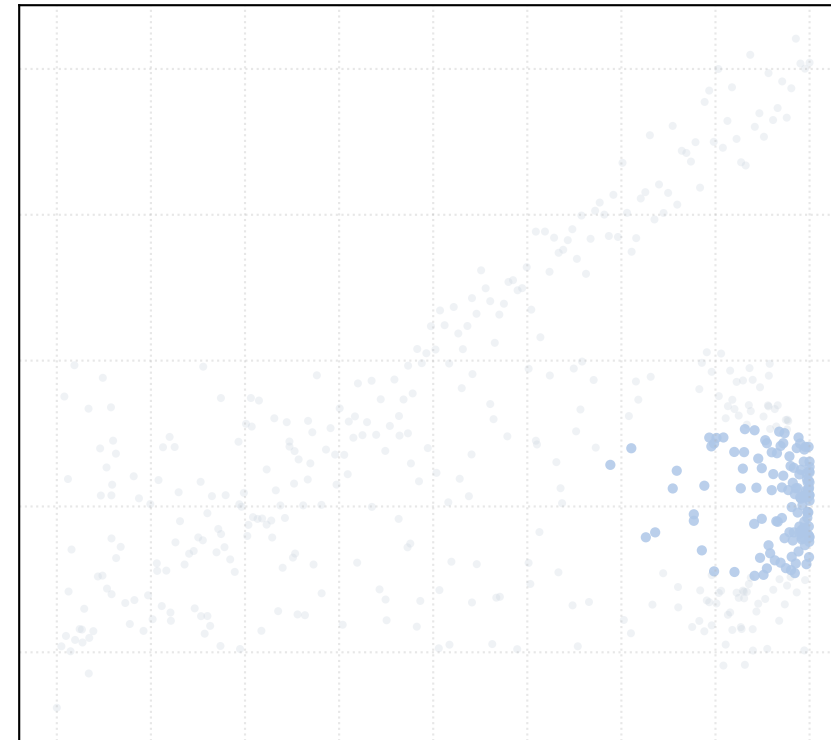
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

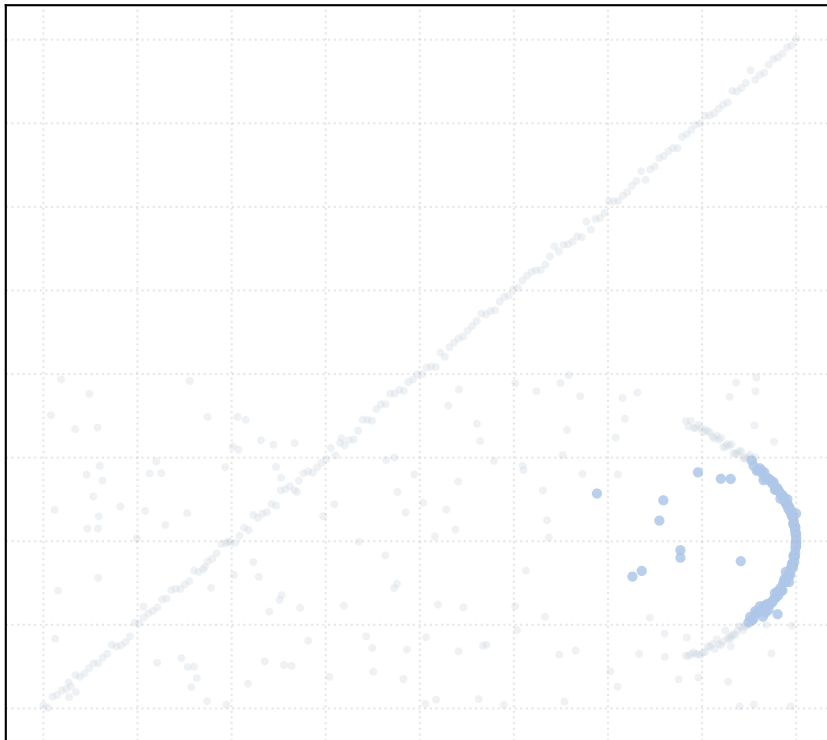


t_obs = 1000 (Max) [1 Clusters]

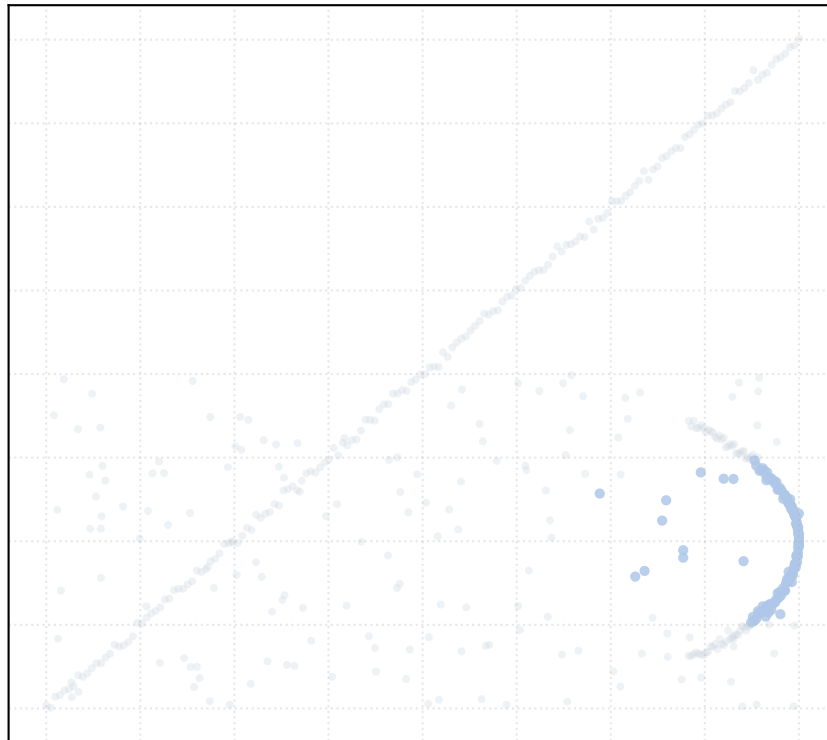


Dataset: mixed_generators_noise_low | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

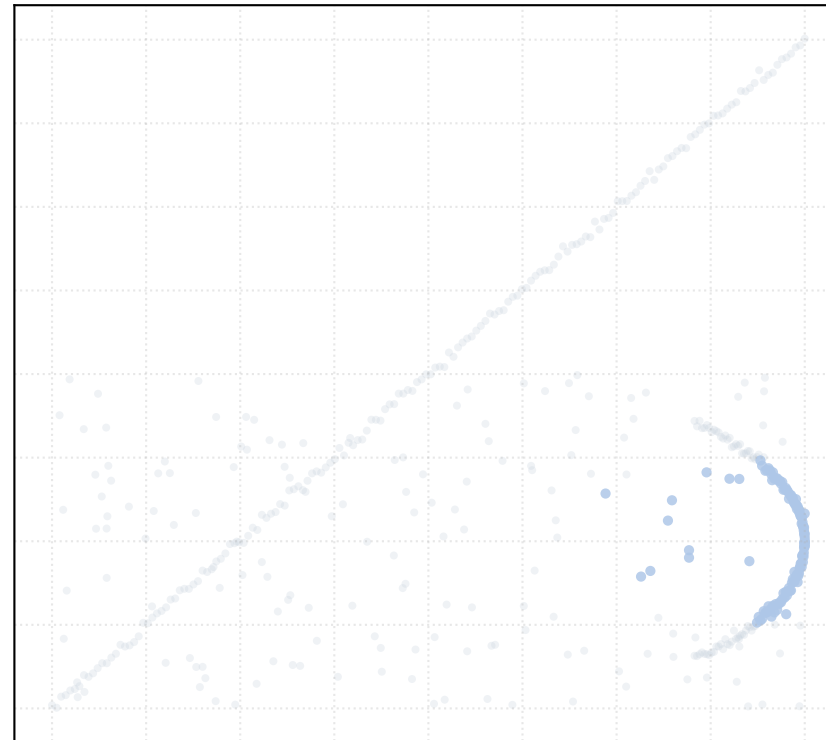
t_obs = Default [1 Clusters]



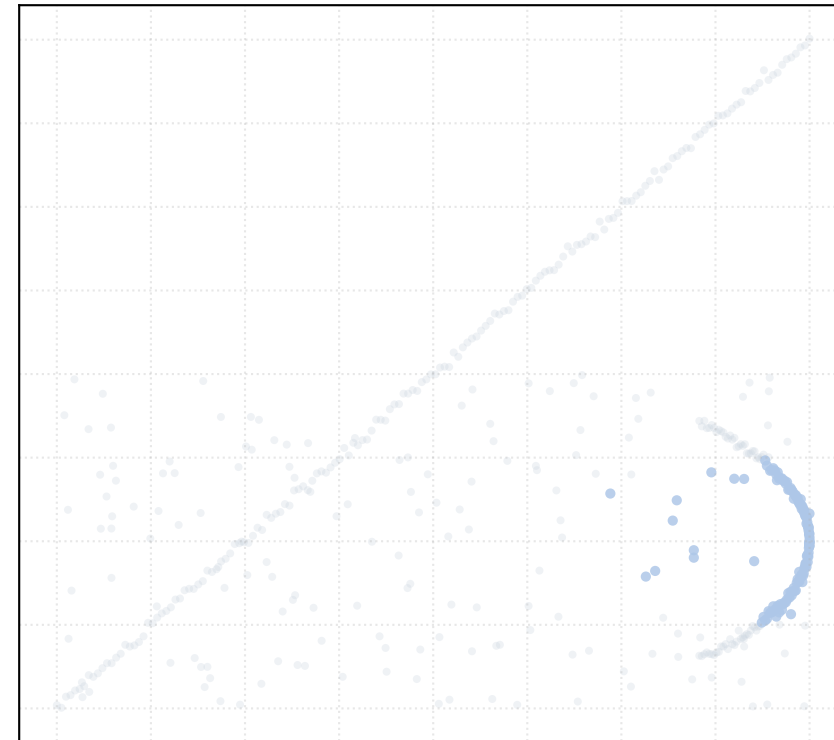
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

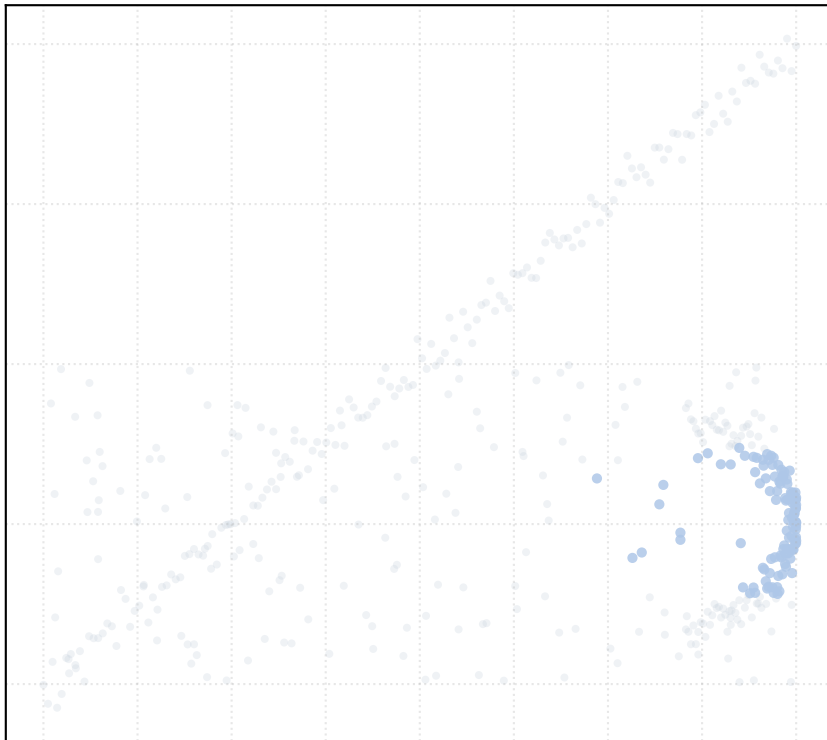


t_obs = 1000 (Max) [1 Clusters]

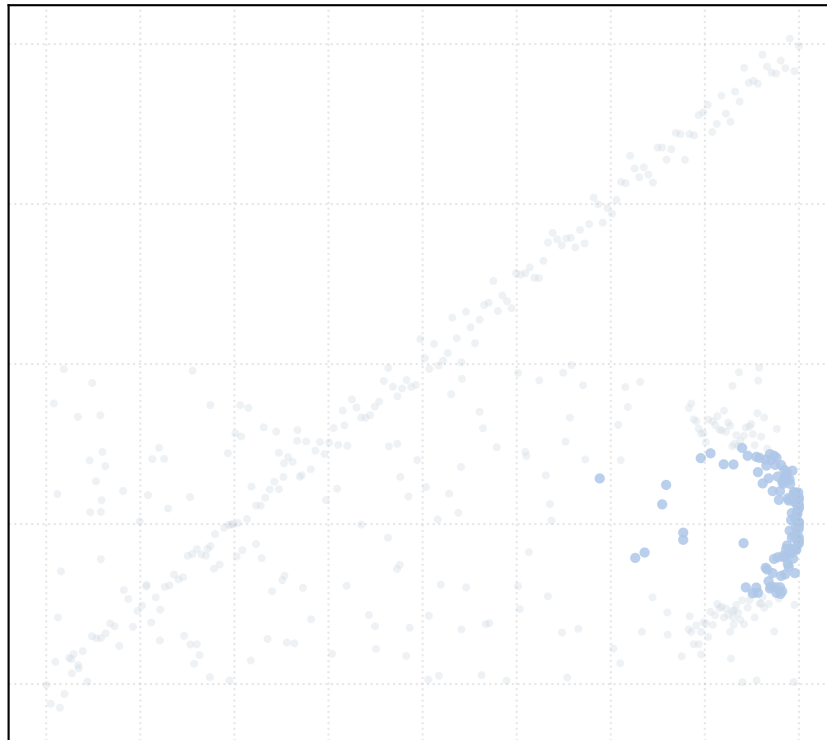


Dataset: mixed_generators_noise_medium | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

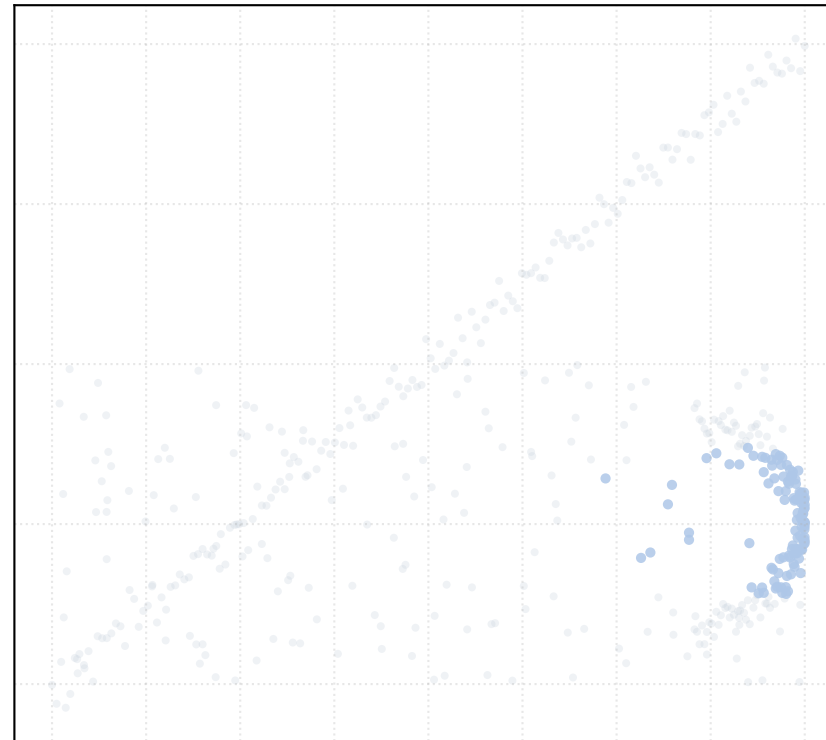
t_obs = Default [1 Clusters]



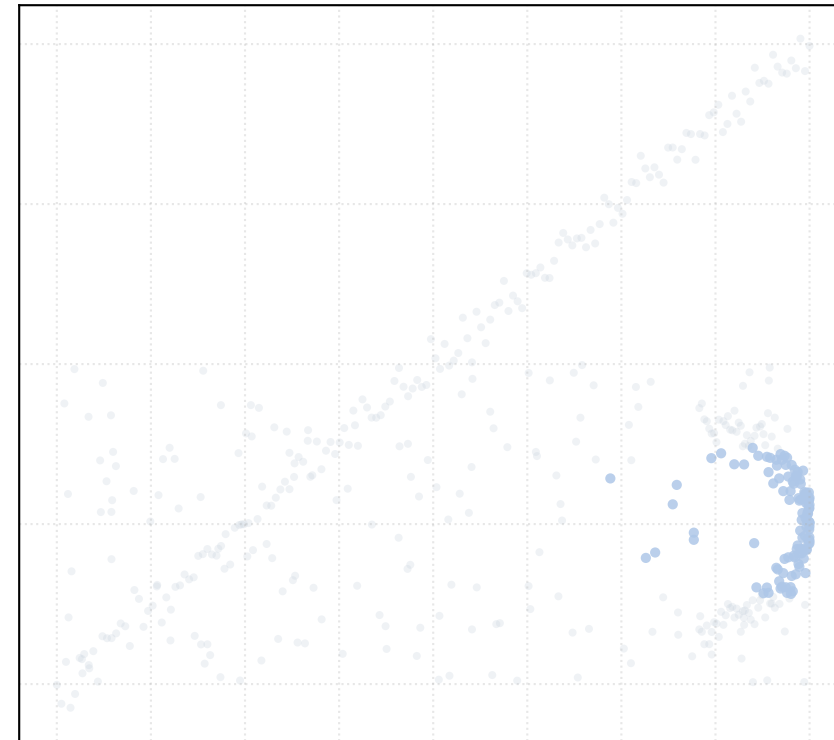
t_obs = 5 [1 Clusters]



t_obs = 50 [1 Clusters]

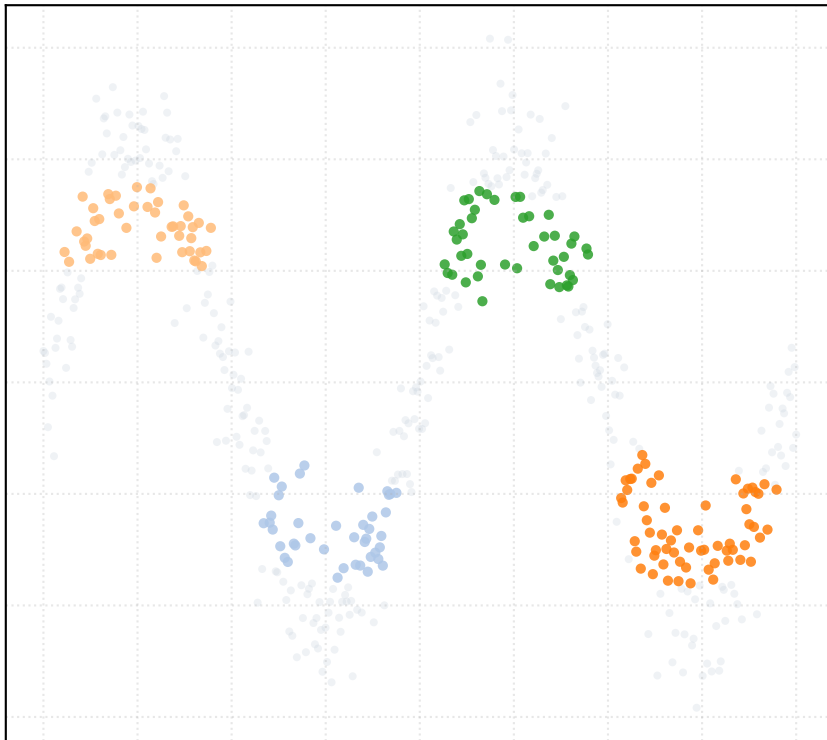


t_obs = 1000 (Max) [1 Clusters]

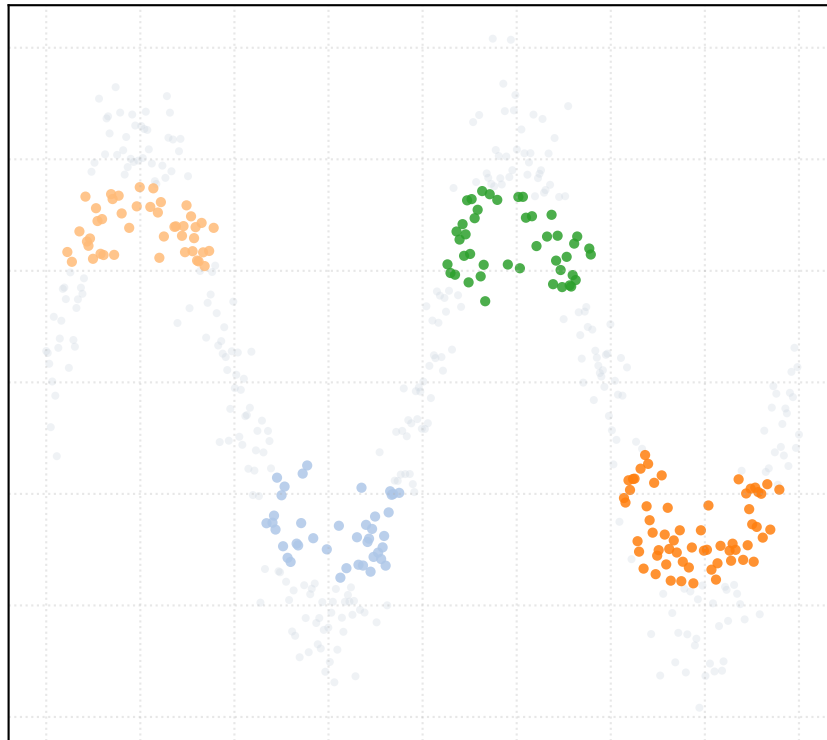


Dataset: s_curve_noise_high | Trajectory Step Threshold Sweep (t_observables in grow_ensemble)

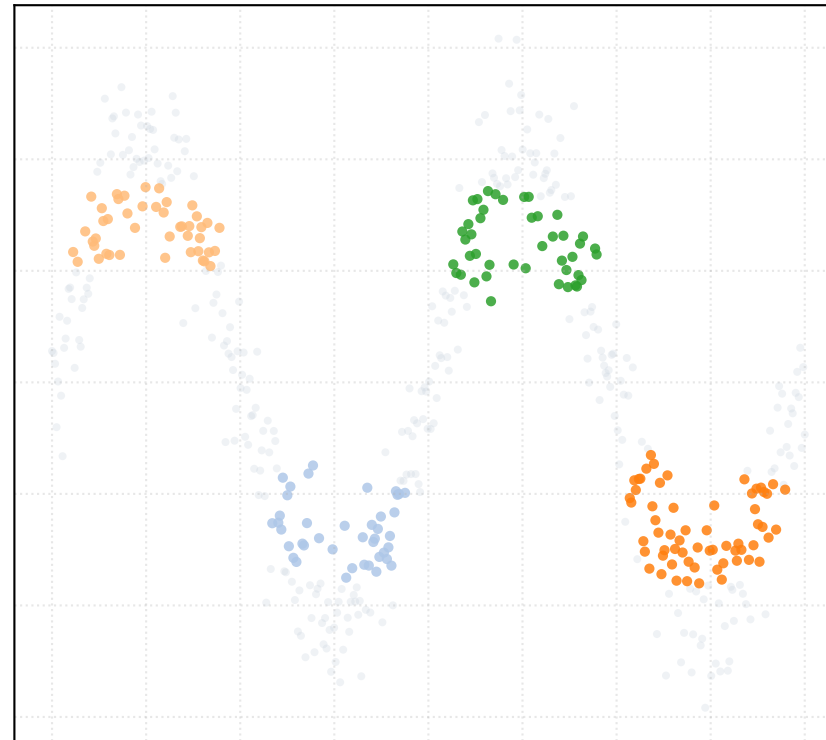
t_obs = Default [4 Clusters]



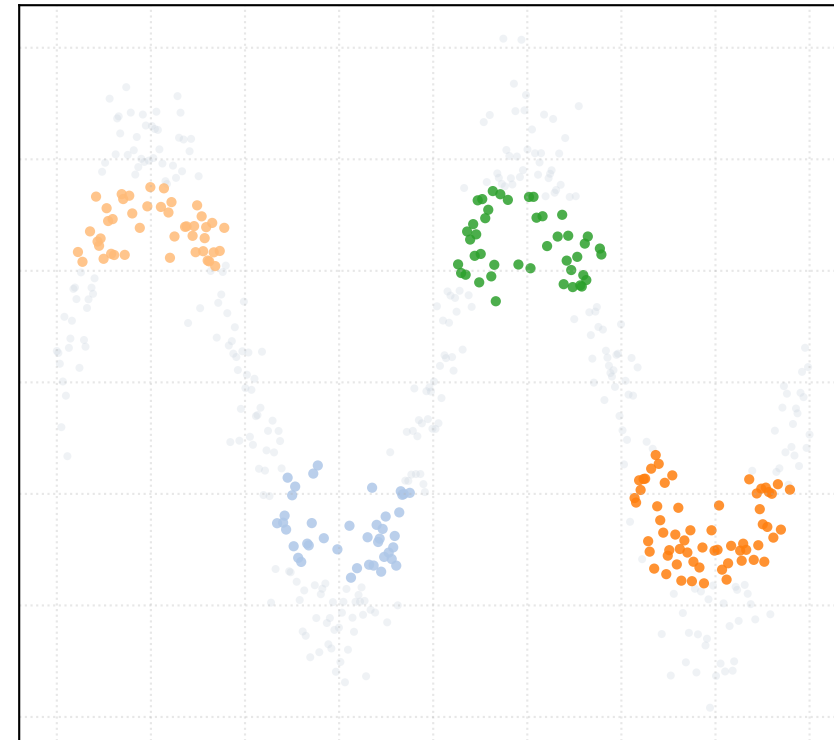
t_obs = 5 [4 Clusters]



t_obs = 50 [4 Clusters]

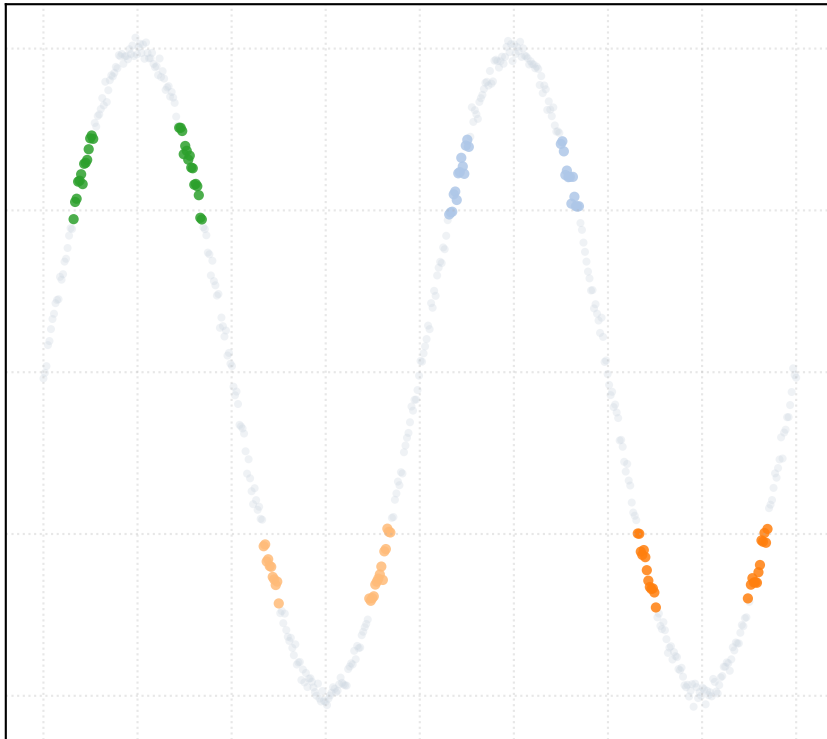


t_obs = 1000 (Max) [4 Clusters]

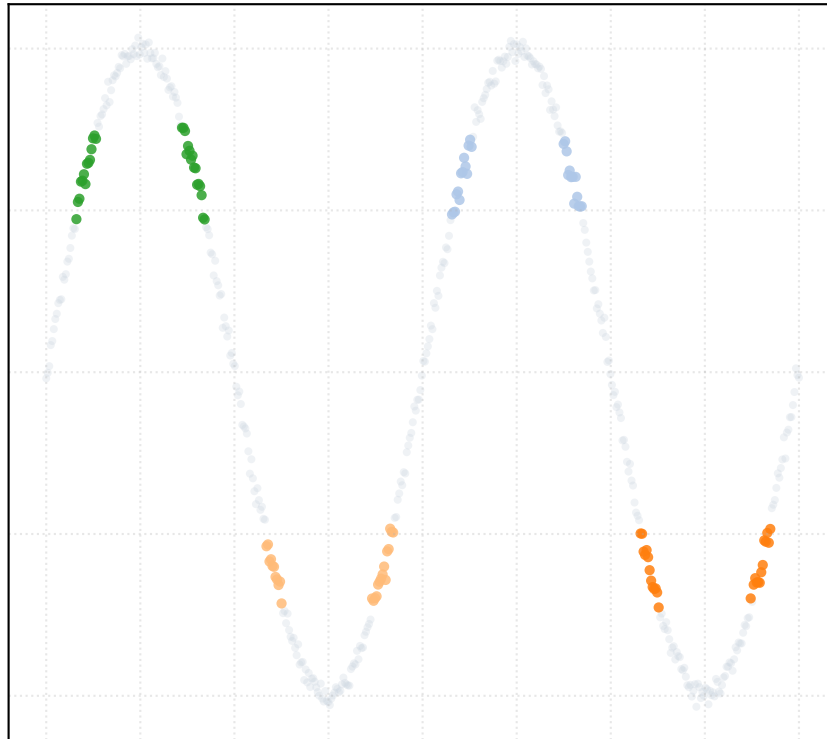


Dataset: s_curve_noise_low | Trajectory Step Threshold Sweep ($t_{\text{observables}}$ in grow_ensemble)

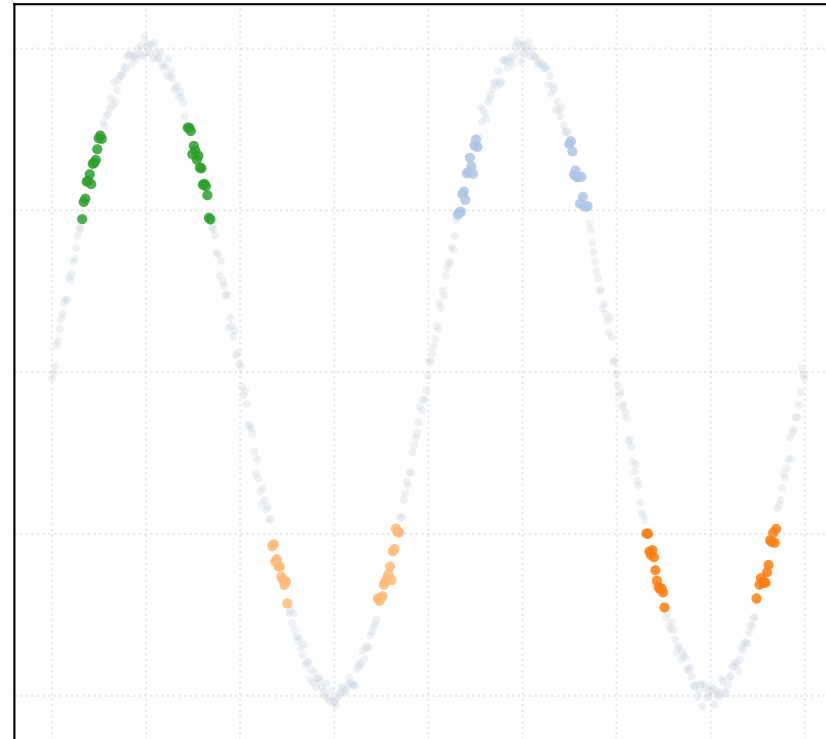
t_{obs} = Default [4 Clusters]



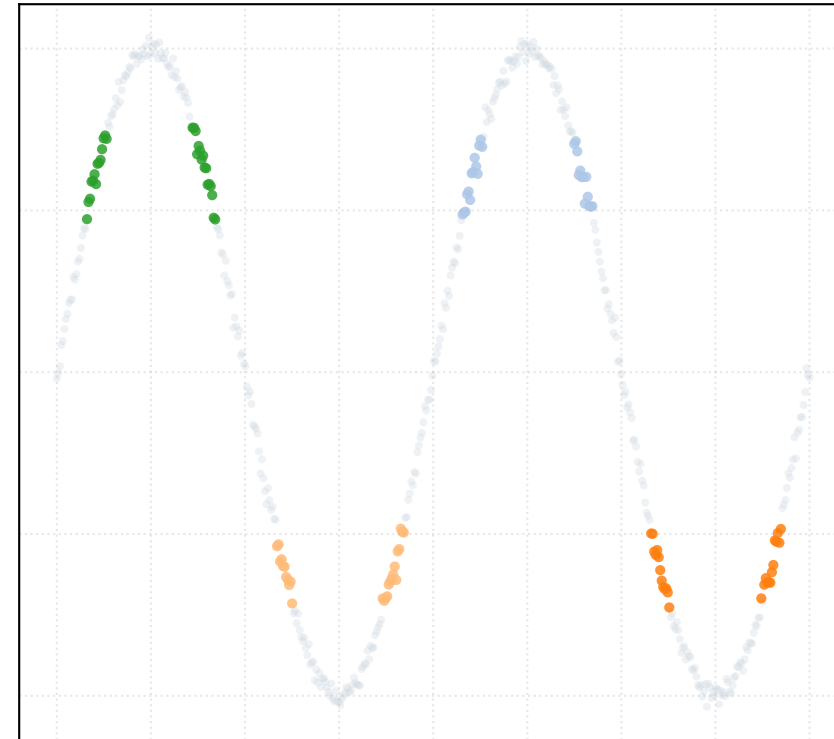
t_{obs} = 5 [4 Clusters]



t_{obs} = 50 [4 Clusters]

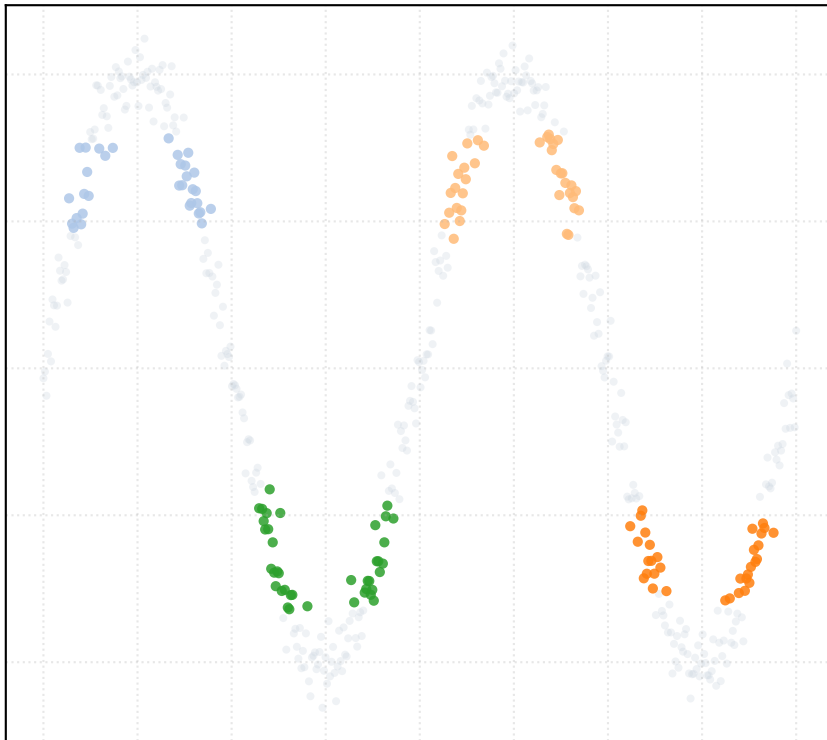


t_{obs} = 1000 (Max) [4 Clusters]

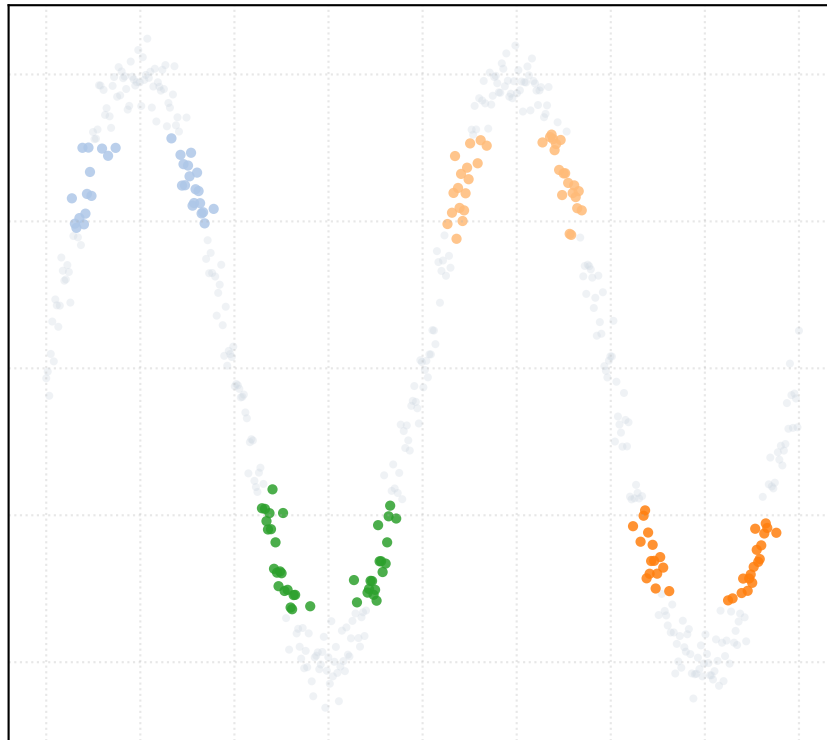


Dataset: s_curve_noise_medium | Trajectory Step Threshold Sweep ($t_{\text{observables}}$ in grow_ensemble)

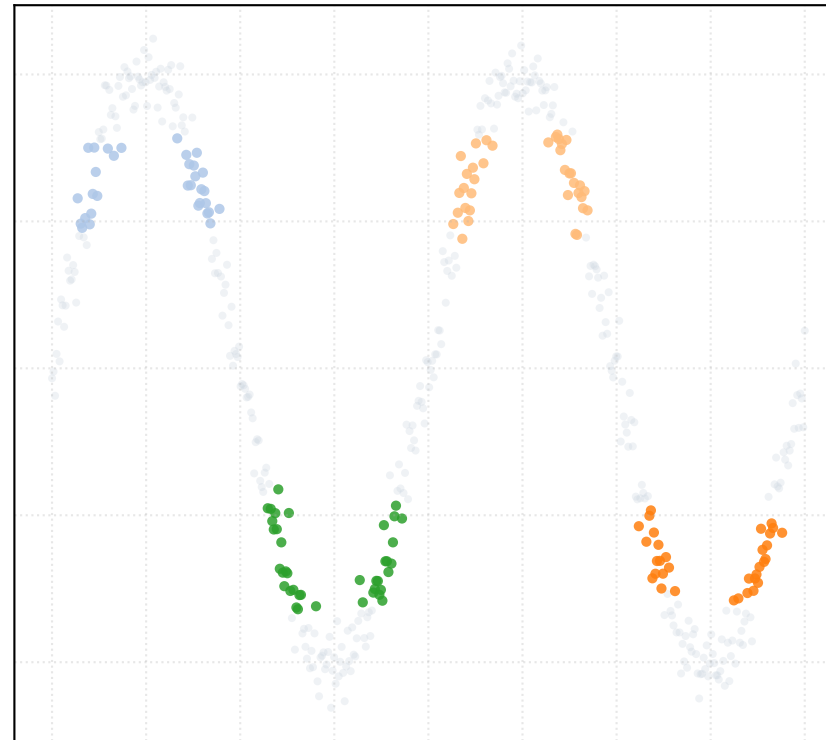
t_{obs} = Default [4 Clusters]



t_{obs} = 5 [4 Clusters]



t_{obs} = 50 [4 Clusters]



t_{obs} = 1000 (Max) [4 Clusters]

